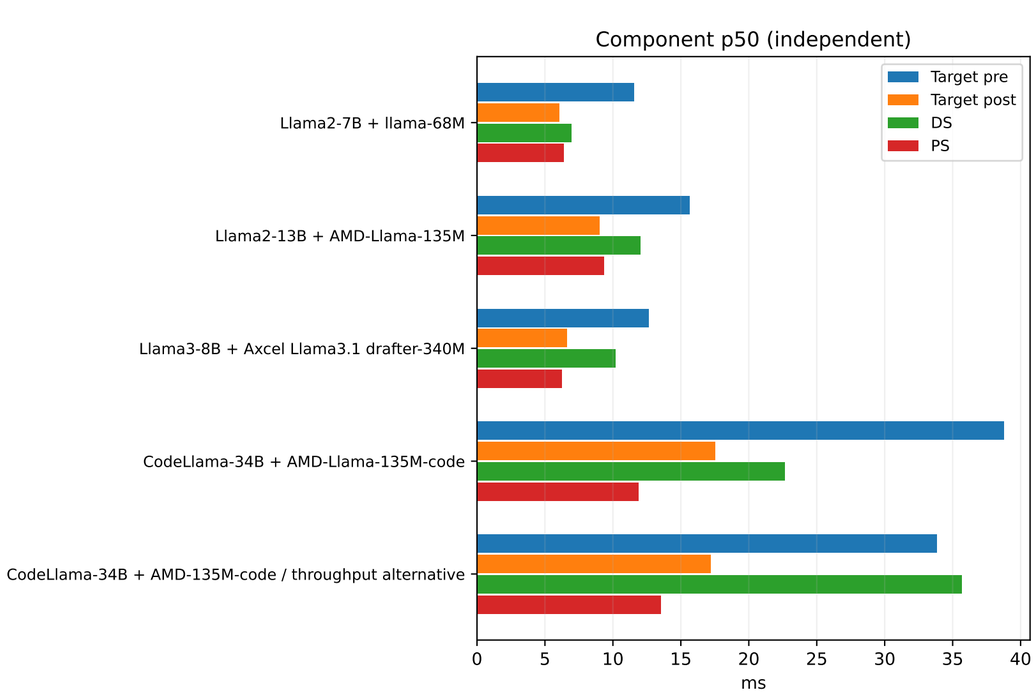
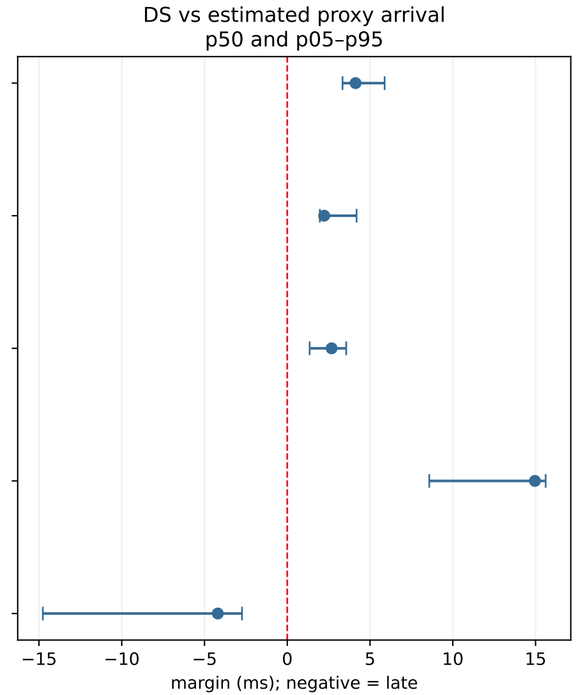
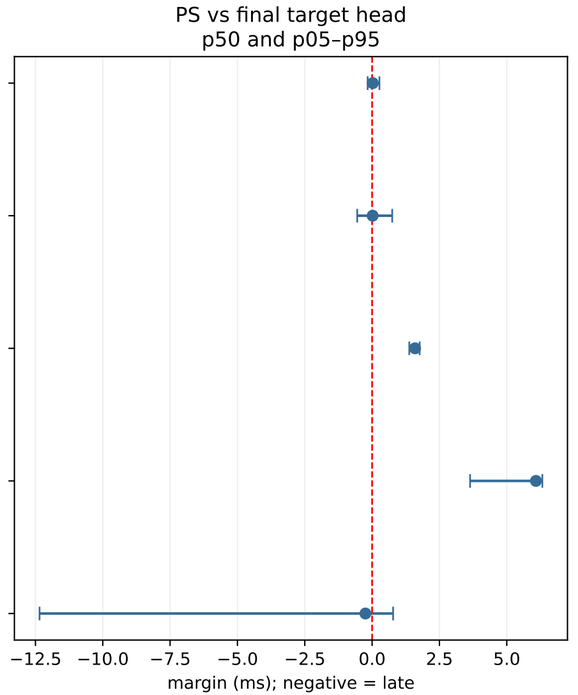
Model-pair latency diagnostics — selected measured v1.0 settings





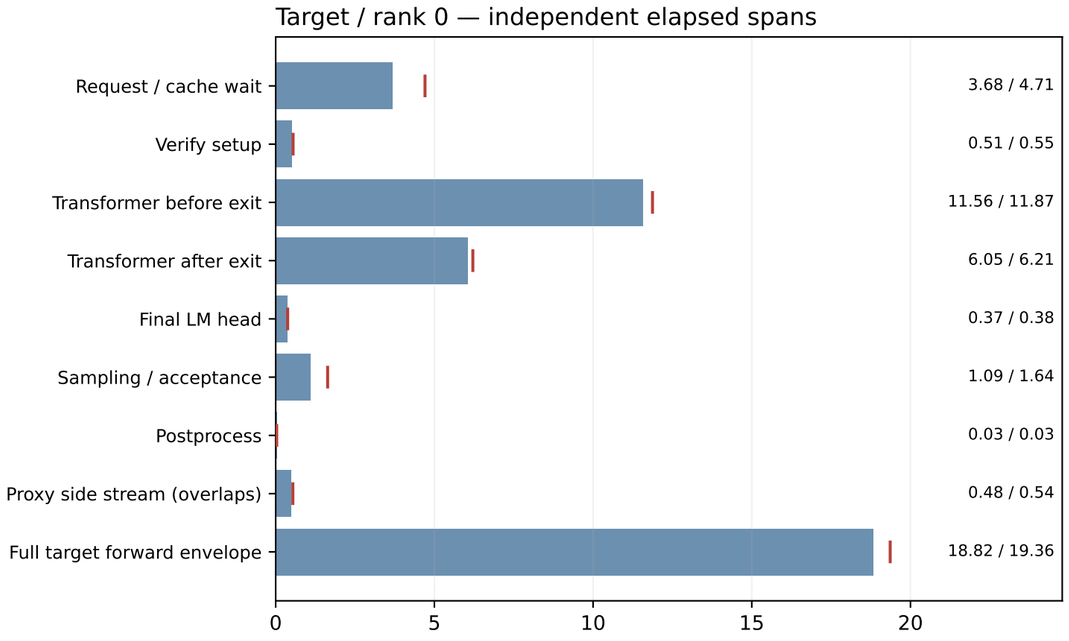


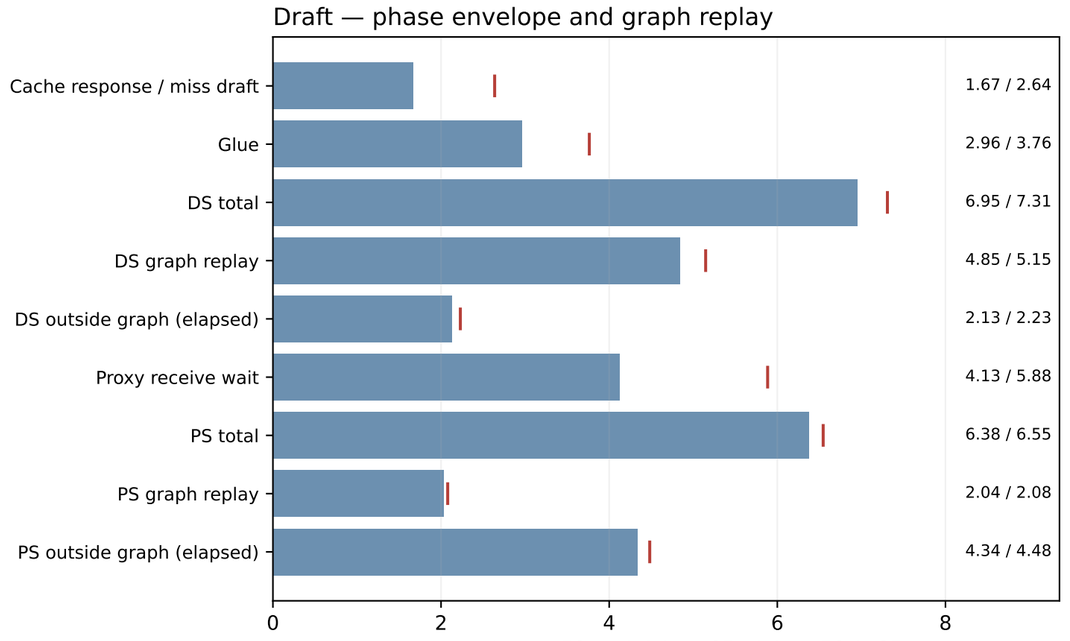
eslab18 RTX3090; B1 BF16 T0.7. Four profiled turns per setting. No model inference rerun; not ver1.1 latency. Target GPU counts differ.

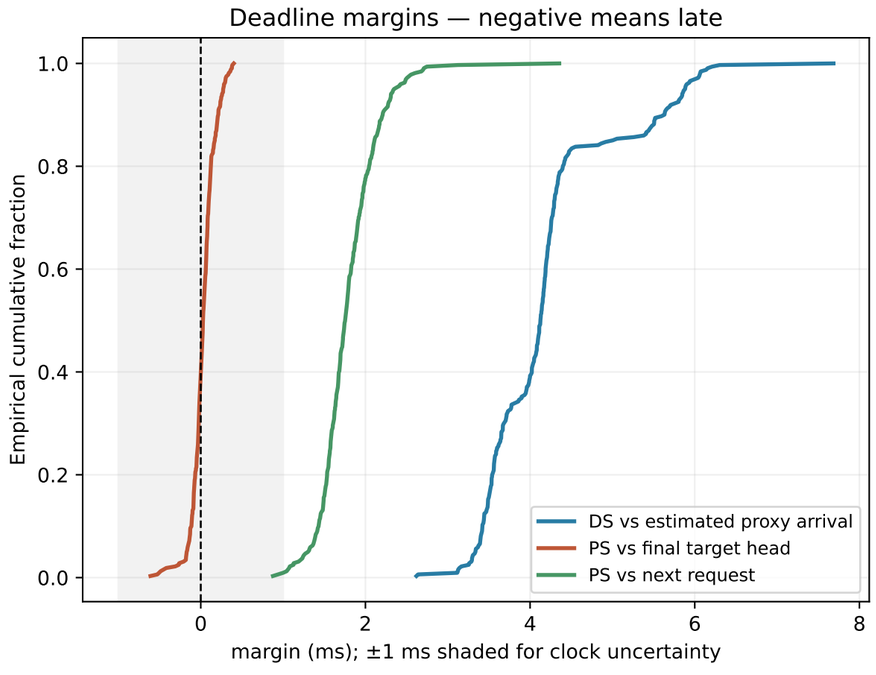
LayerSkip Llama2-7B + llama-68M
Component latency, phase deadlines, and every measured step

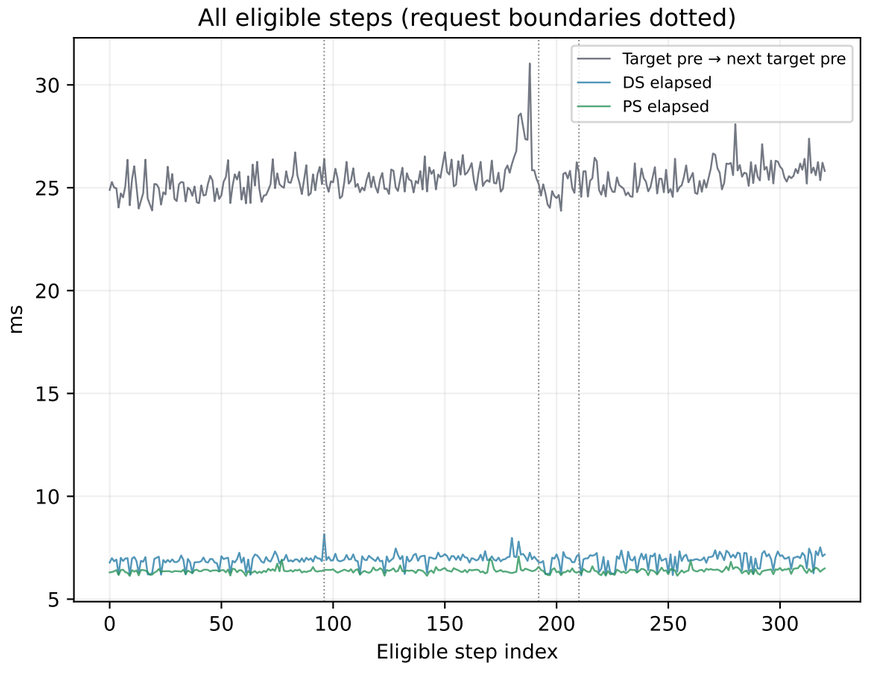
exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 82d964e0 / eslab18 RTX3090 / 4 profiled turns, 321 eligible steps / output cap 256







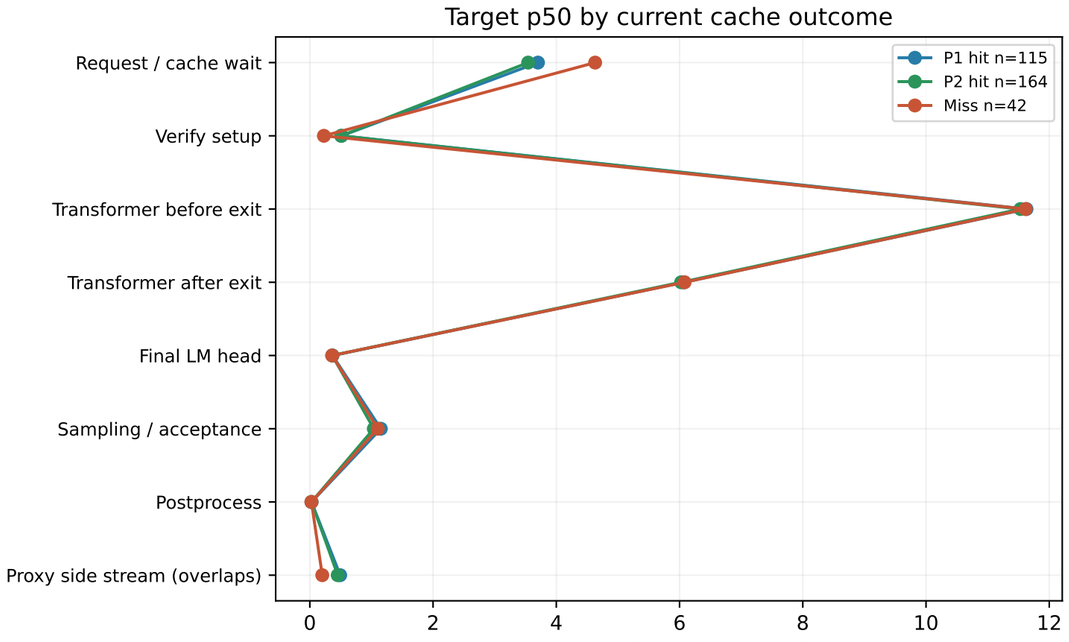


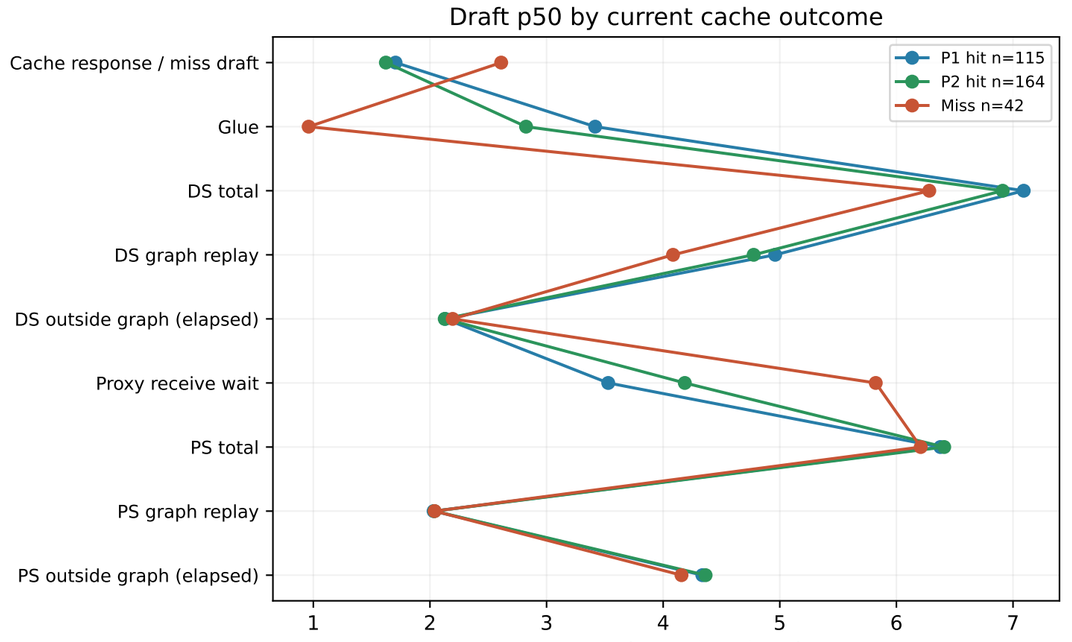
CUDA-event elapsed spans include queue/host stalls; not GPU busy time. Nested/overlapping components are NOT additive.

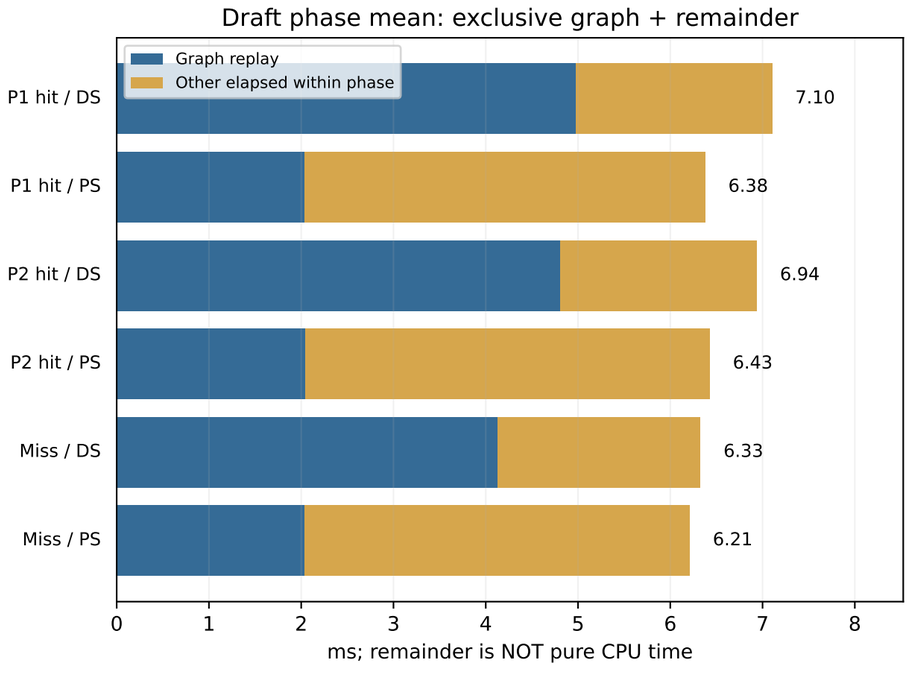
LayerSkip Llama2-7B + llama-68M
P1 hit / P2 hit / miss — component distributions and phase composition

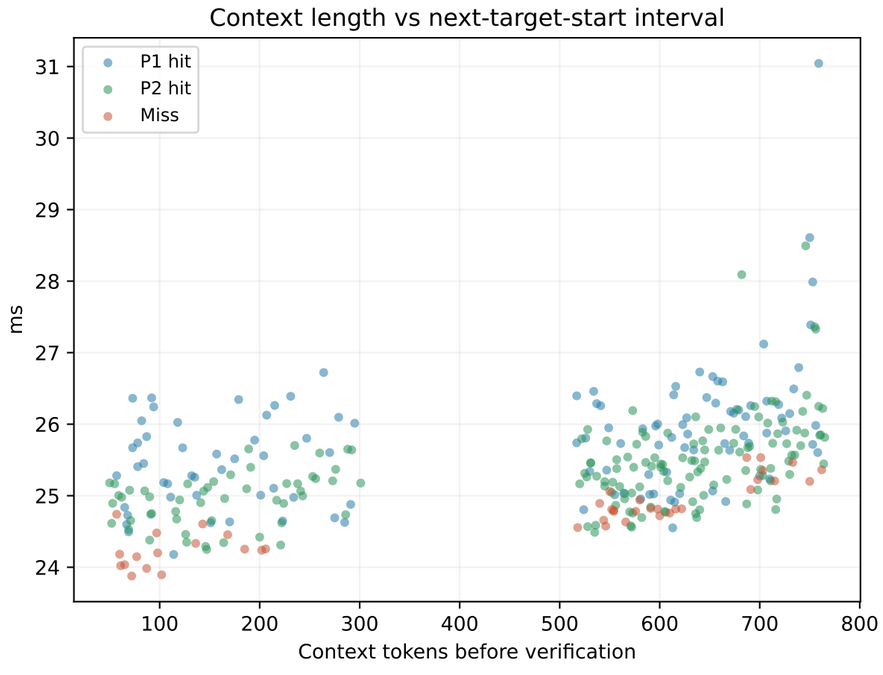
exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 82d964e0 / eslab18 RTX3090 / 4 profiled turns, 321 eligible steps / output cap 256









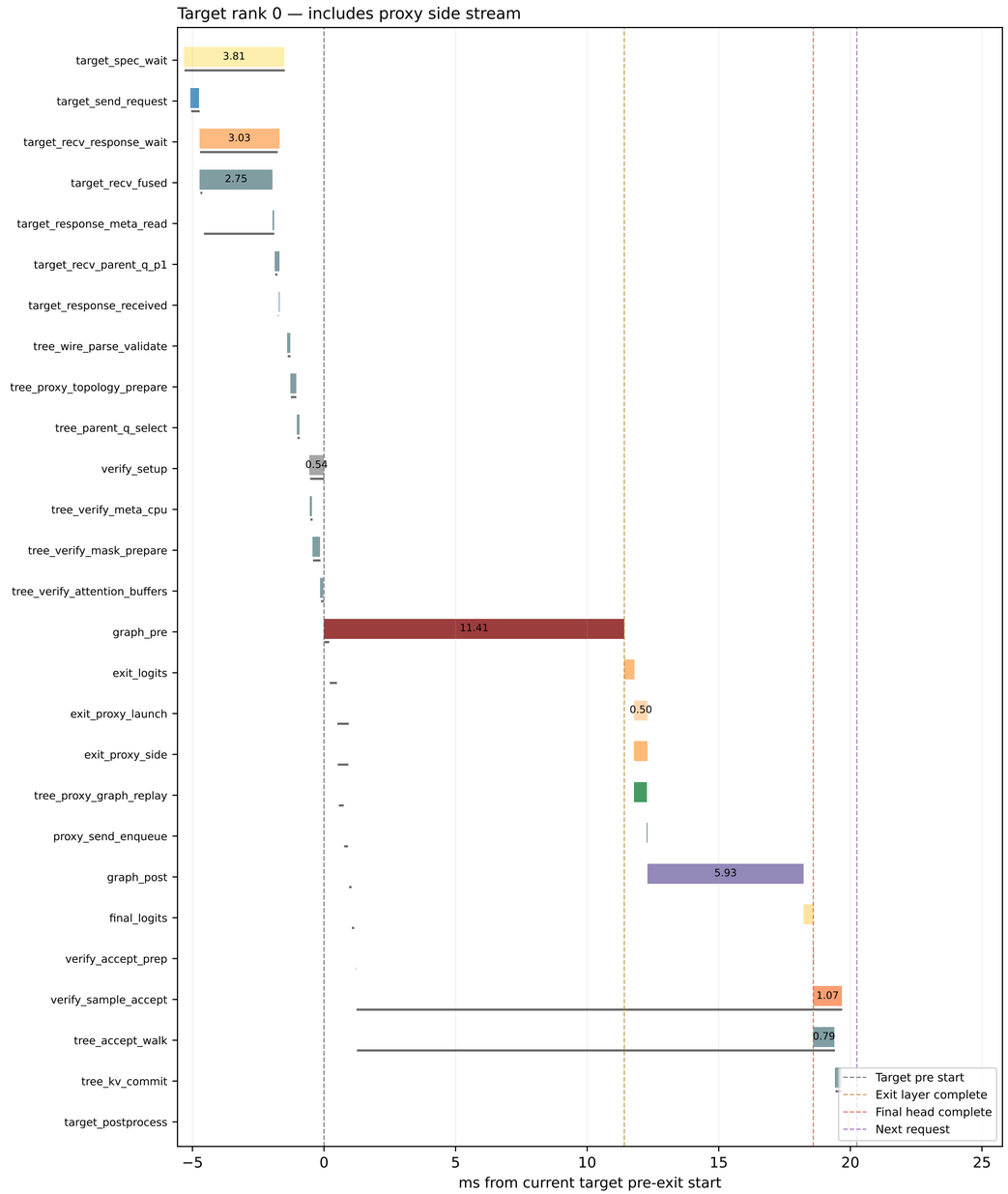
CUDA-event elapsed spans include queue/host stalls; not GPU busy time. Nested/overlapping components are NOT additive.

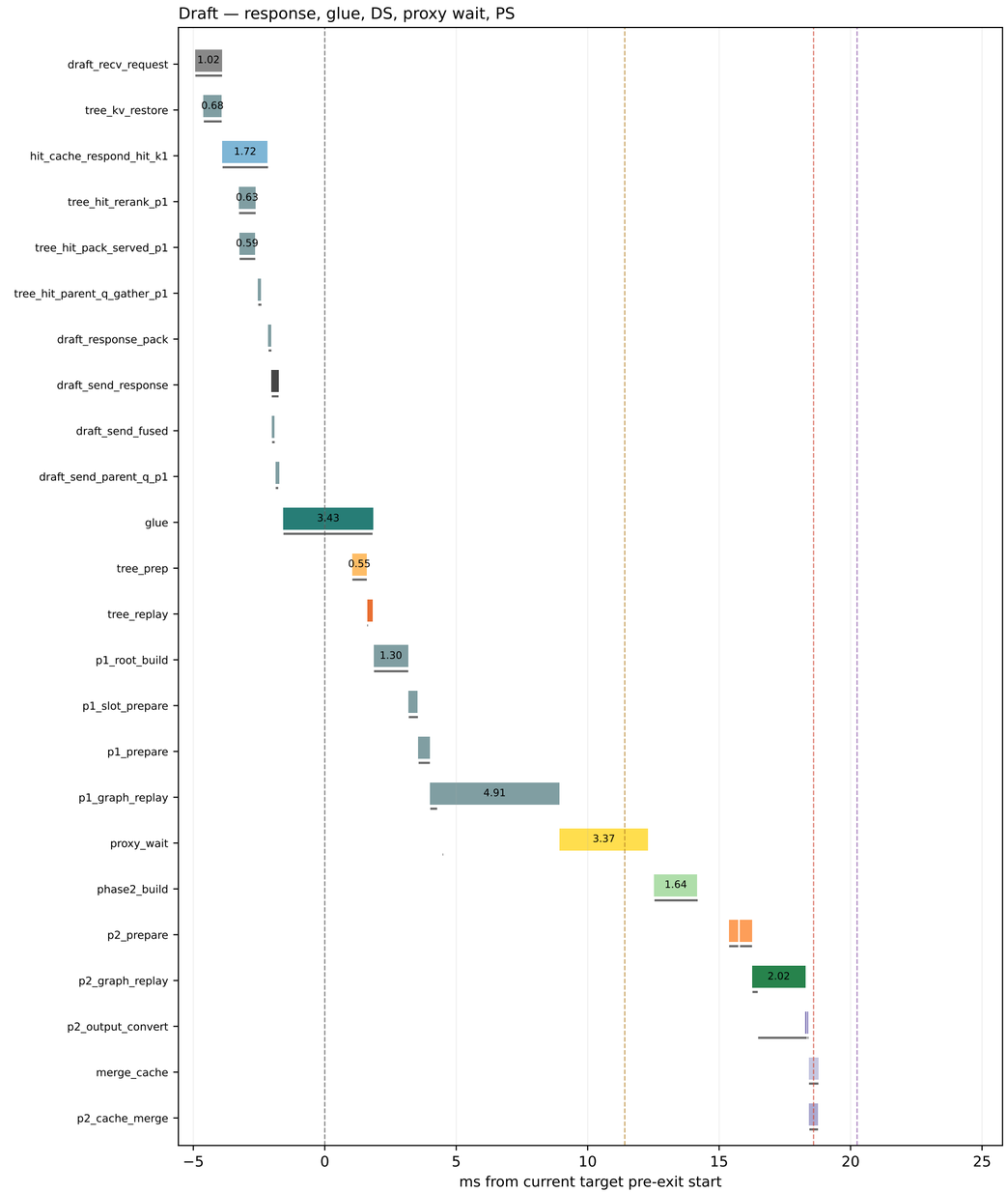
LayerSkip Llama2-7B + llama-68M
P1 hit / p25 target-step duration / request 1, UID 84_t0, step 25

exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 82d964e0 / eslab18 RTX3090 / 4 profiled turns, 321 eligible steps / output cap 256

valid proposal nodes=12, AL incl. recovery=3; DS margin +3.368 ms (estimated); PS margin -0.174 ms; next outcome P1 hit





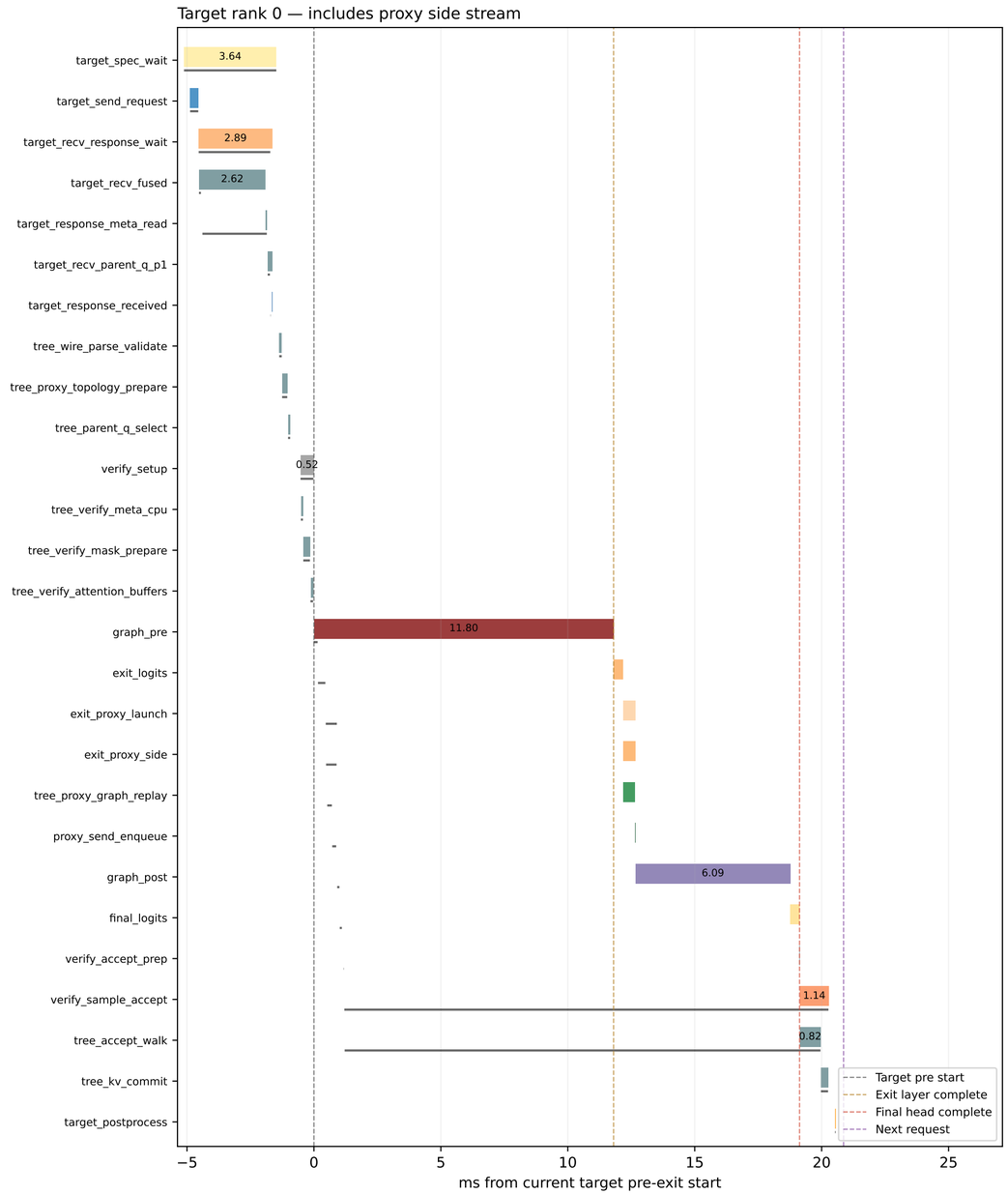
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

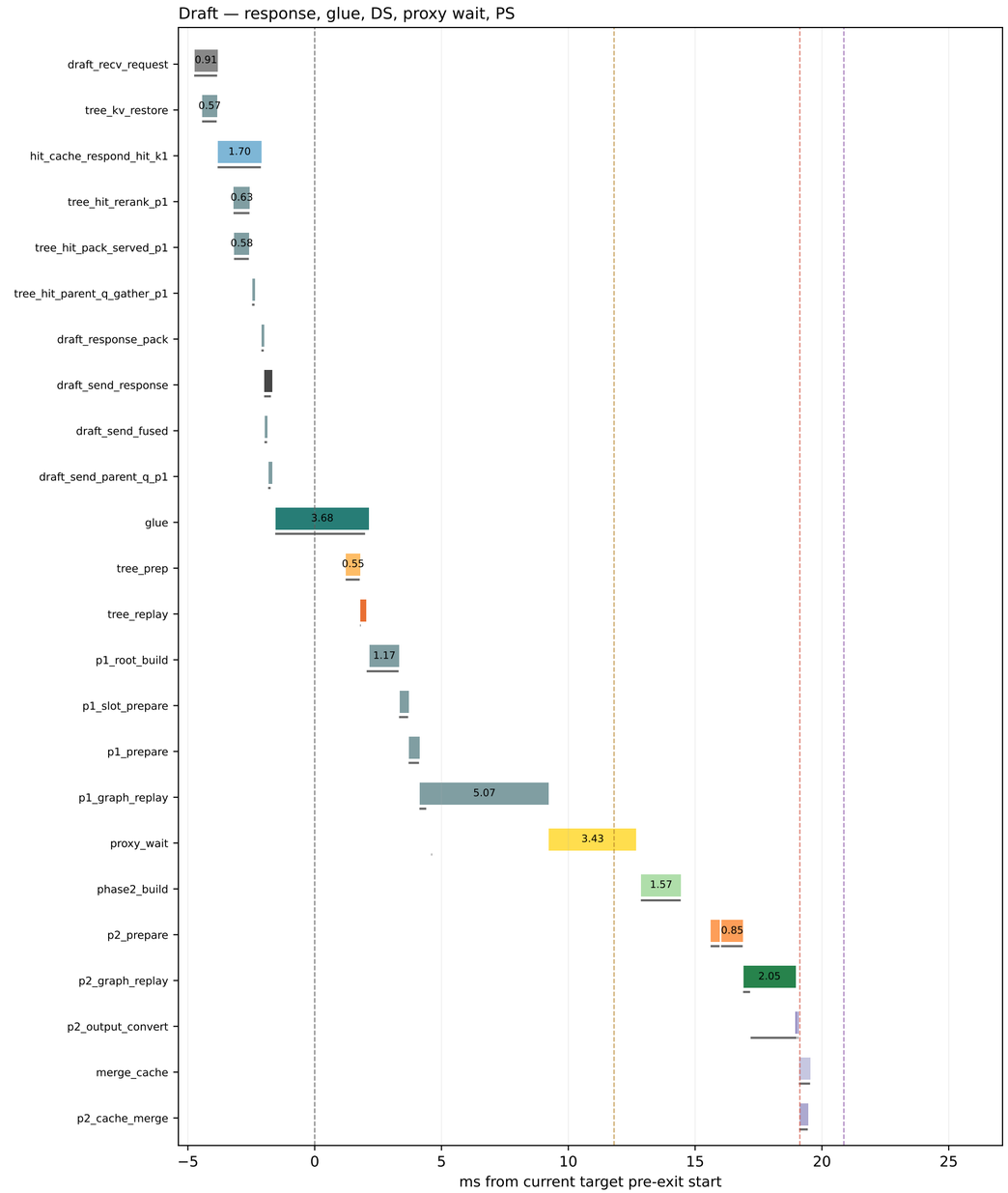
LayerSkip Llama2-7B + llama-68M
P1 hit / p50 target-step duration / request 2, UID 244_t0, step 78

exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 82d964e0 / eslab18 RTX3090 / 4 profiled turns, 321 eligible steps / output cap 256

valid proposal nodes=12, AL incl. recovery=3; DS margin +3.433 ms (estimated); PS margin -0.419 ms; next outcome P1 hit





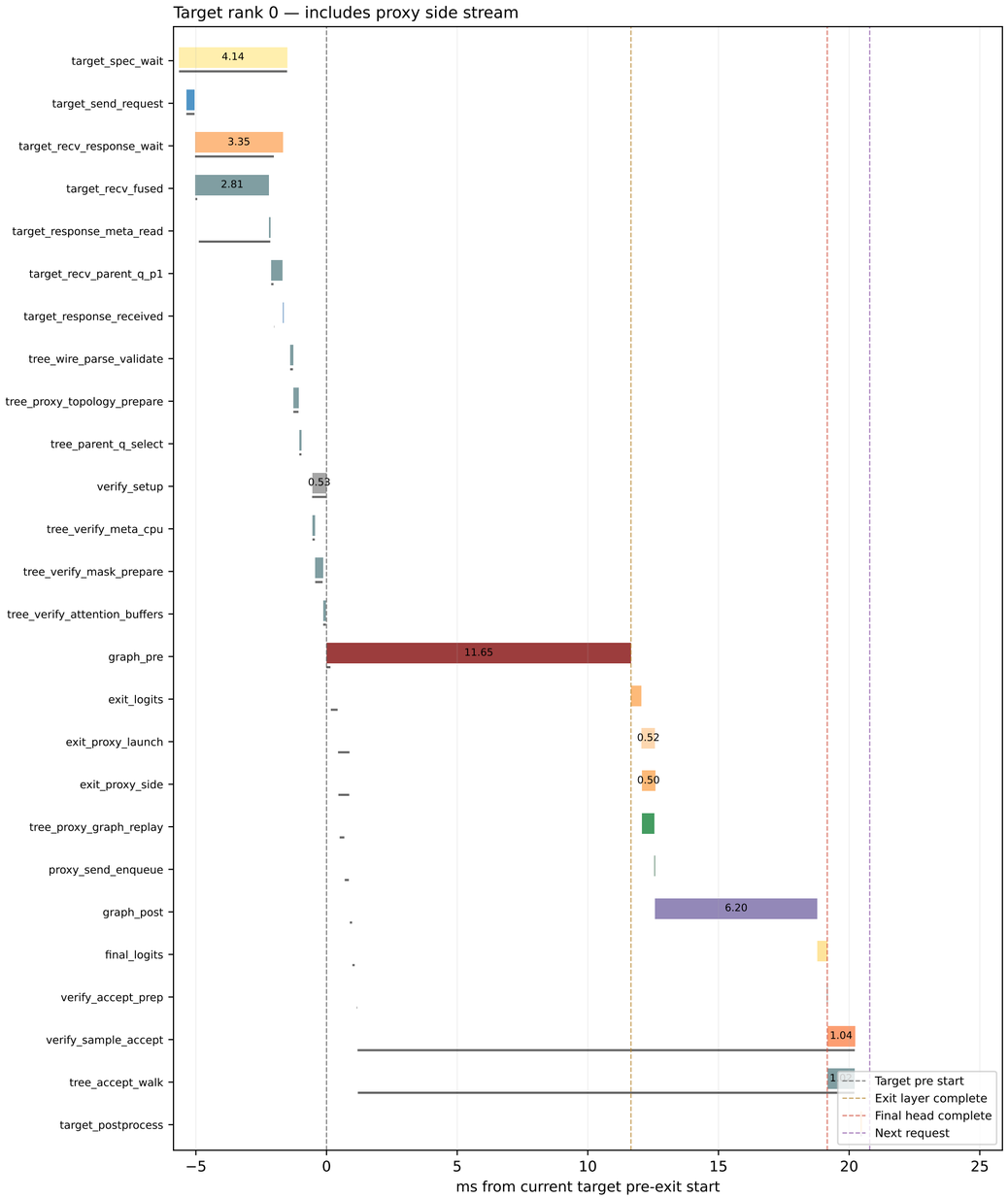
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

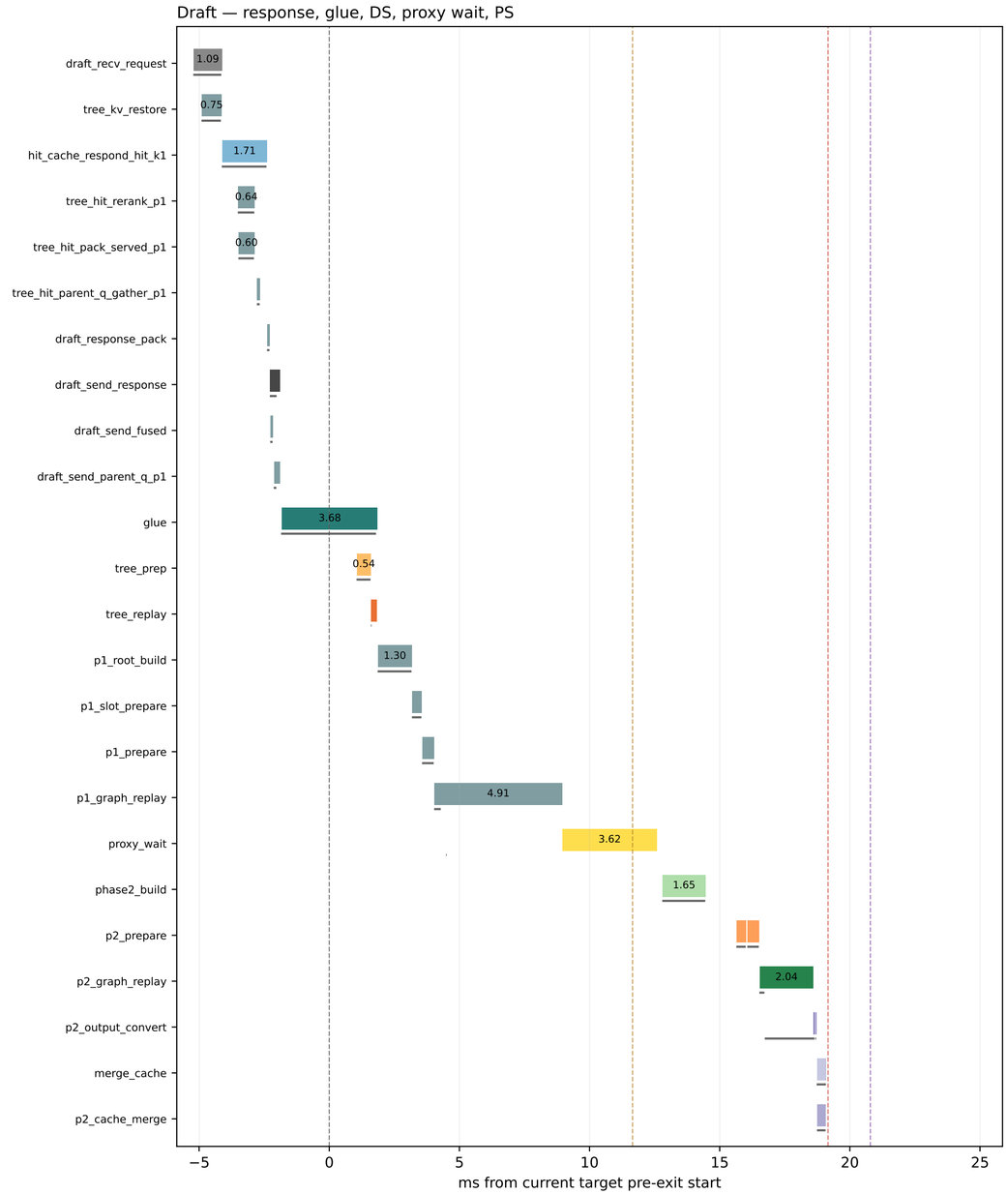
LayerSkip Llama2-7B + llama-68M
P1 hit / p75 target-step duration / request 2, UID 244_t0, step 61

exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 82d964e0 / eslab18 RTX3090 / 4 profiled turns, 321 eligible steps / output cap 256

valid proposal nodes=12, AL incl. recovery=1; DS margin +3.618 ms (estimated); PS margin +0.071 ms; next outcome P2 hit





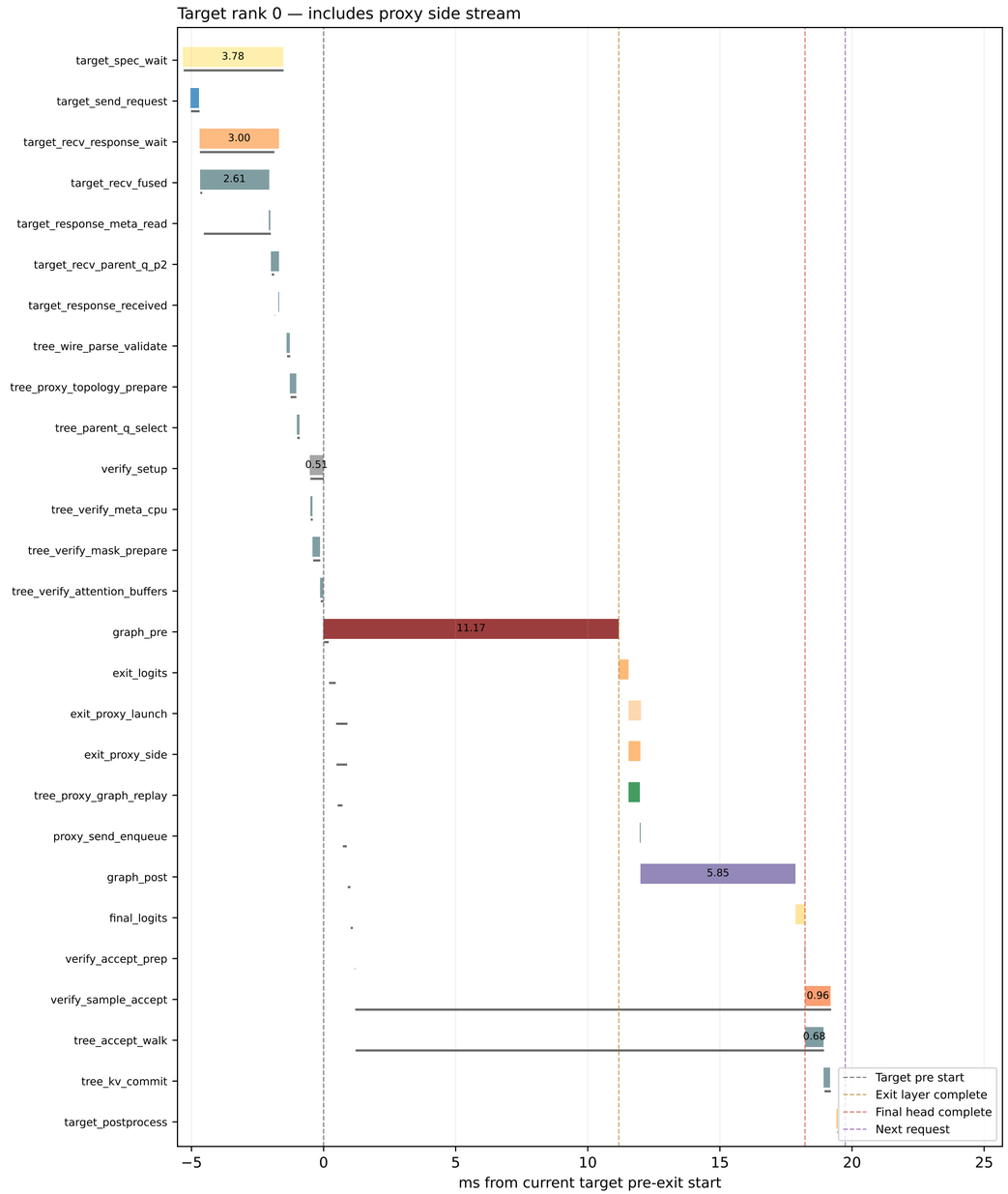
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

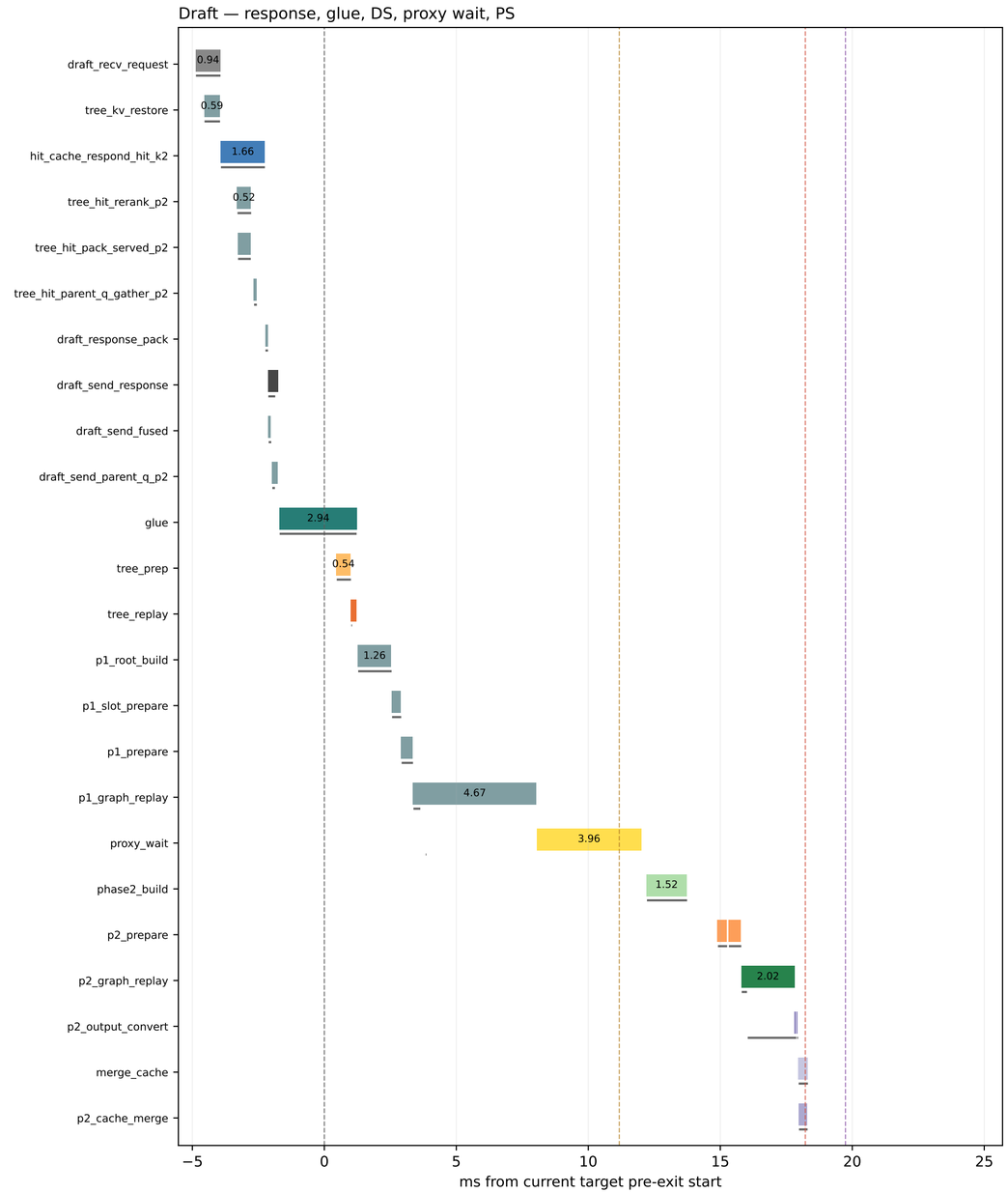
LayerSkip Llama2-7B + llama-68M
P2 hit / p25 target-step duration / request 1, UID 84_t0, step 3

exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 82d964e0 / eslab18 RTX3090 / 4 profiled turns, 321 eligible steps / output cap 256

valid proposal nodes=8, AL incl. recovery=4; DS margin +3.959 ms (estimated); PS margin -0.085 ms; next outcome P1 hit





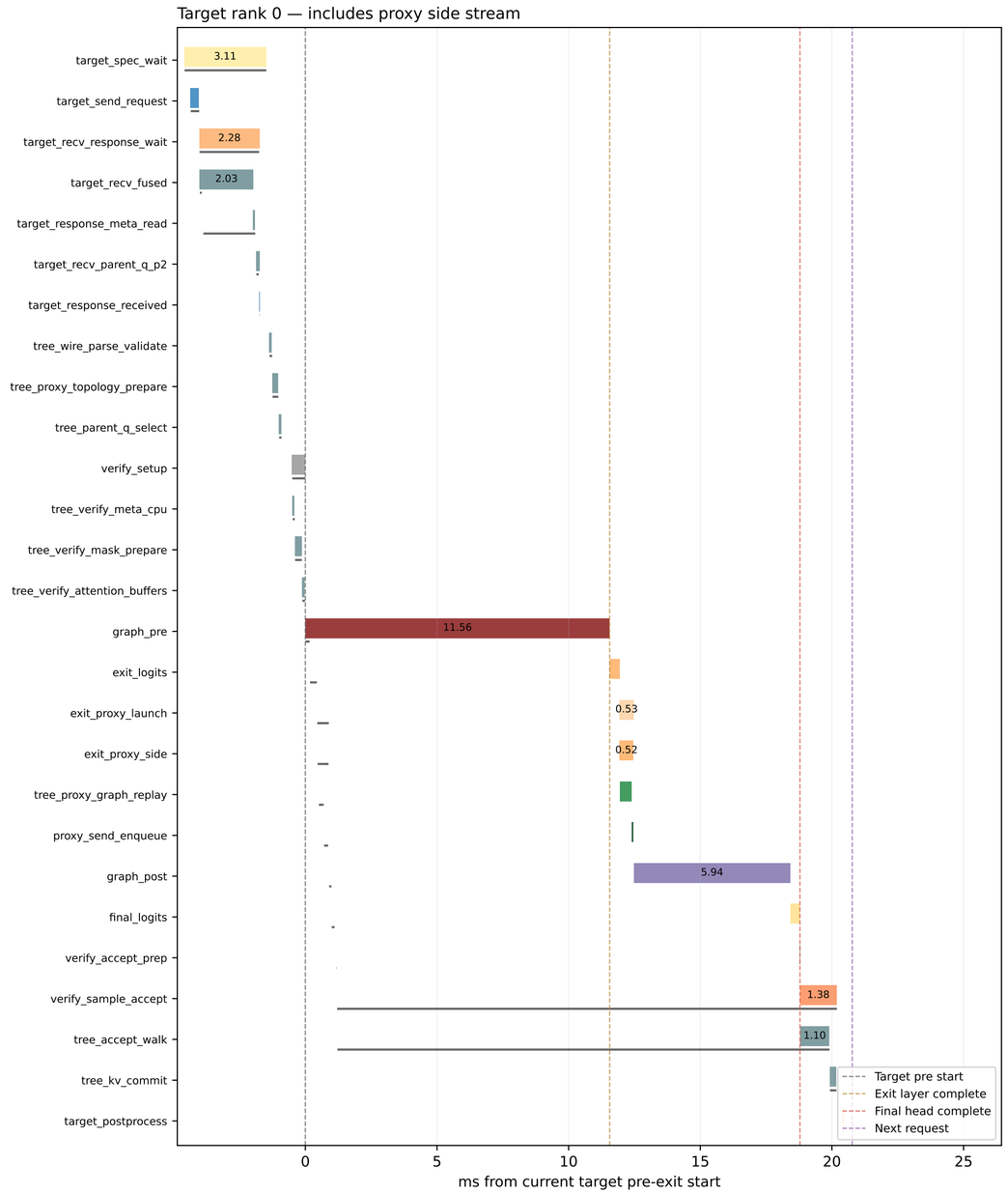
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

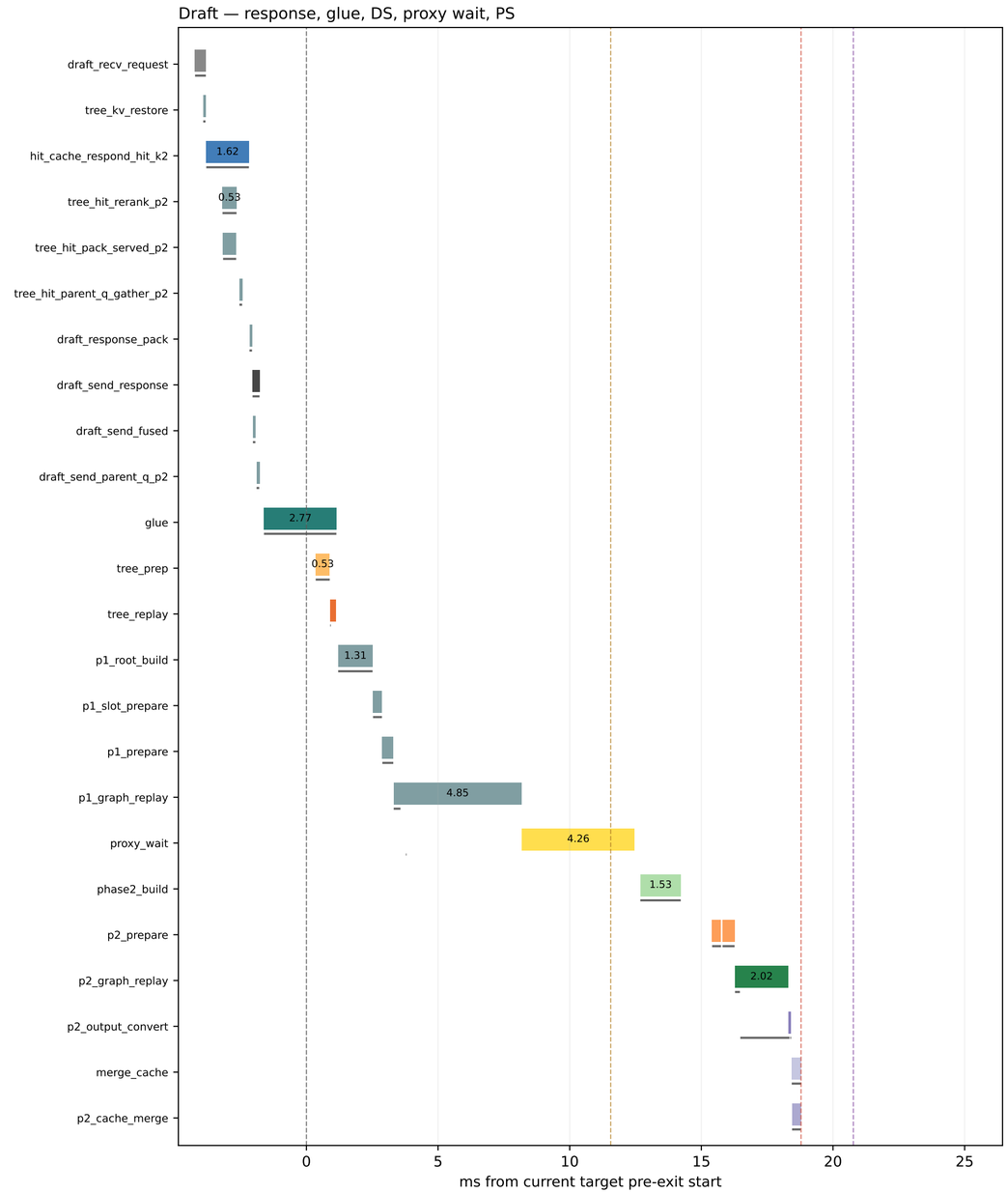
LayerSkip Llama2-7B + llama-68M
P2 hit / p50 target-step duration / request 1, UID 84_t0, step 96

exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 82d964e0 / eslab18 RTX3090 / 4 profiled turns, 321 eligible steps / output cap 256

valid proposal nodes=8, AL incl. recovery=3; DS margin +4.265 ms (estimated); PS margin -0.011 ms; next outcome P1 hit





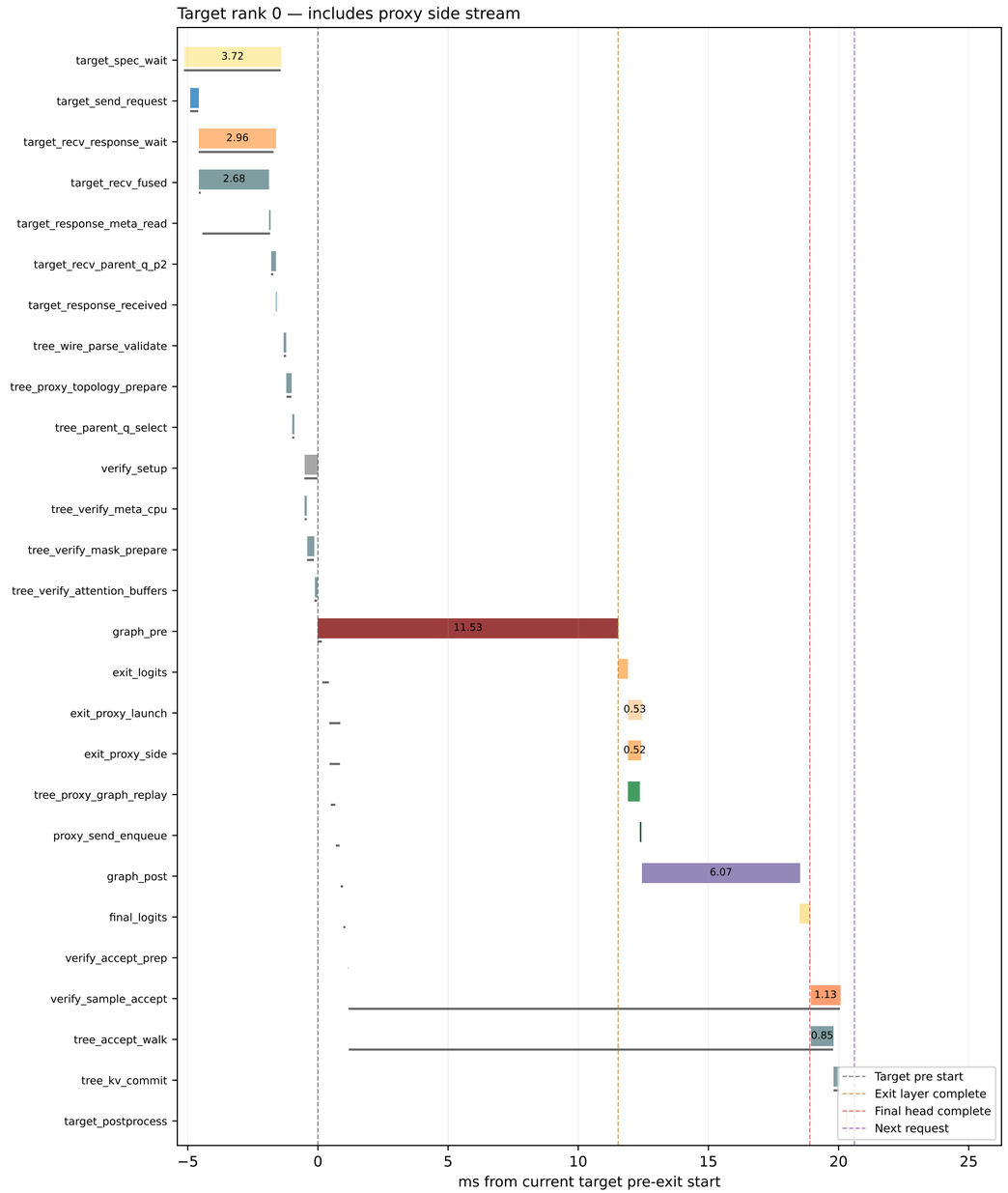
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

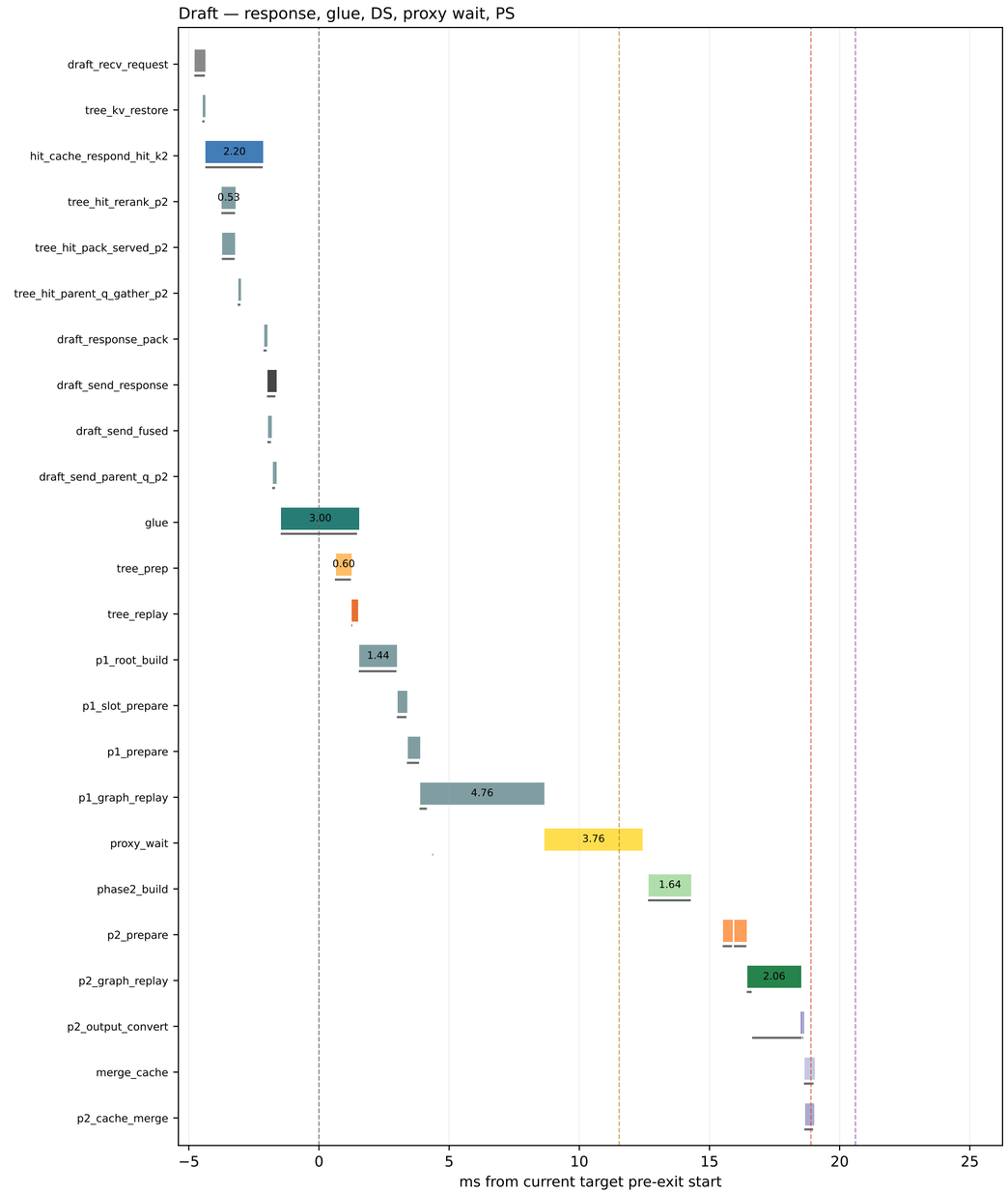
LayerSkip Llama2-7B + llama-68M
P2 hit / p75 target-step duration / request 4, UID 288_t0, step 9

exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 82d964e0 / eslab18 RTX3090 / 4 profiled turns, 321 eligible steps / output cap 256

valid proposal nodes=8, AL incl. recovery=3; DS margin +3.763 ms (estimated); PS margin -0.128 ms; next outcome P1 hit





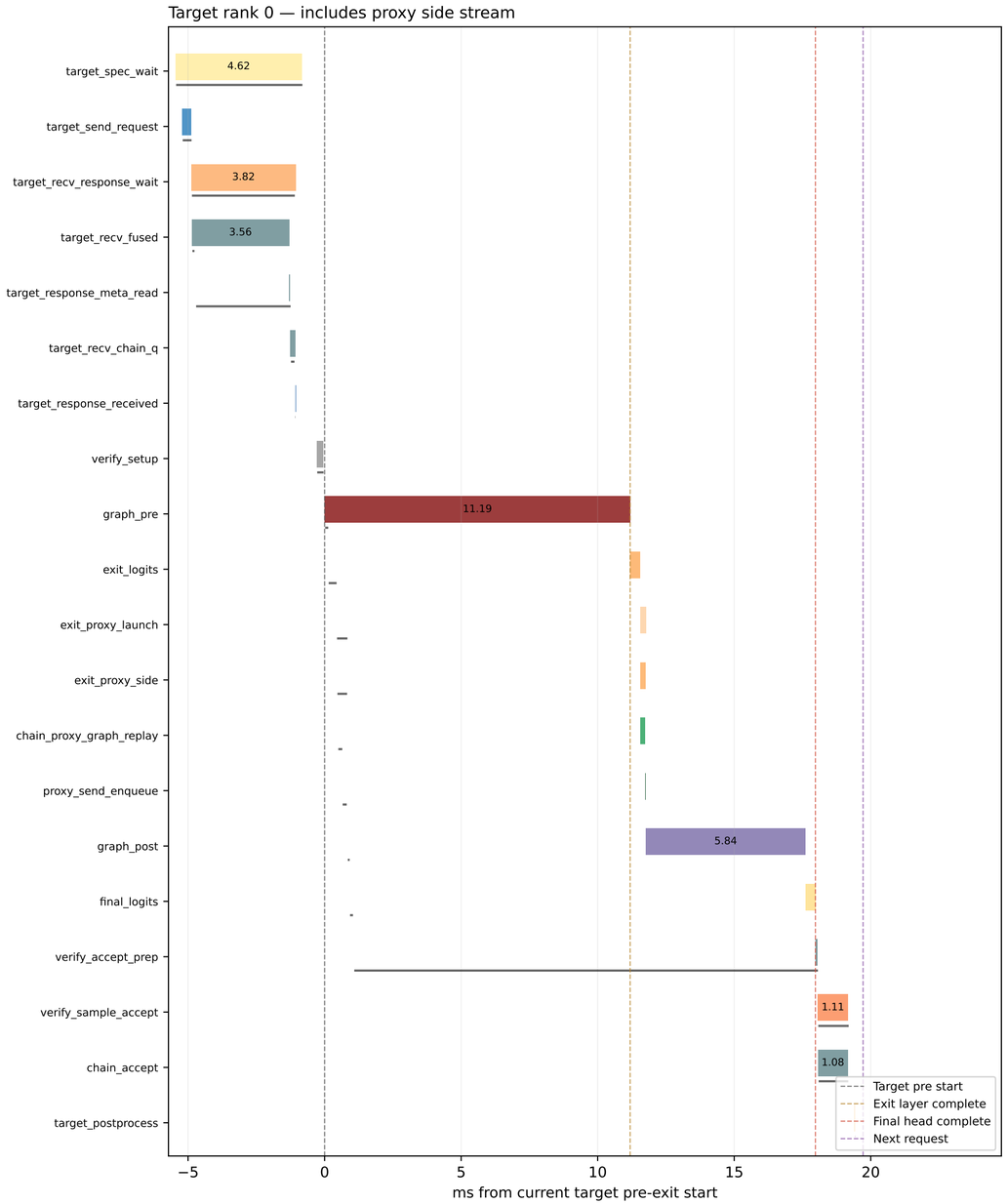
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

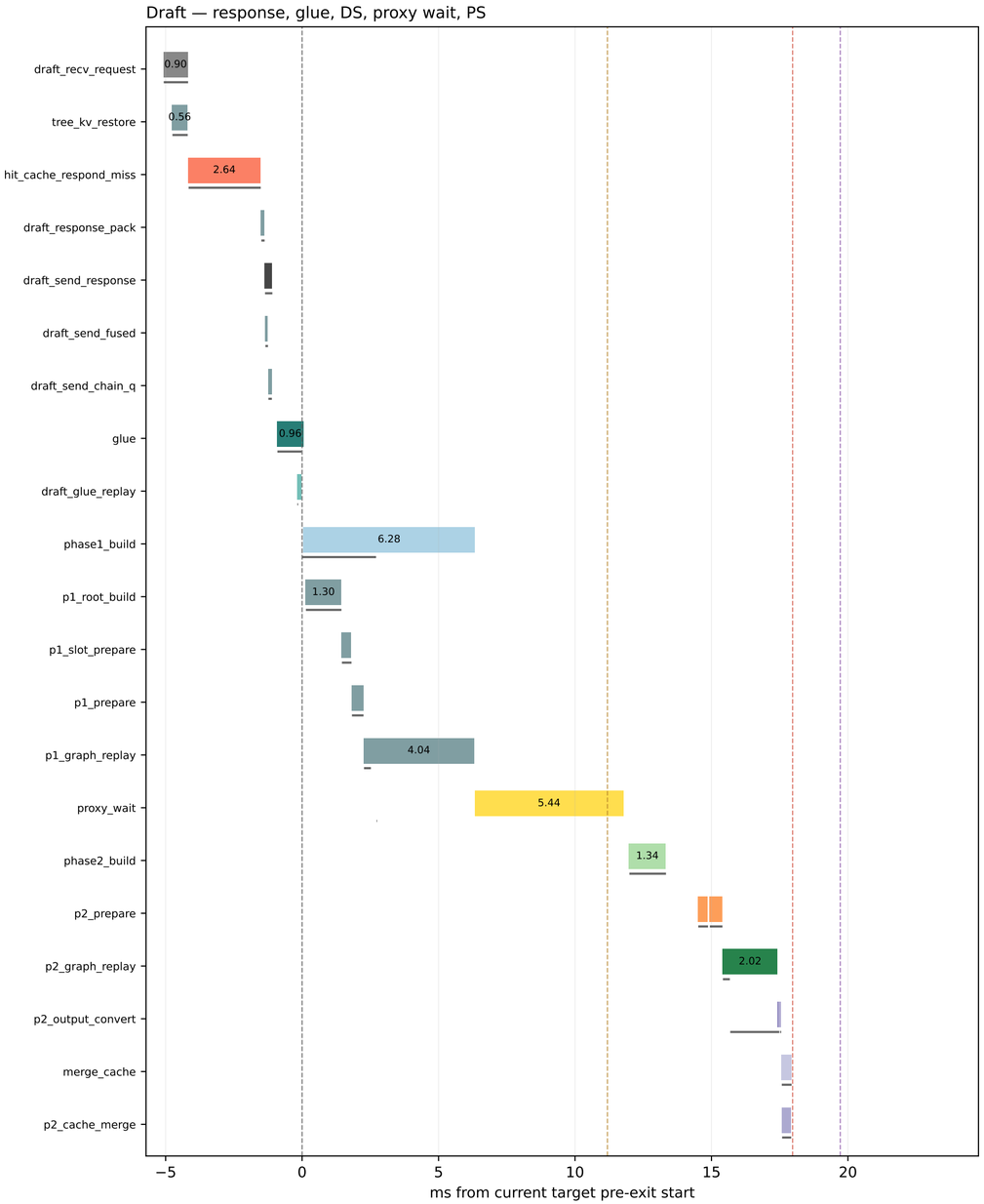
LayerSkip Llama2-7B + llama-68M
Miss / p25 target-step duration / request 1, UID 84_t0, step 16

exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 82d964e0 / eslab18 RTX3090 / 4 profiled turns, 321 eligible steps / output cap 256

valid proposal nodes=3, AL incl. recovery=3; DS margin +5.439 ms (estimated); PS margin +0.049 ms; next outcome P2 hit





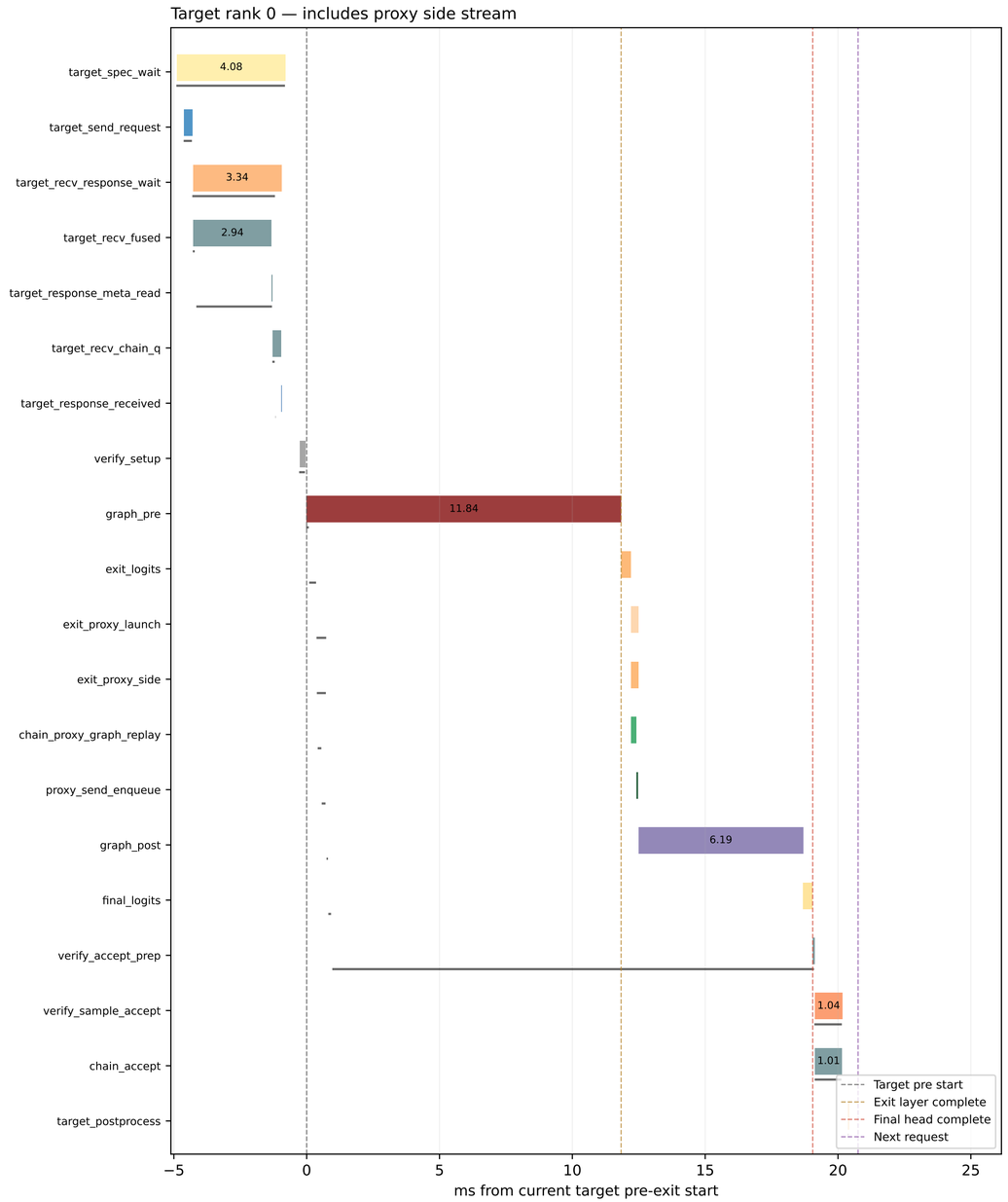
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

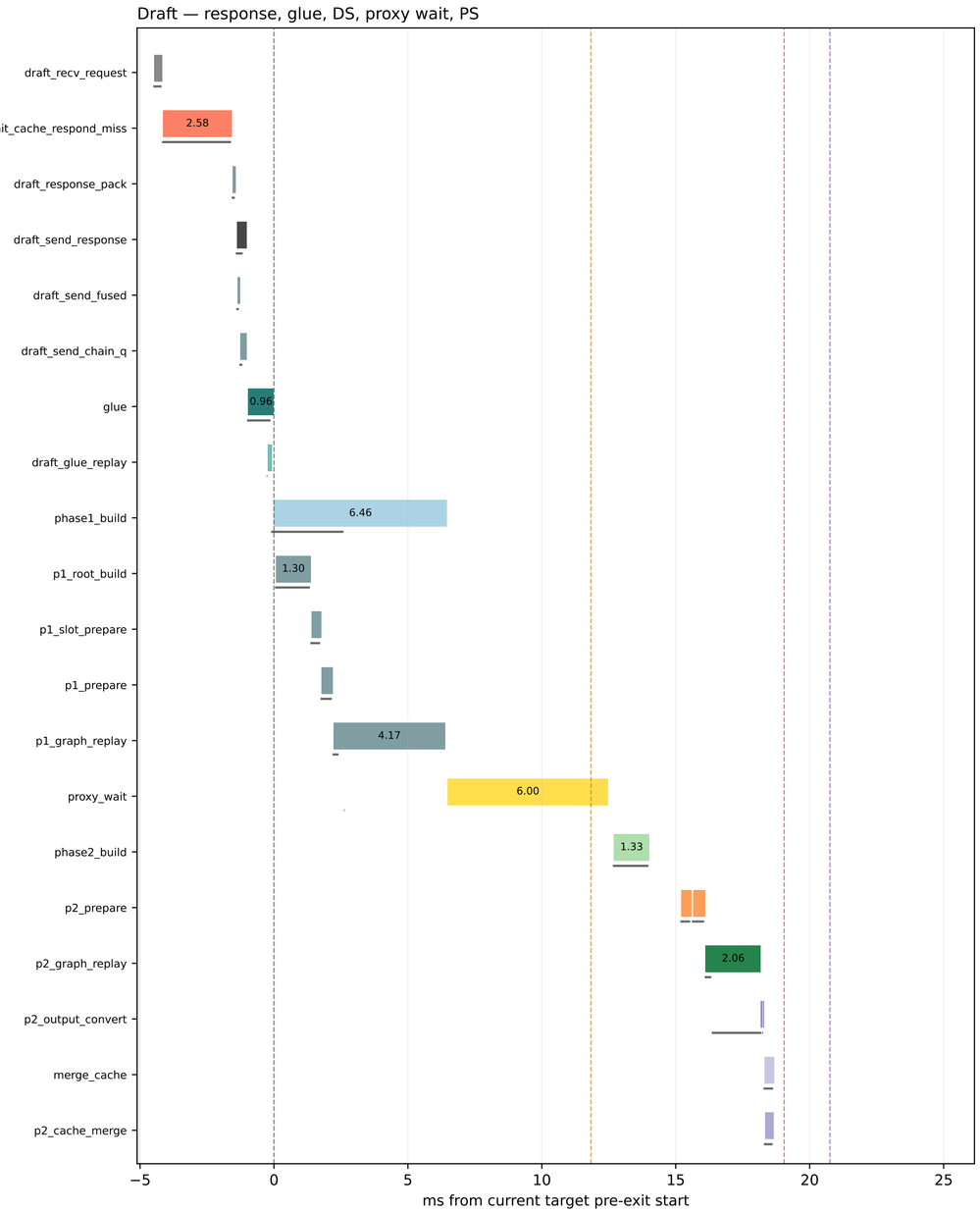
LayerSkip Llama2-7B + llama-68M
Miss / p50 target-step duration / request 4, UID 288_t0, step 84

exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 82d964e0 / eslab18 RTX3090 / 4 profiled turns, 321 eligible steps / output cap 256

valid proposal nodes=3, AL incl. recovery=1; DS margin +5.994 ms (estimated); PS margin +0.379 ms; next outcome P1 hit





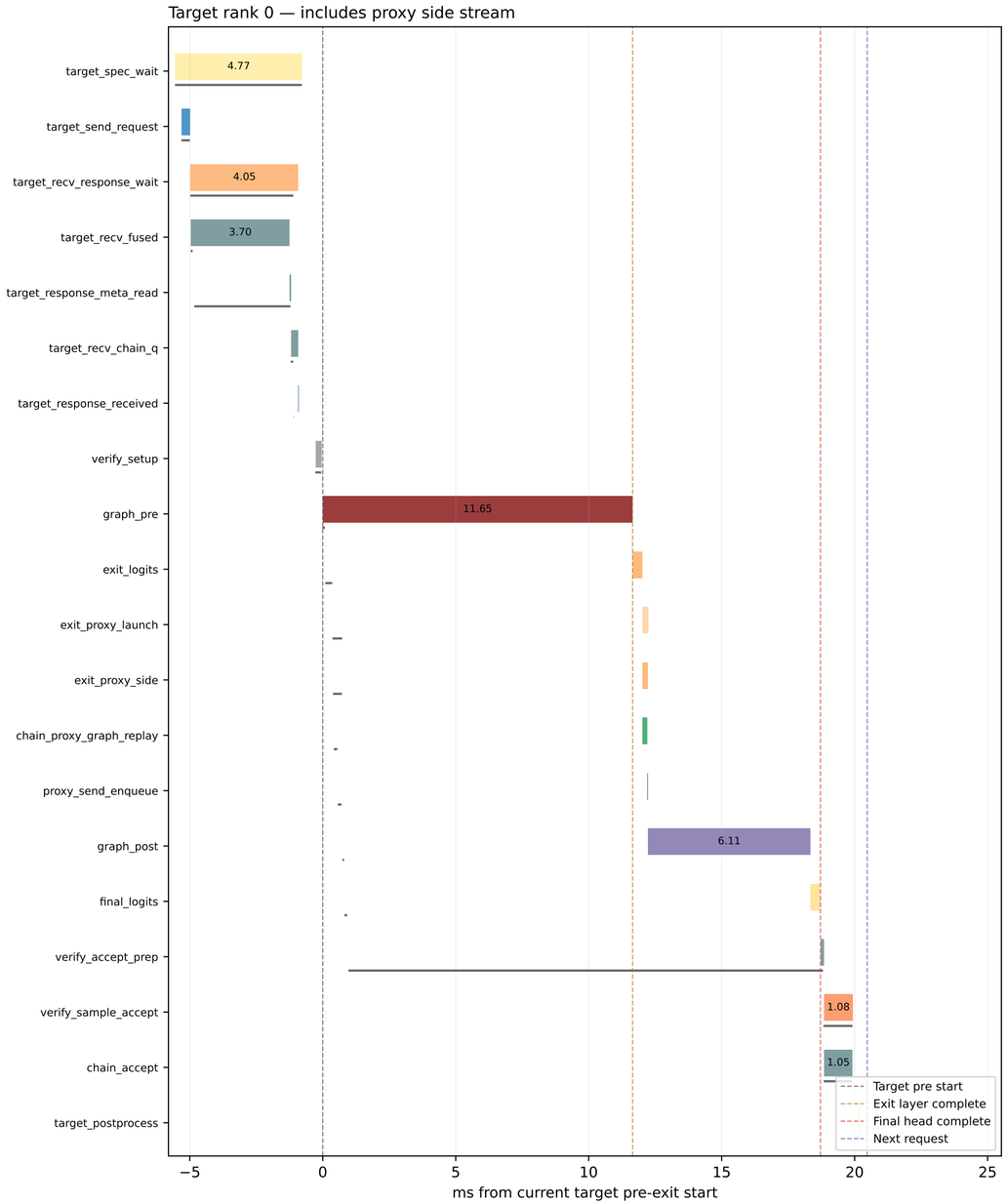
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

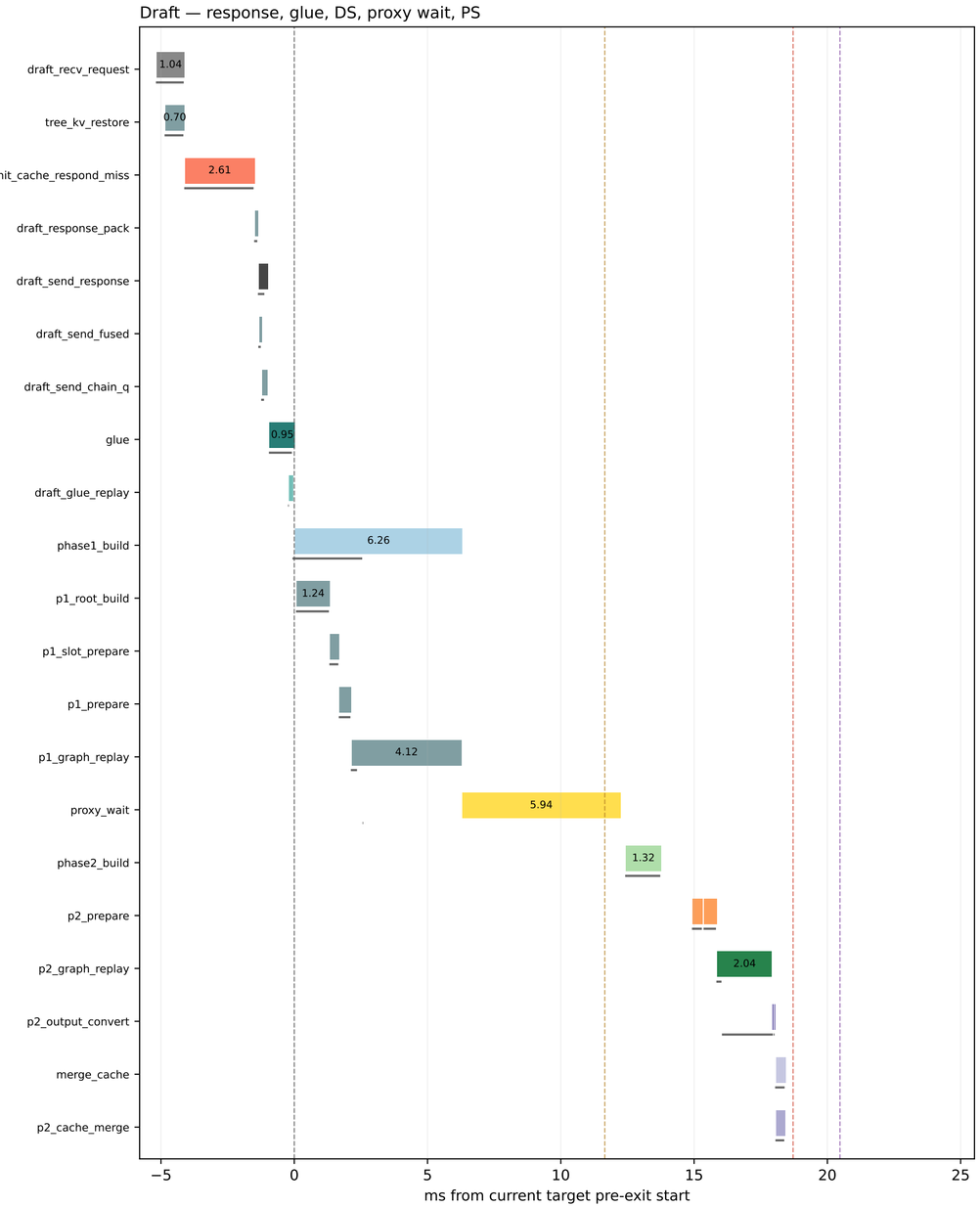
LayerSkip Llama2-7B + llama-68M
Miss / p75 target-step duration / request 4, UID 288_t0, step 38

exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 82d964e0 / eslab18 RTX3090 / 4 profiled turns, 321 eligible steps / output cap 256

valid proposal nodes=3, AL incl. recovery=1; DS margin +5.941 ms (estimated); PS margin +0.281 ms; next outcome P2 hit



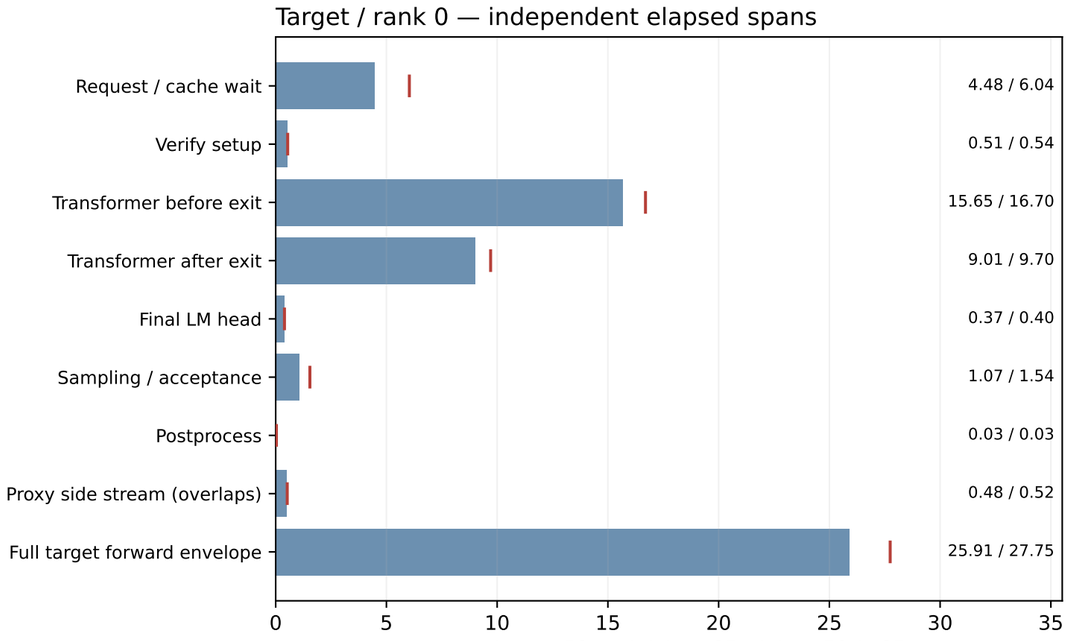


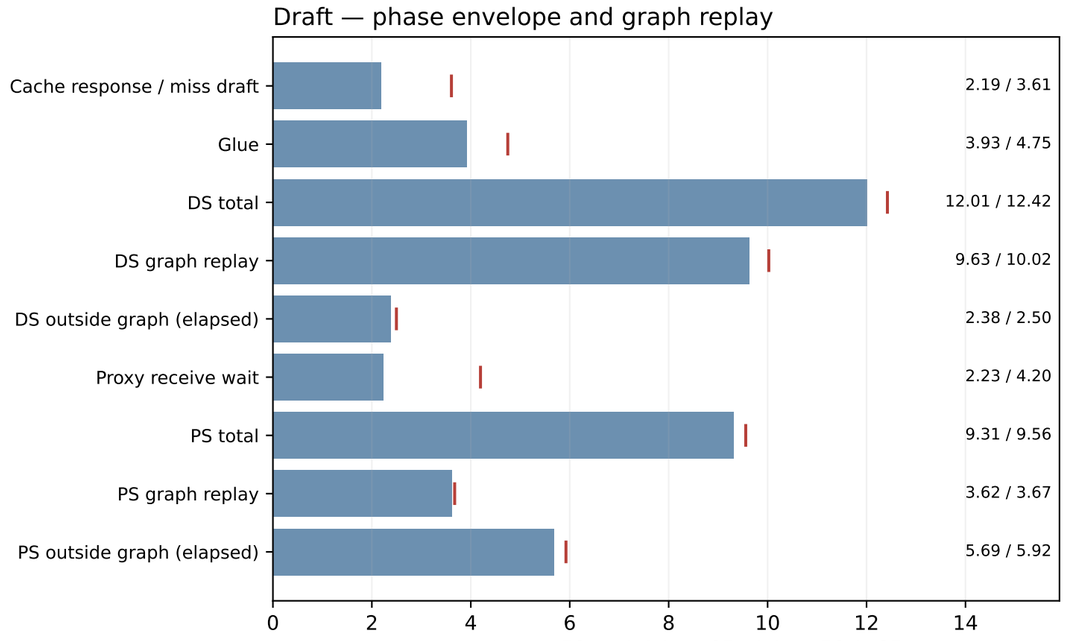
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

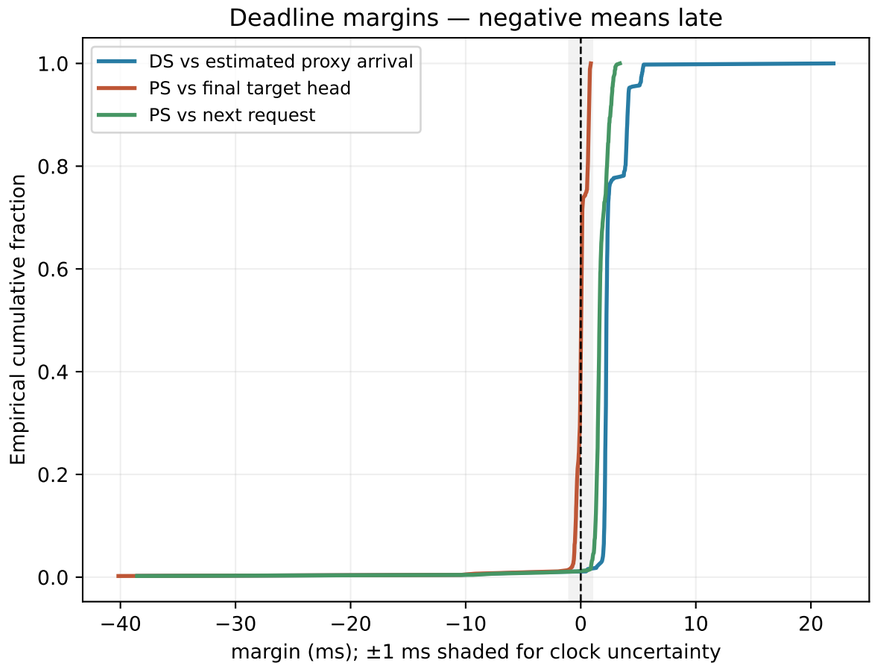
LayerSkip Llama2-13B + AMD-Llama-135M
Component latency, phase deadlines, and every measured step

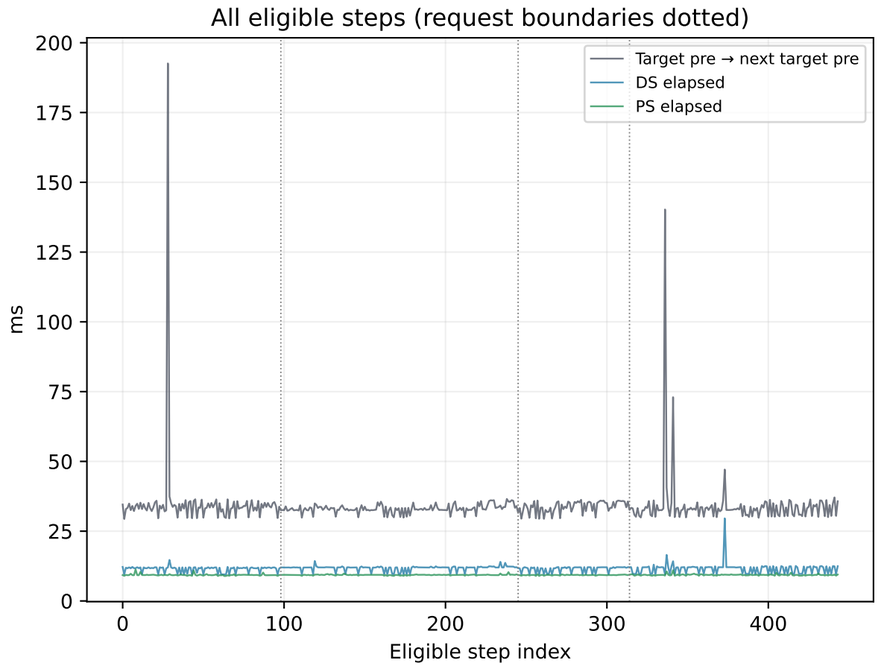
exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 59f46511 / eslab18 RTX3090 / 4 profiled turns, 444 eligible steps / output cap 256







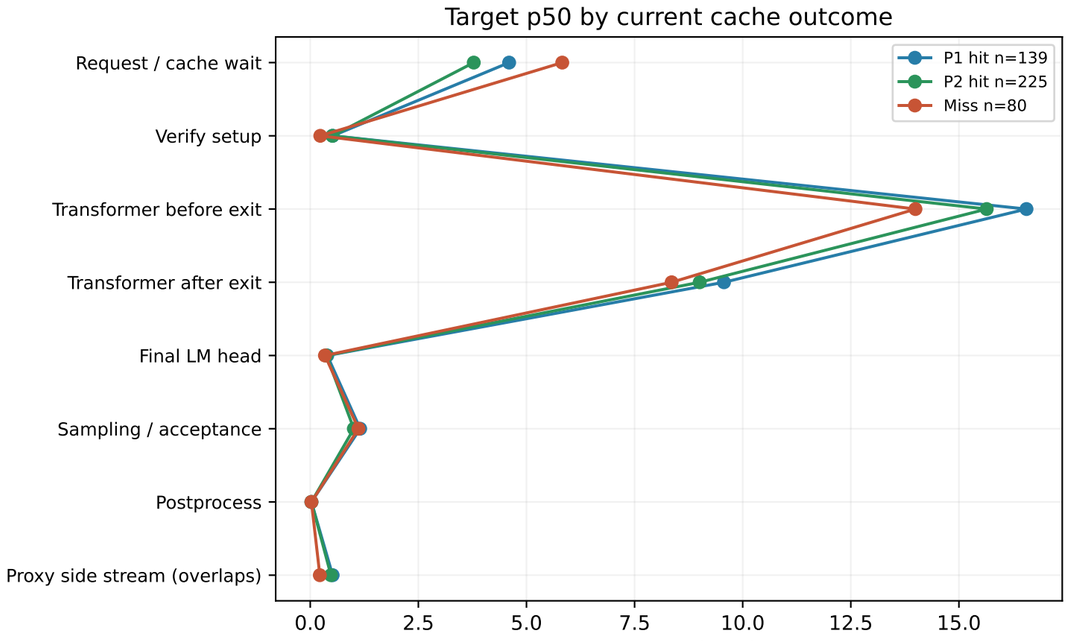


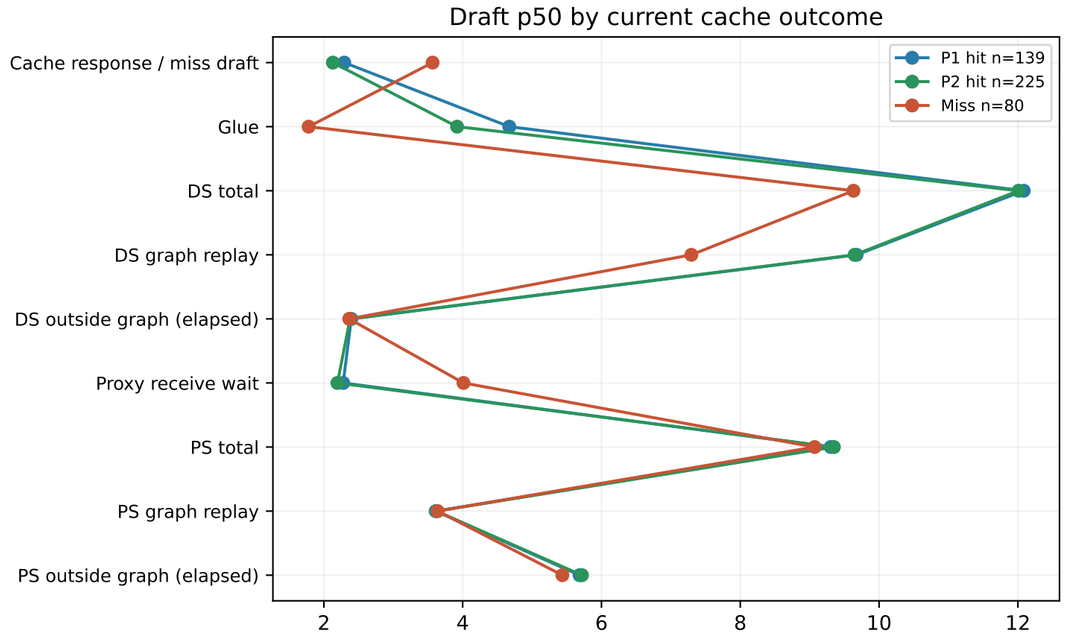
CUDA-event elapsed spans include queue/host stalls; not GPU busy time. Nested/overlapping components are NOT additive.

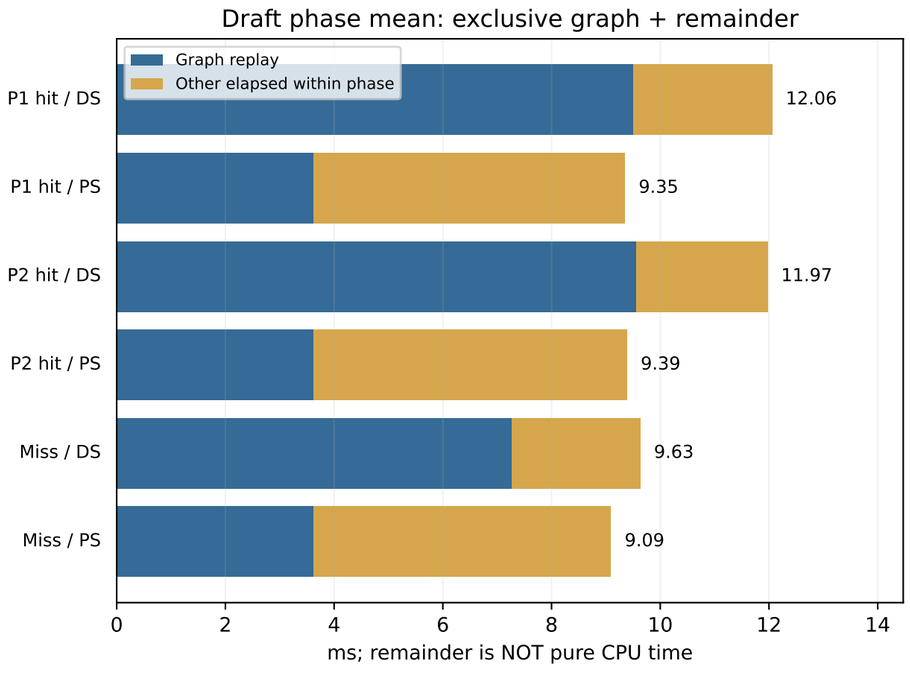
LayerSkip Llama2-13B + AMD-Llama-135M
P1 hit / P2 hit / miss — component distributions and phase composition

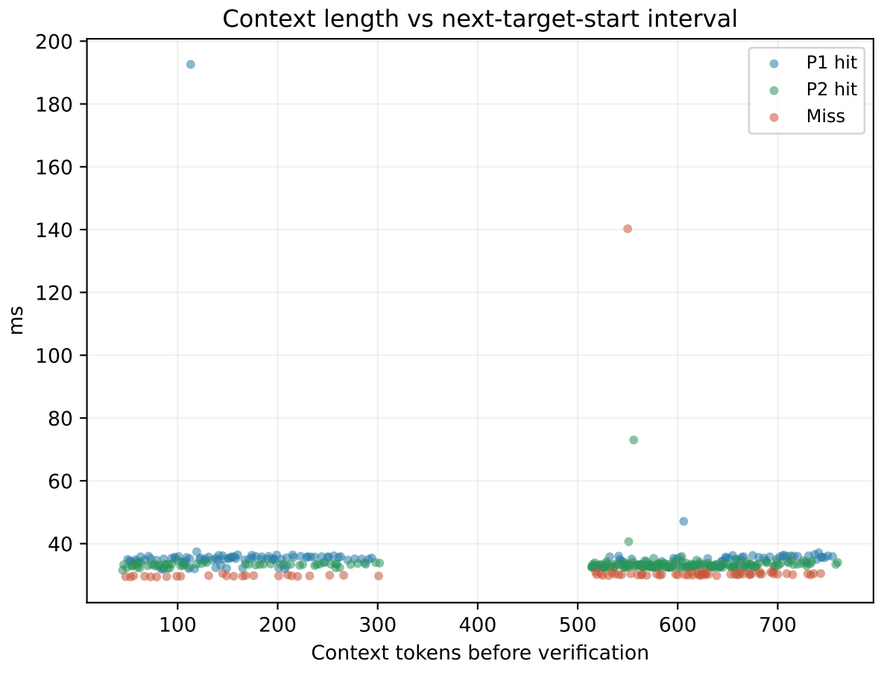
exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 59f46511 / eslab18 RTX3090 / 4 profiled turns, 444 eligible steps / output cap 256









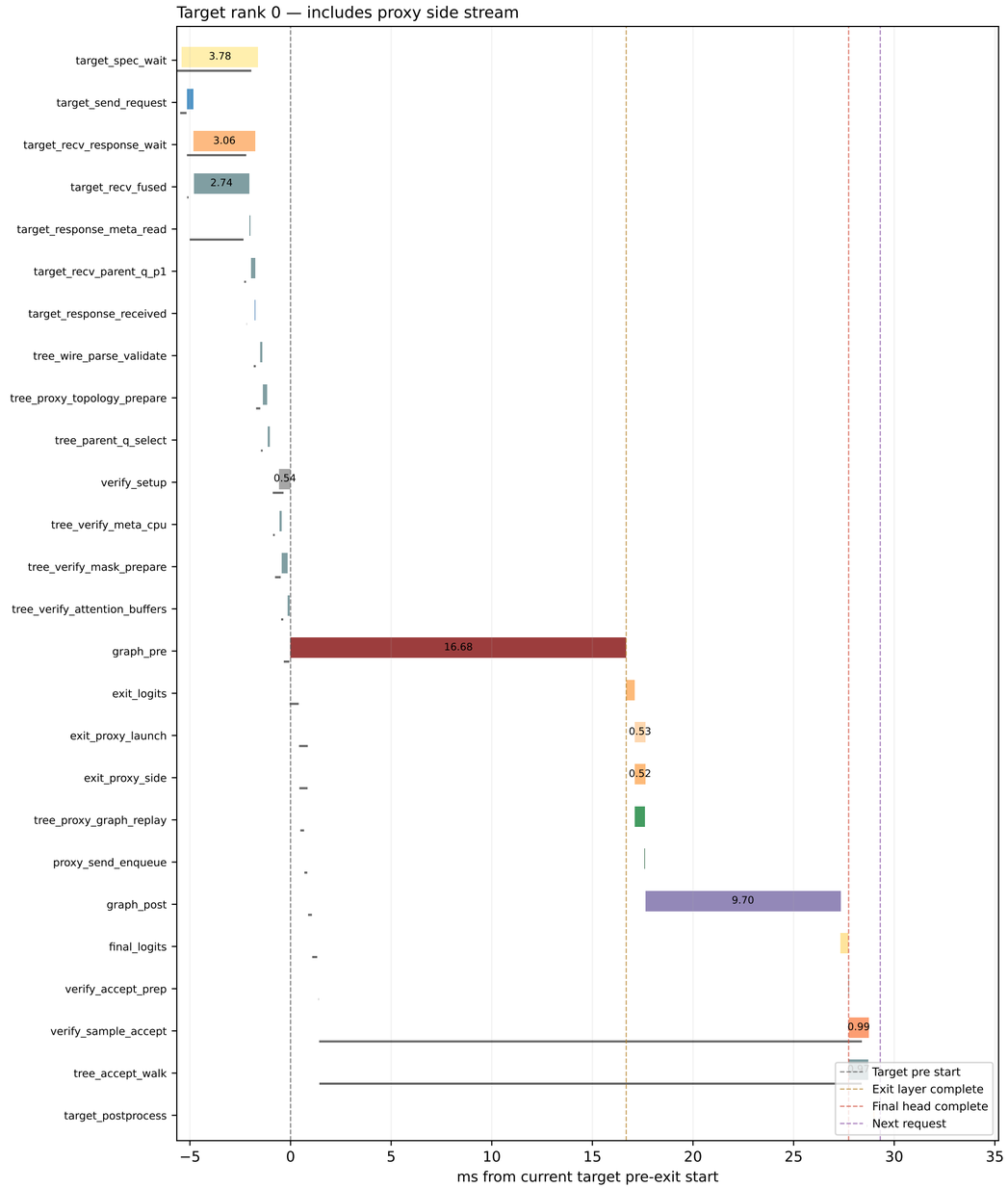
CUDA-event elapsed spans include queue/host stalls; not GPU busy time. Nested/overlapping components are NOT additive.

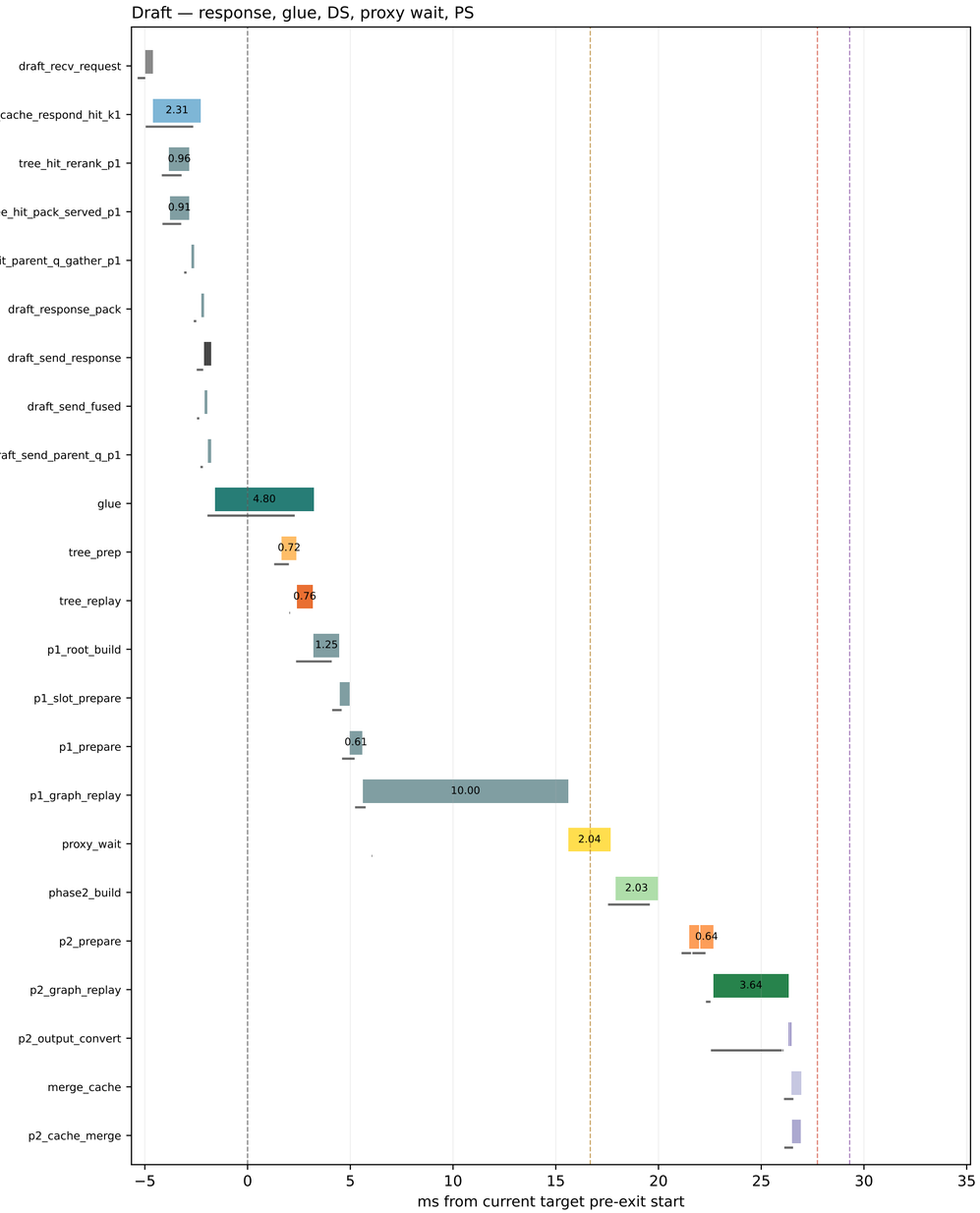
LayerSkip Llama2-13B + AMD-Llama-135M
P1 hit / p25 target-step duration / request 4, UID 288_t0, step 20

exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 59f46511 / eslab18 RTX3090 / 4 profiled turns, 444 eligible steps / output cap 256

valid proposal nodes=12, AL incl. recovery=1; DS margin +2.028 ms (estimated); PS margin +0.797 ms; next outcome P2 hit





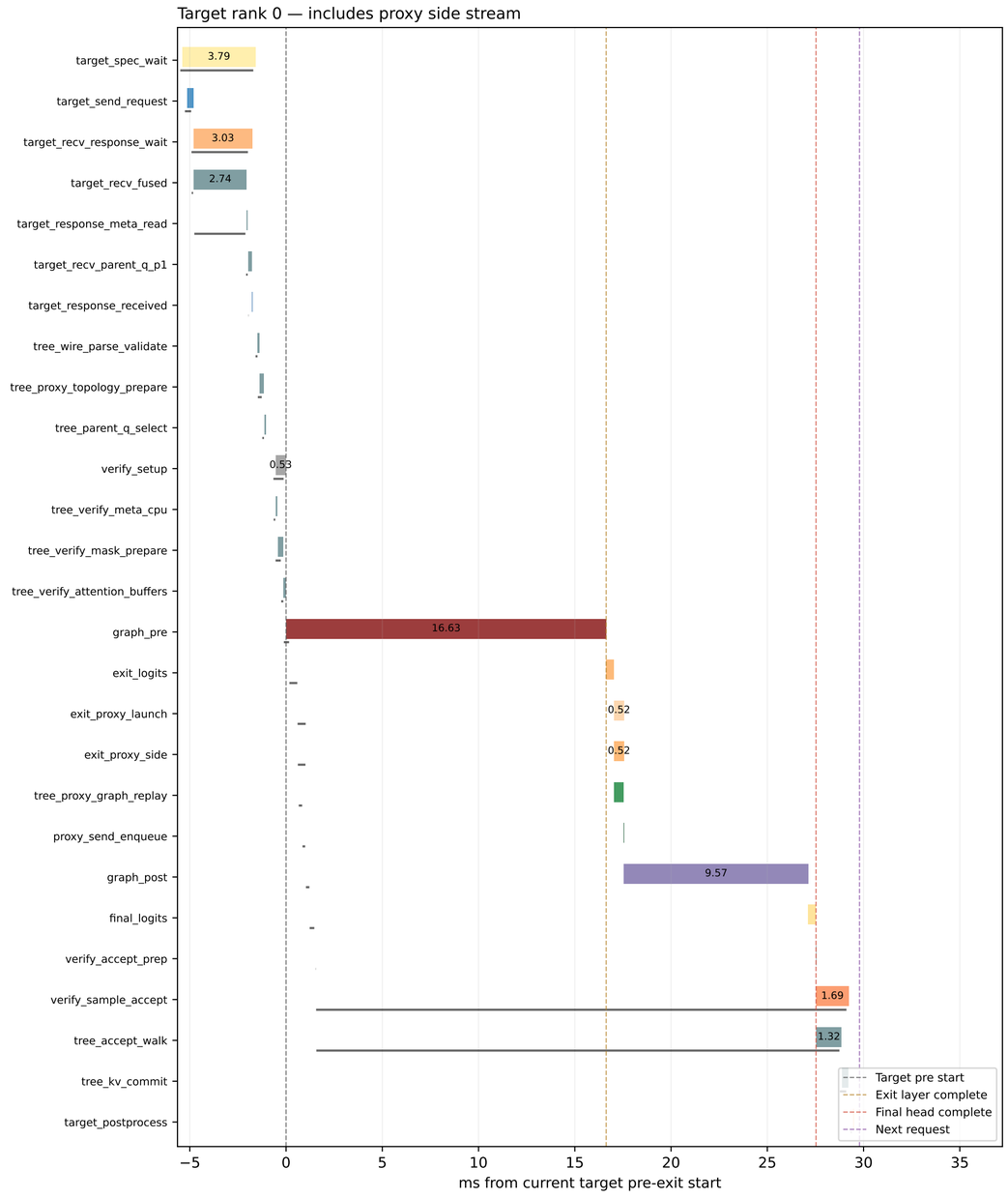
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

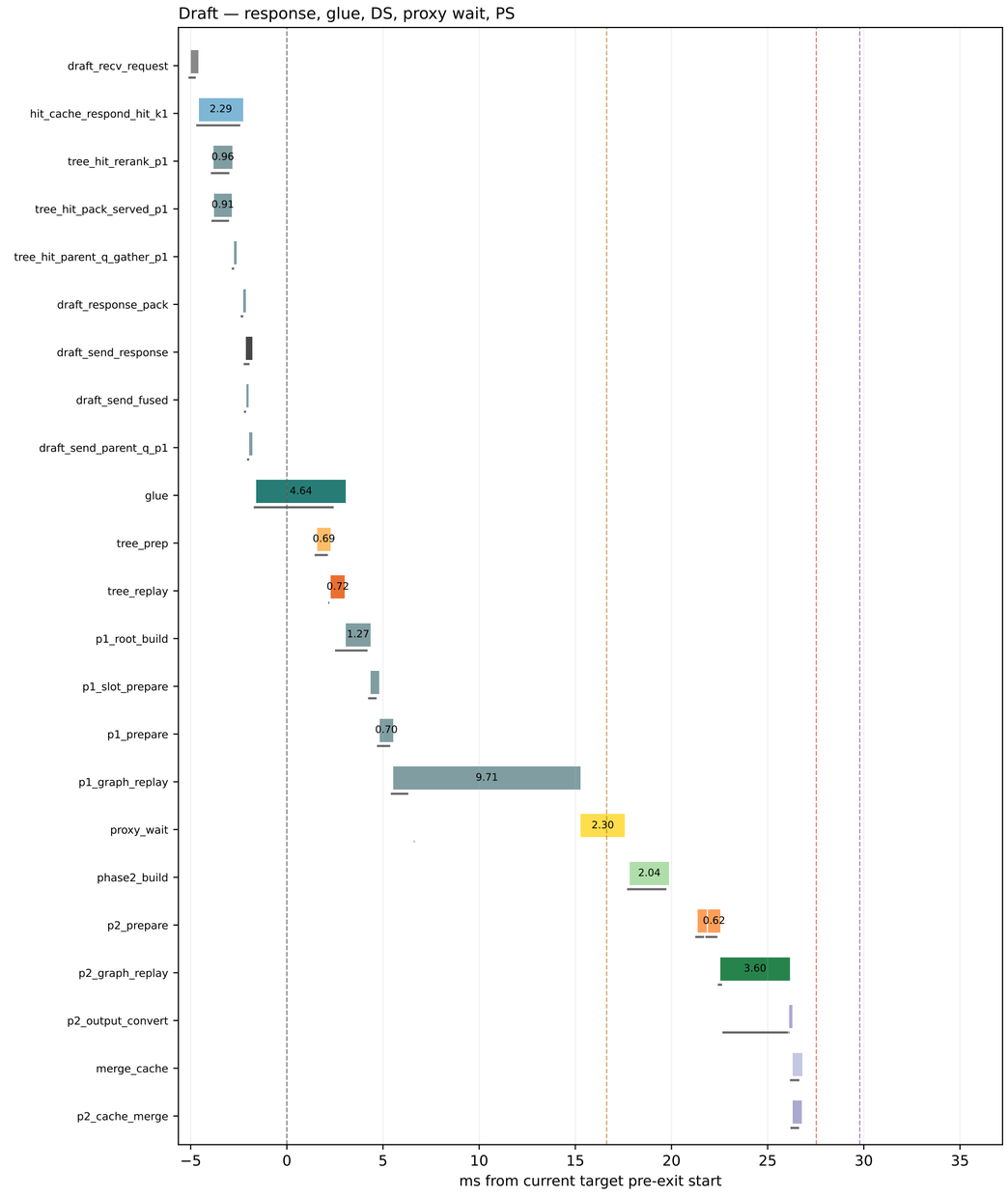
LayerSkip Llama2-13B + AMD-Llama-135M
P1 hit / p50 target-step duration / request 1, UID 84_t0, step 68

exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 59f46511 / eslab18 RTX3090 / 4 profiled turns, 444 eligible steps / output cap 256

valid proposal nodes=12, AL incl. recovery=5; DS margin +2.302 ms (estimated); PS margin +0.743 ms; next outcome Miss





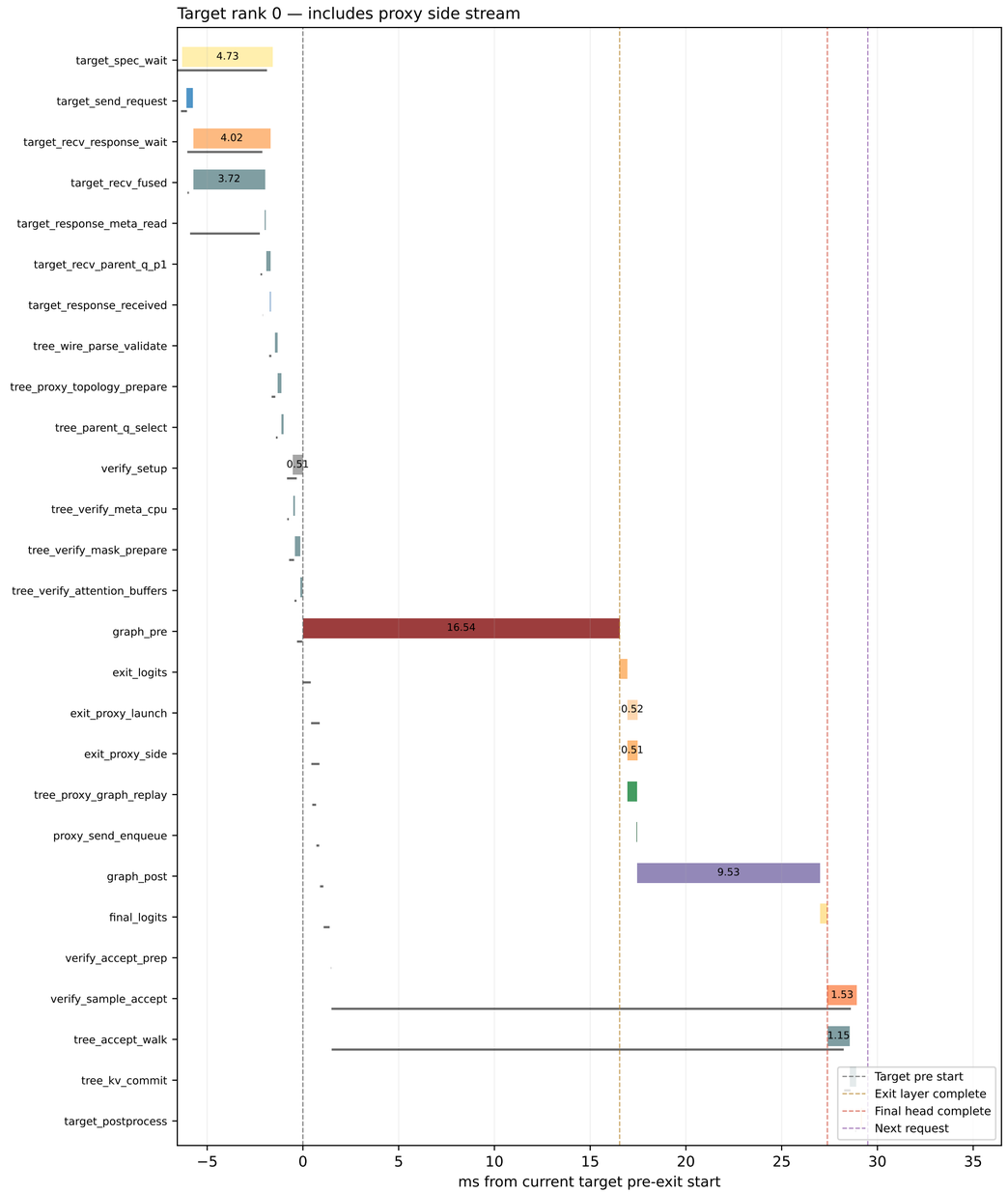
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

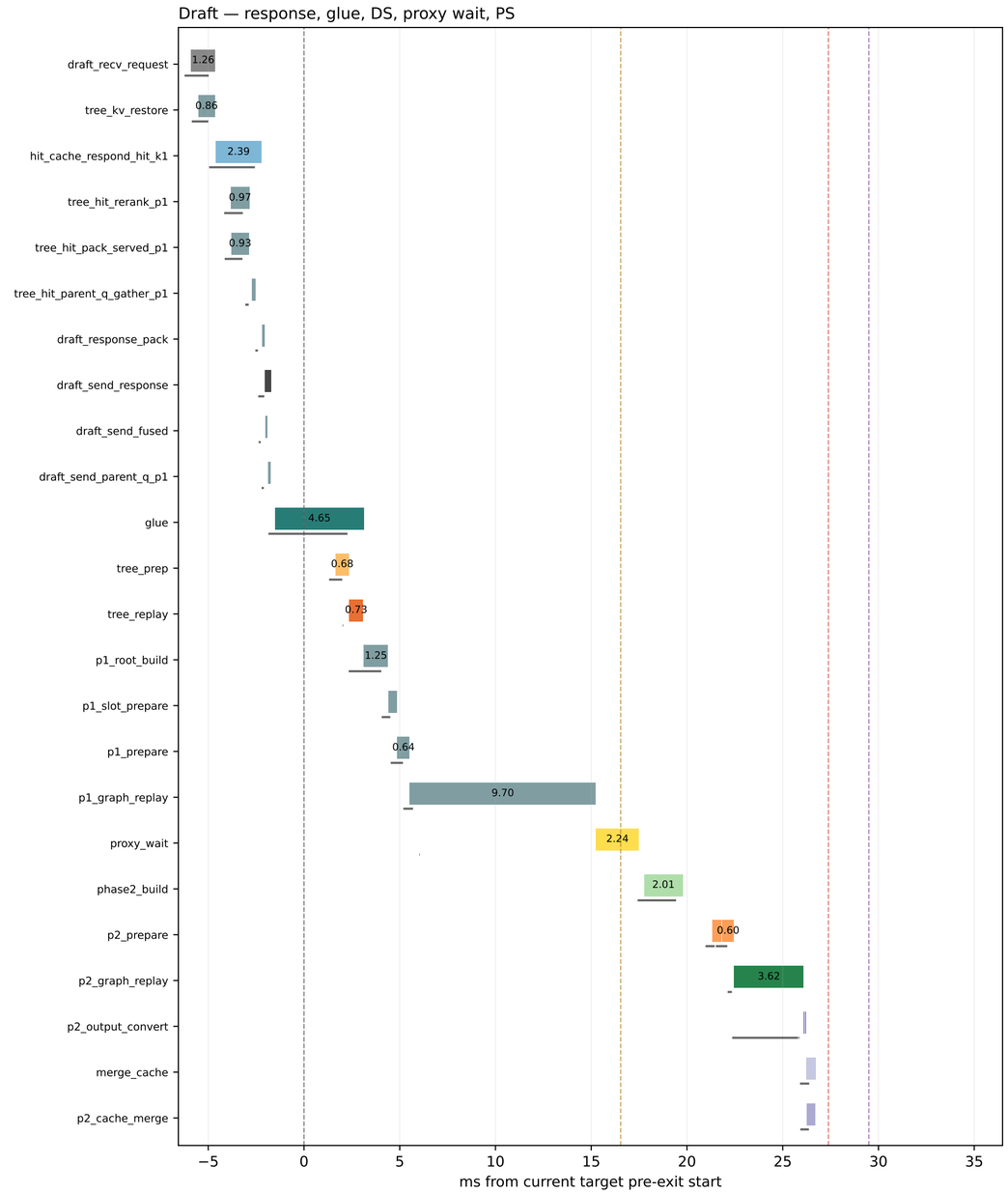
LayerSkip Llama2-13B + AMD-Llama-135M
P1 hit / p75 target-step duration / request 3, UID 128_t0, step 64

exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 59f46511 / eslab18 RTX3090 / 4 profiled turns, 444 eligible steps / output cap 256

valid proposal nodes=12, AL incl. recovery=7; DS margin +2.235 ms (estimated); PS margin +0.660 ms; next outcome P1 hit





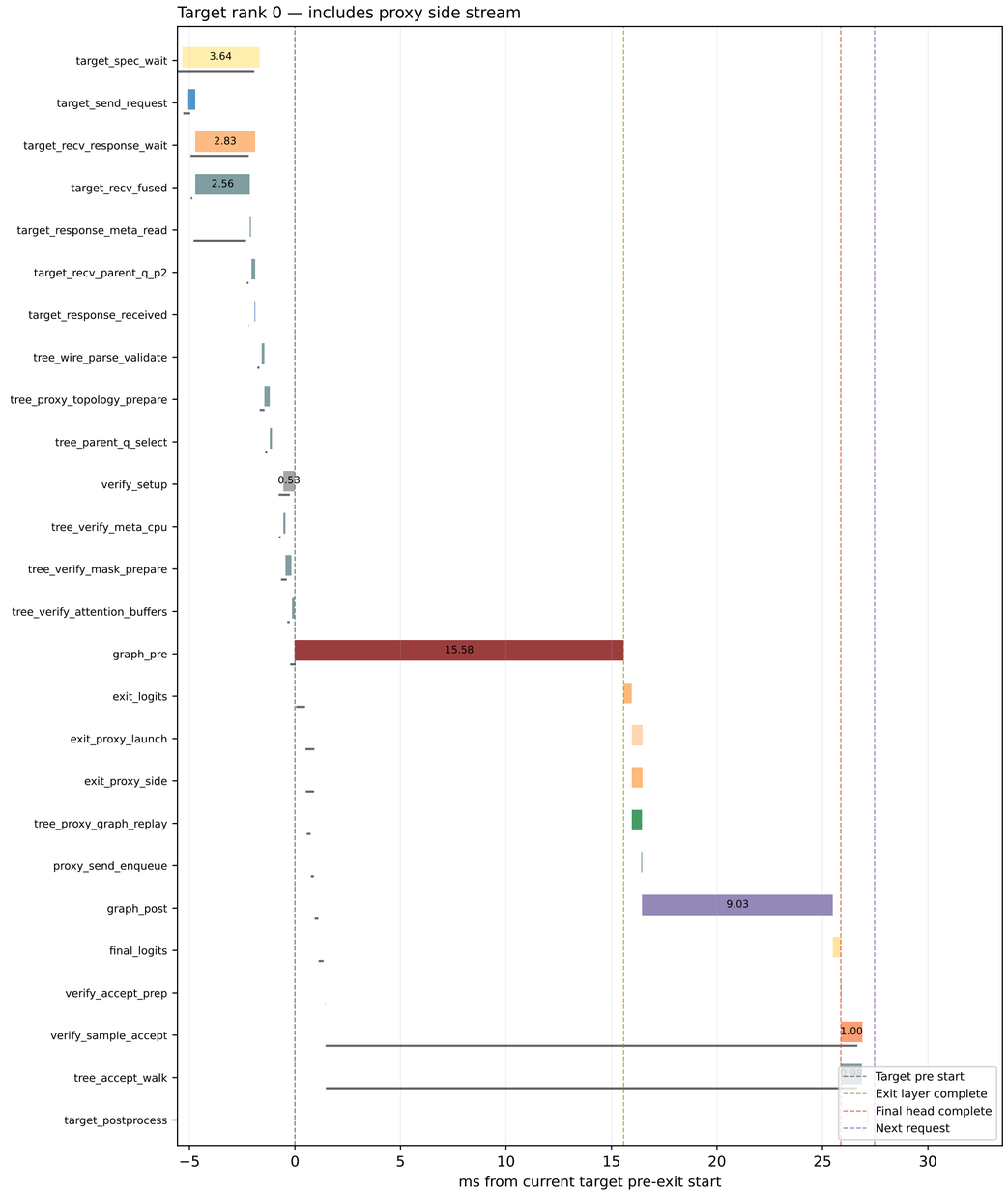
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

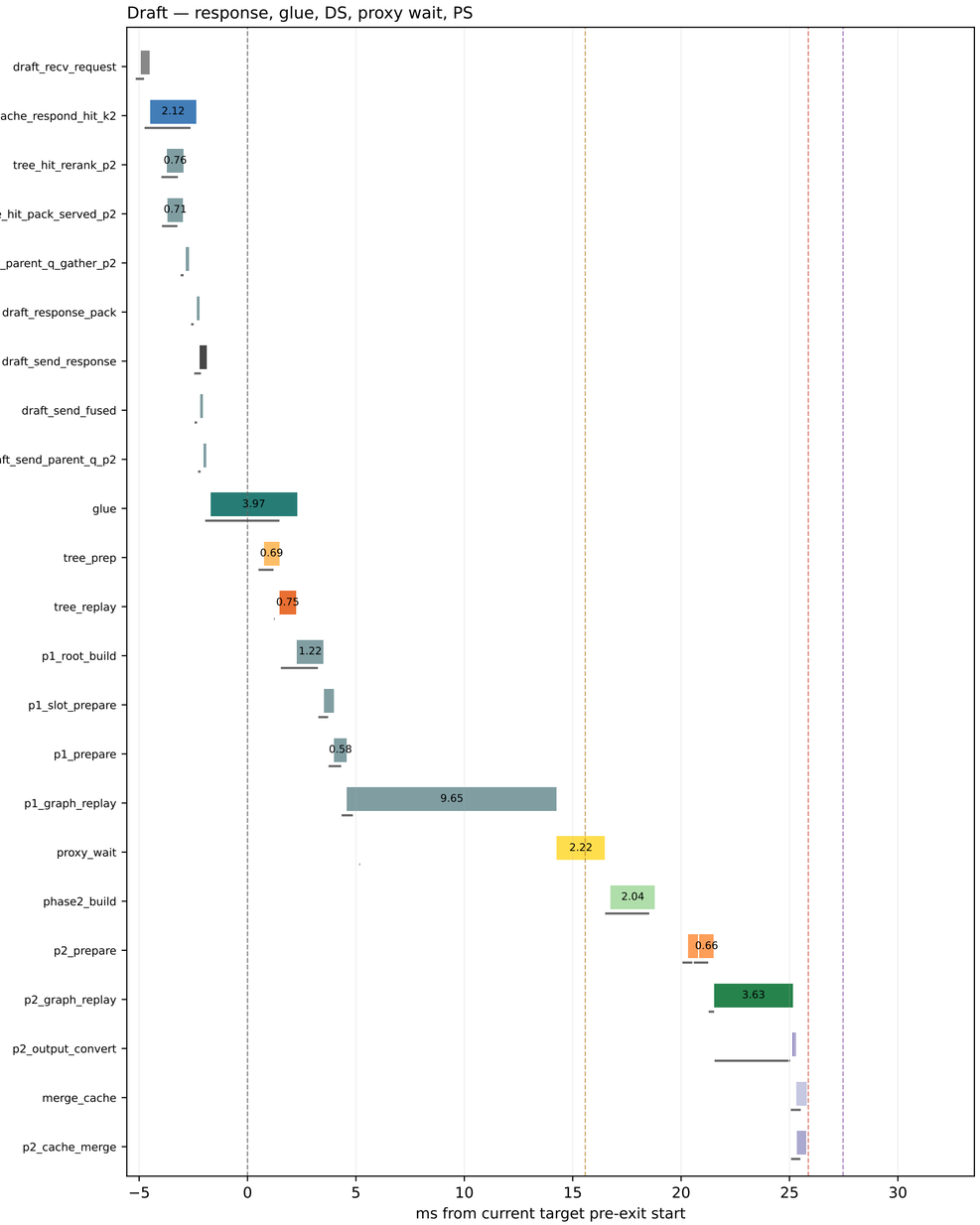
LayerSkip Llama2-13B + AMD-Llama-135M
P2 hit / p25 target-step duration / request 2, UID 244_t0, step 109

exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 59f46511 / eslab18 RTX3090 / 4 profiled turns, 444 eligible steps / output cap 256

valid proposal nodes=8, AL incl. recovery=1; DS margin +2.215 ms (estimated); PS margin +0.092 ms; next outcome P1 hit





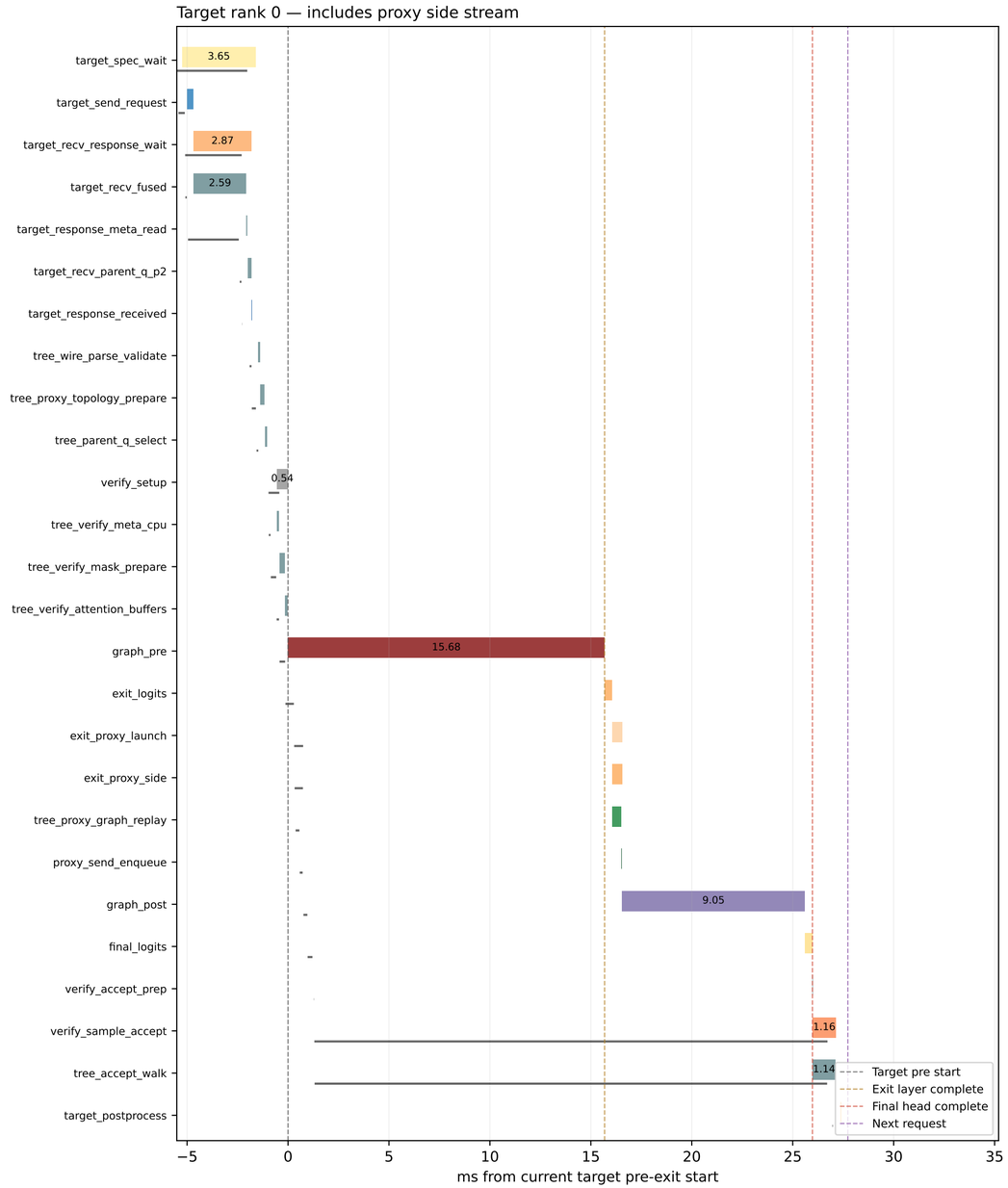
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

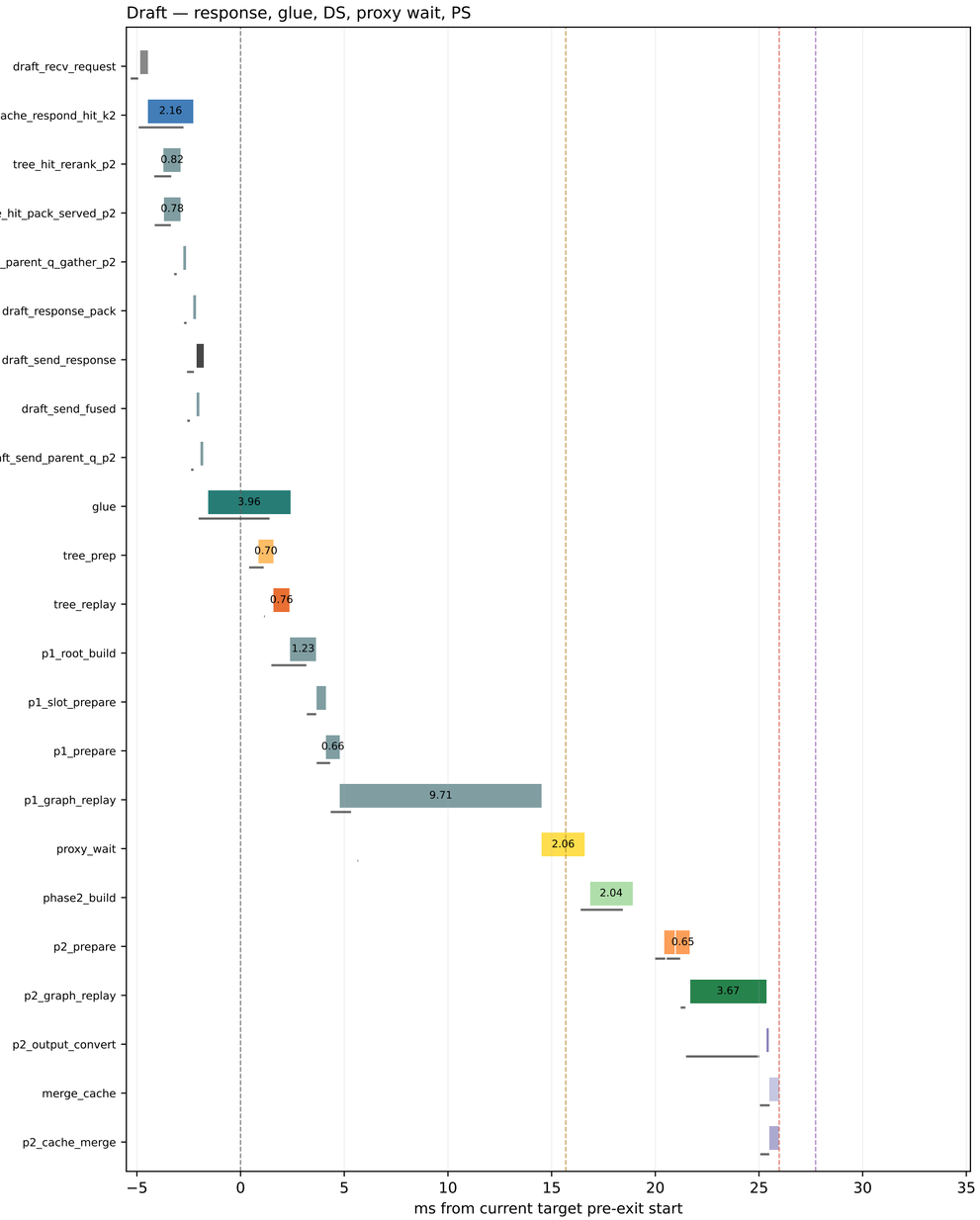
LayerSkip Llama2-13B + AMD-Llama-135M
P2 hit / p50 target-step duration / request 4, UID 288_t0, step 127

exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 59f46511 / eslab18 RTX3090 / 4 profiled turns, 444 eligible steps / output cap 256

valid proposal nodes=8, AL incl. recovery=2; DS margin +2.046 ms (estimated); PS margin +0.002 ms; next outcome Miss





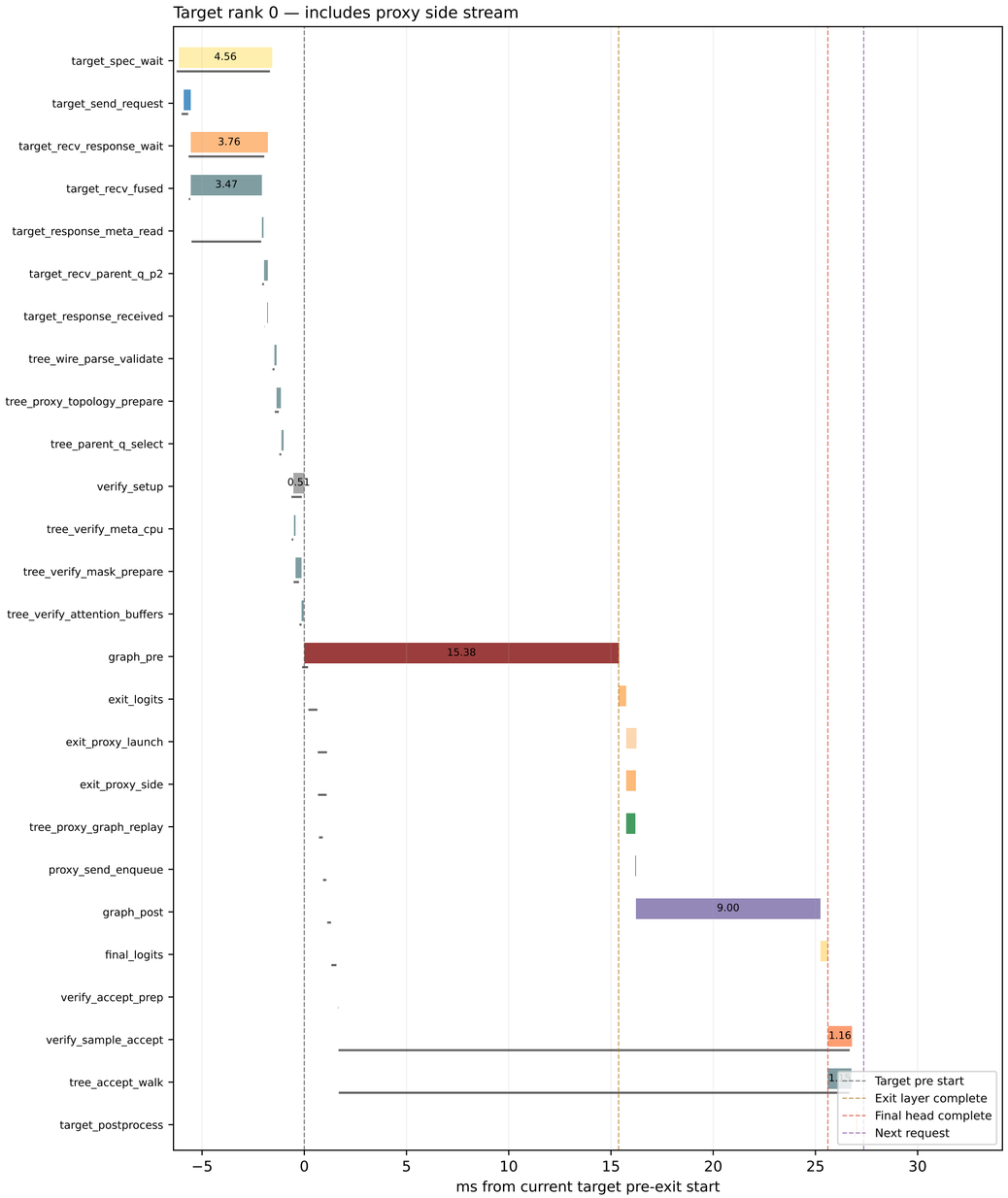
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

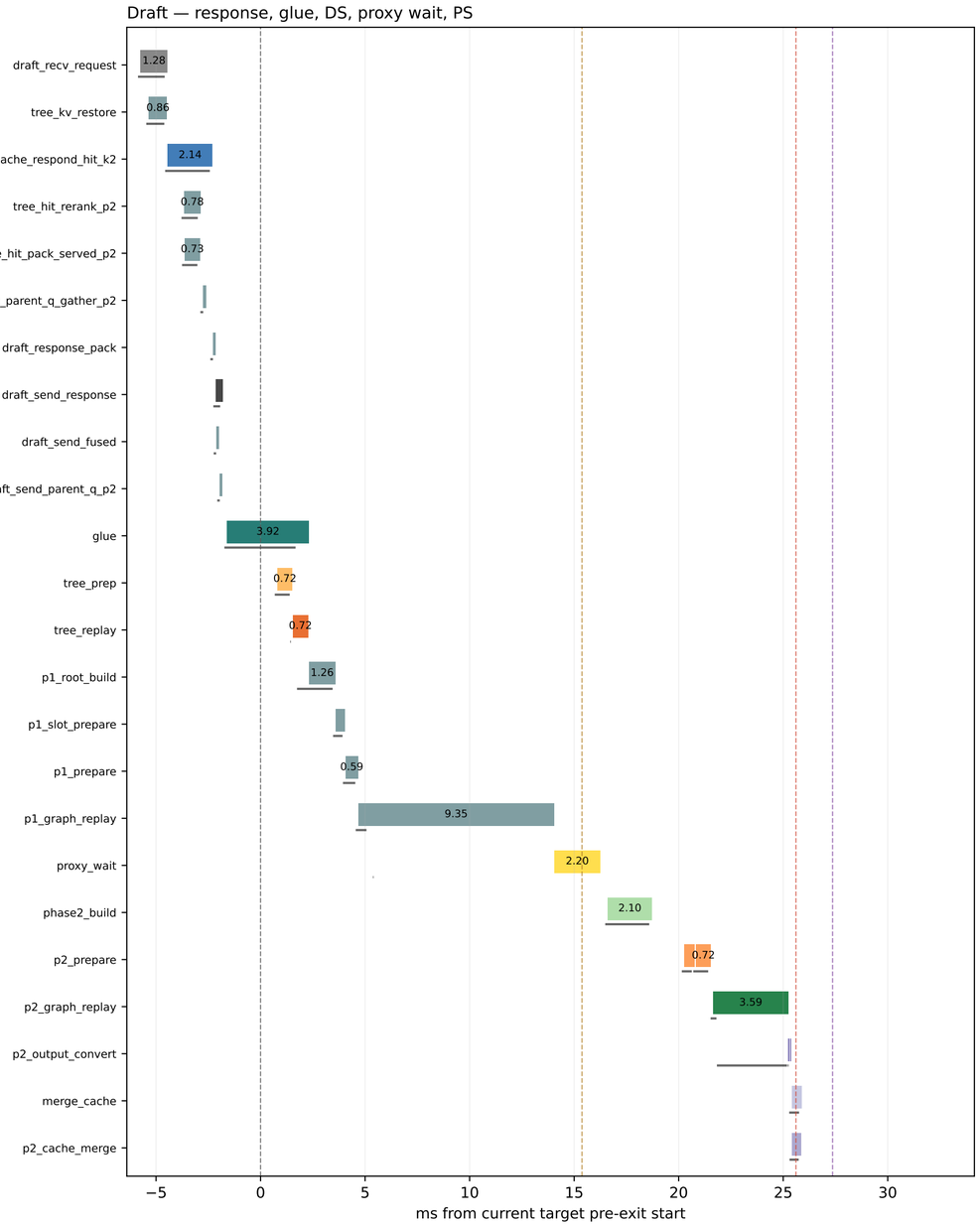
LayerSkip Llama2-13B + AMD-Llama-135M
P2 hit / p75 target-step duration / request 1, UID 84_t0, step 55

exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 59f46511 / eslab18 RTX3090 / 4 profiled turns, 444 eligible steps / output cap 256

valid proposal nodes=8, AL incl. recovery=2; DS margin +2.202 ms (estimated); PS margin -0.277 ms; next outcome P1 hit





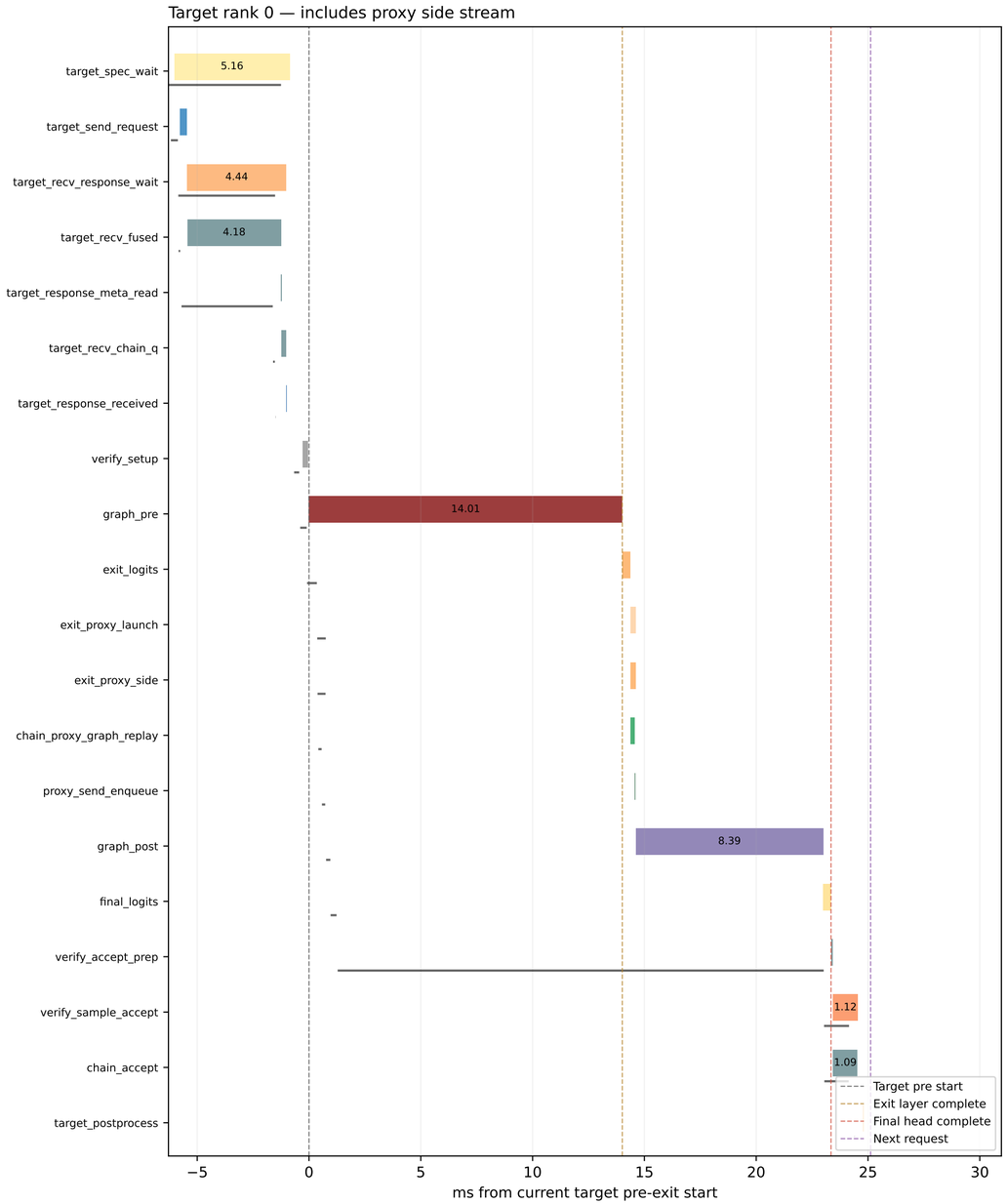
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

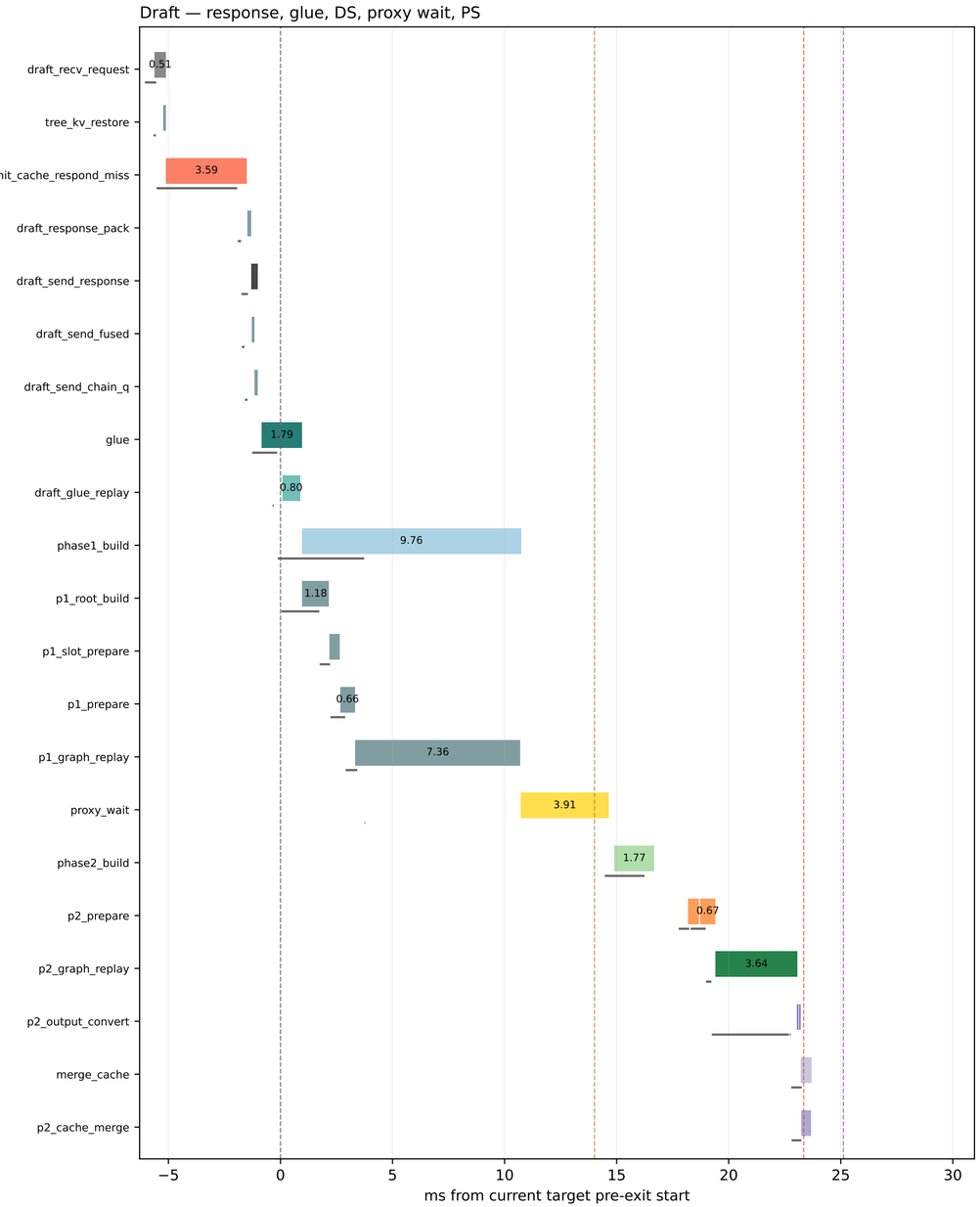
LayerSkip Llama2-13B + AMD-Llama-135M
Miss / p25 target-step duration / request 4, UID 288_t0, step 93

exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 59f46511 / eslab18 RTX3090 / 4 profiled turns, 444 eligible steps / output cap 256

valid proposal nodes=3, AL incl. recovery=1; DS margin +3.901 ms (estimated); PS margin -0.342 ms; next outcome P2 hit





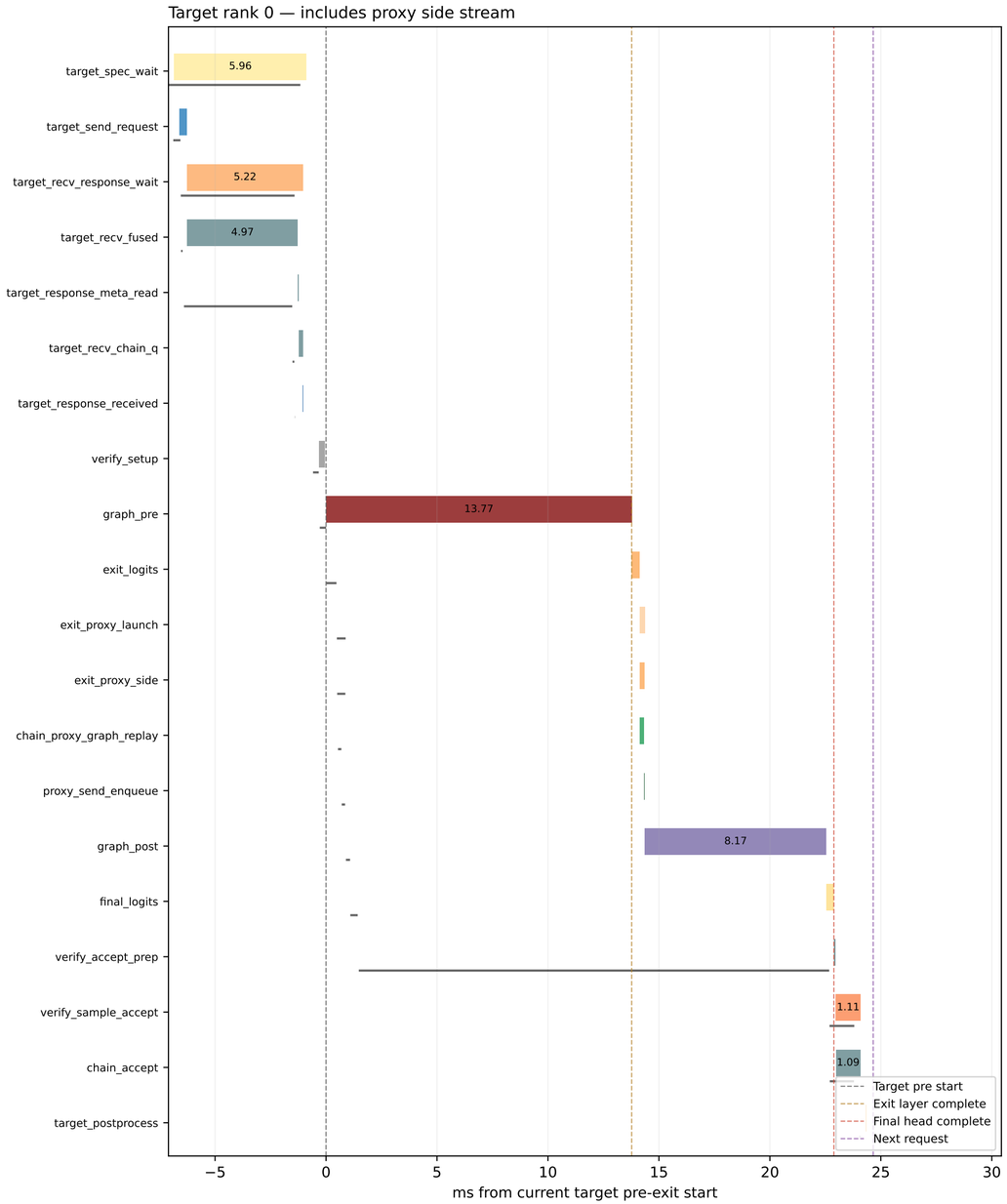
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

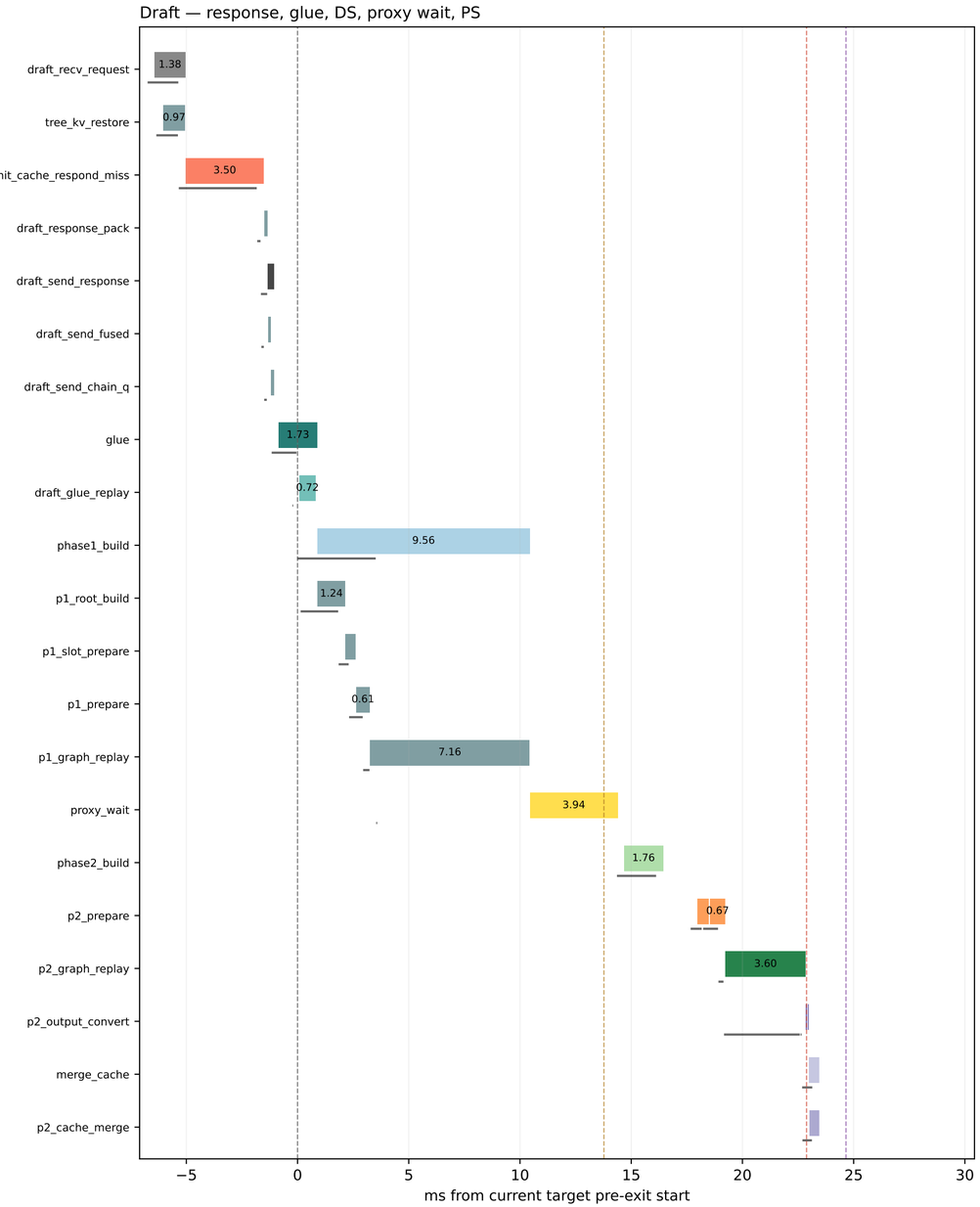
LayerSkip Llama2-13B + AMD-Llama-135M
Miss / p50 target-step duration / request 3, UID 128_t0, step 29

exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 59f46511 / eslab18 RTX3090 / 4 profiled turns, 444 eligible steps / output cap 256

valid proposal nodes=3, AL incl. recovery=1; DS margin +3.939 ms (estimated); PS margin -0.572 ms; next outcome P2 hit





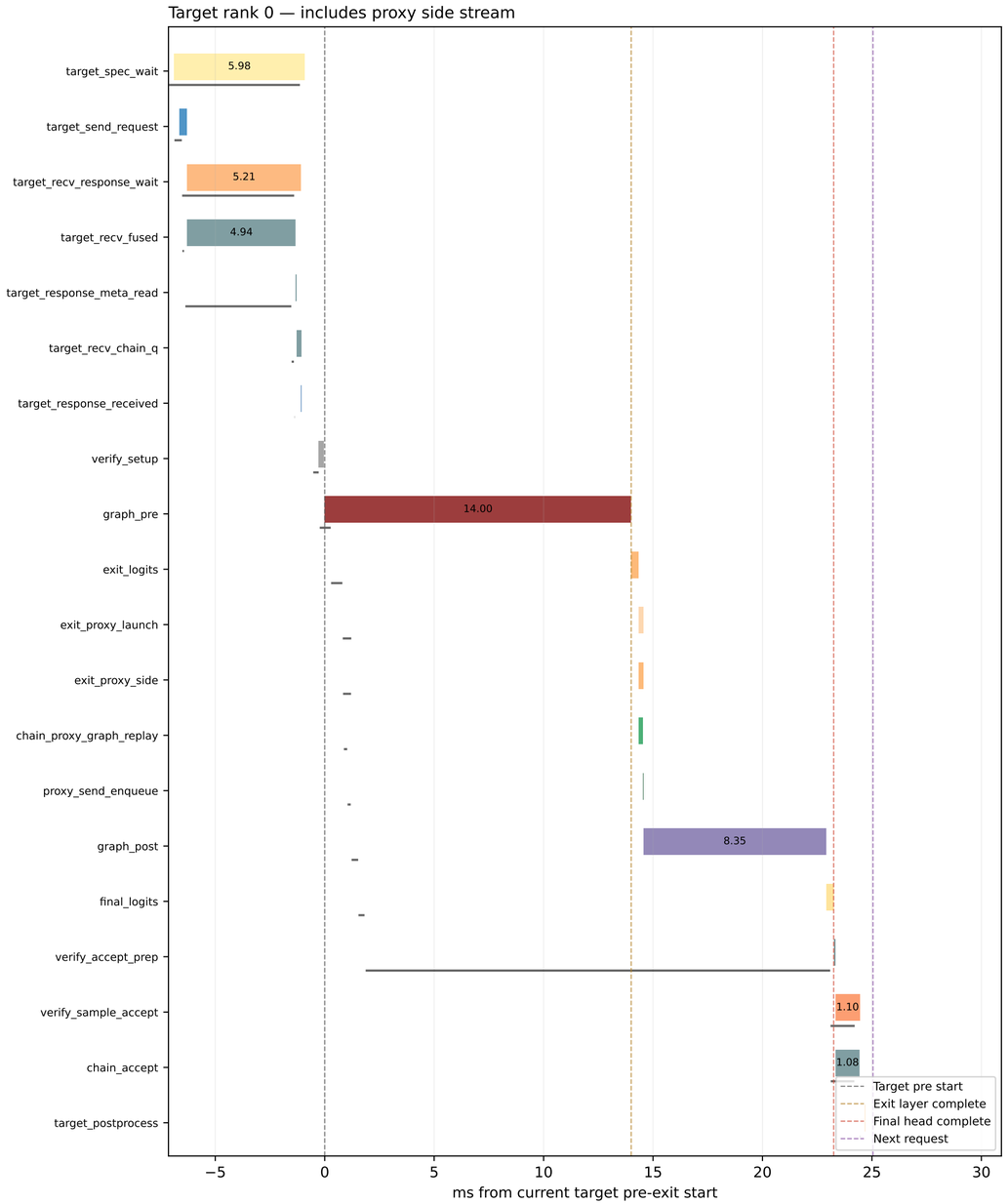
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

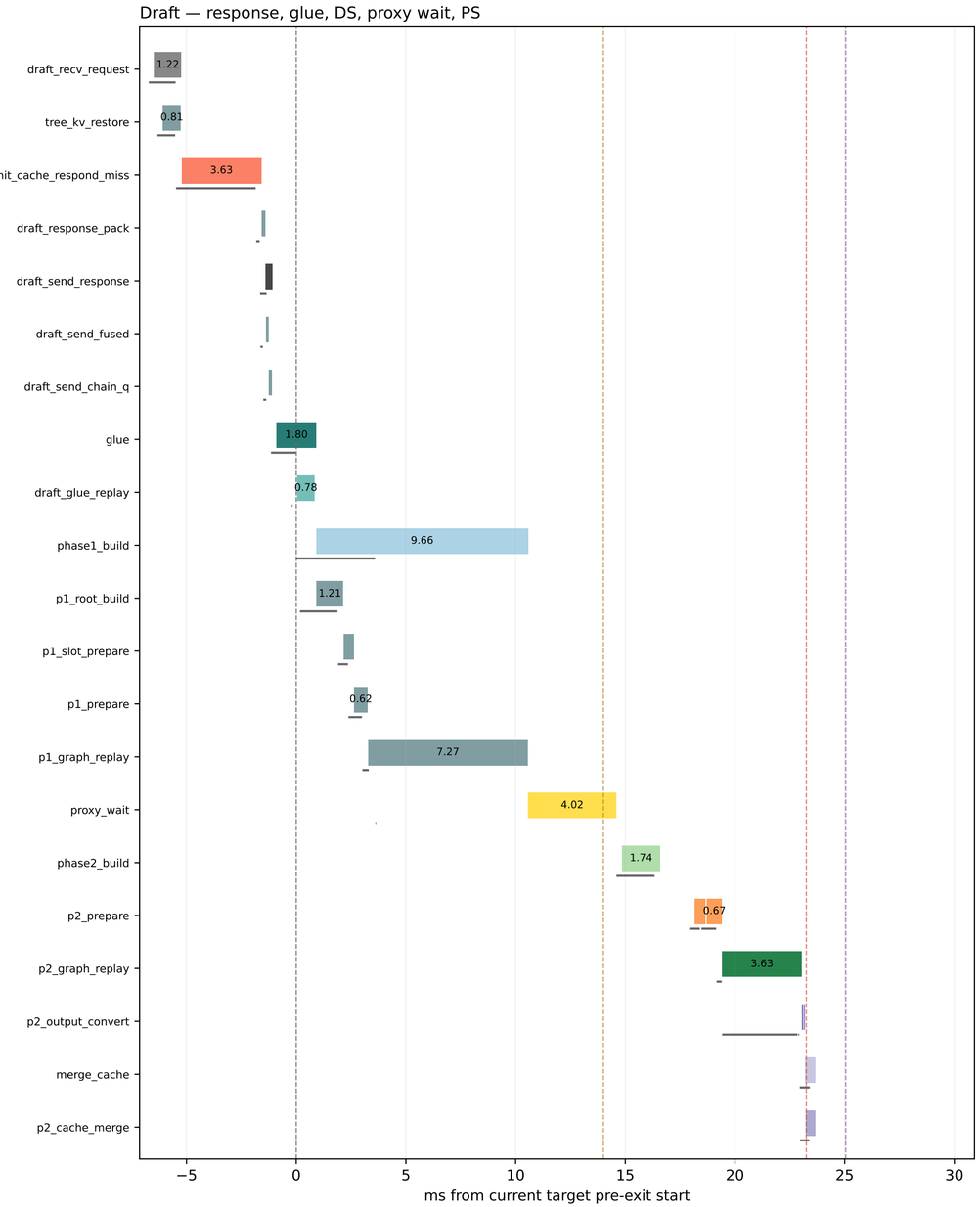
LayerSkip Llama2-13B + AMD-Llama-135M
Miss / p75 target-step duration / request 2, UID 244_t0, step 108

exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 59f46511 / eslab18 RTX3090 / 4 profiled turns, 444 eligible steps / output cap 256

valid proposal nodes=3, AL incl. recovery=1; DS margin +4.015 ms (estimated); PS margin -0.415 ms; next outcome P2 hit



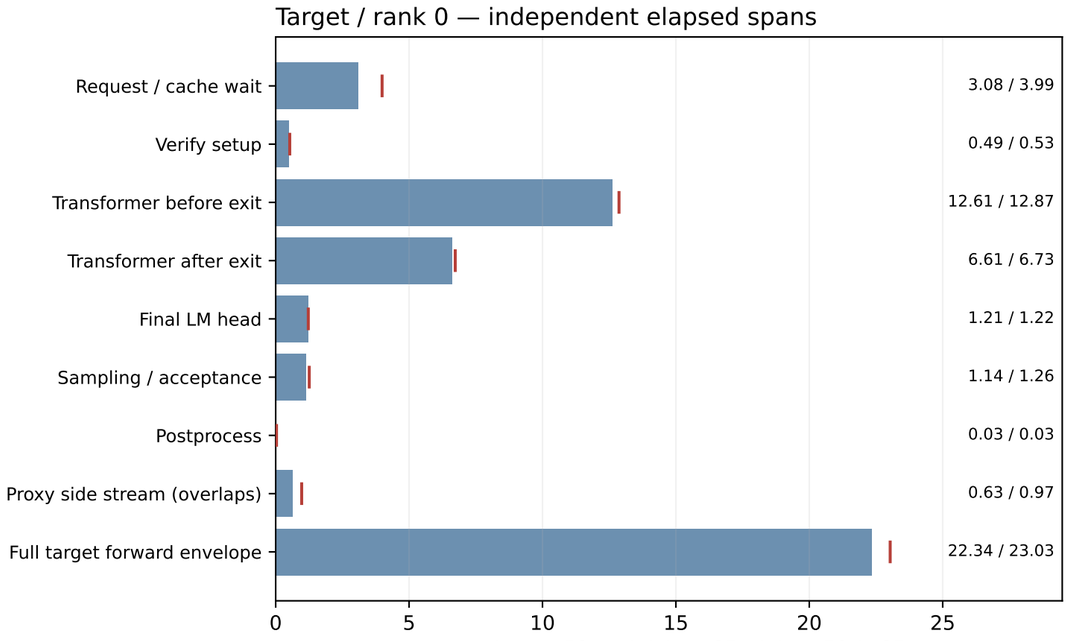


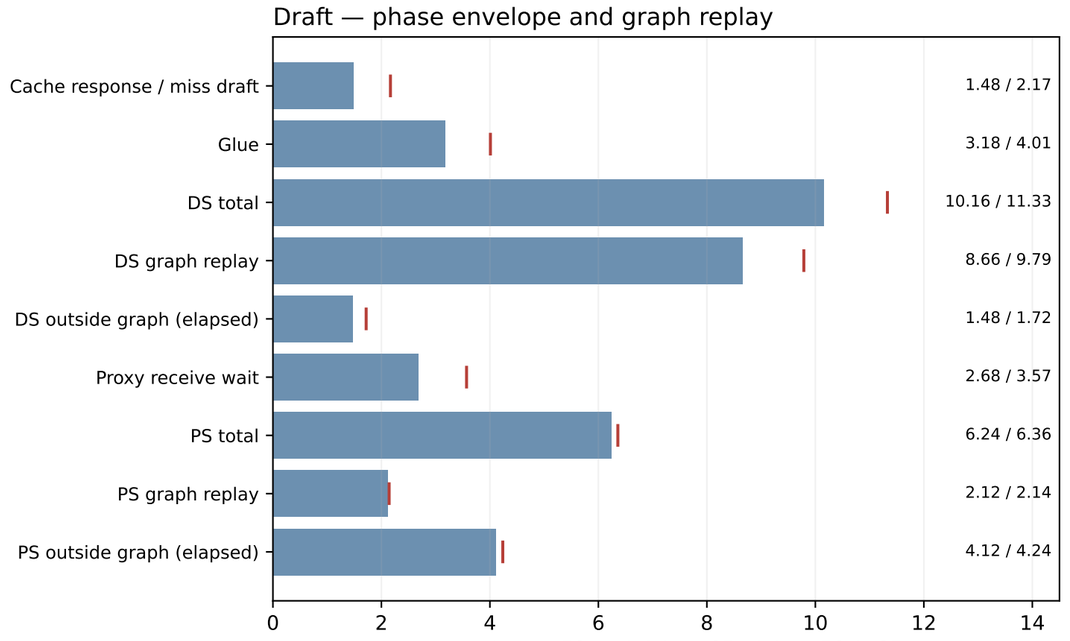
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

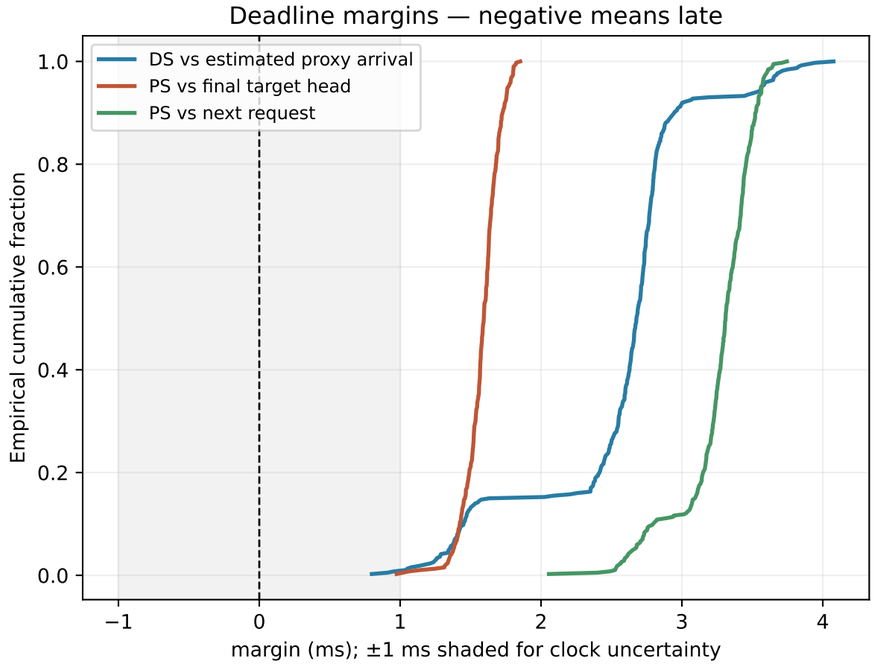
LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M
Component latency, phase deadlines, and every measured step

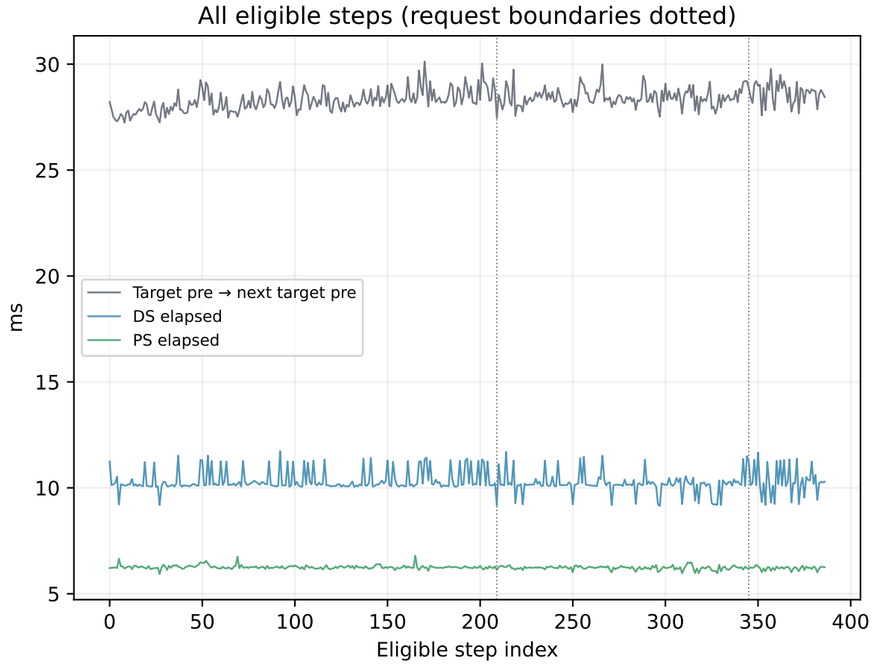
exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 16dd4198 / eslab18 RTX3090 / 4 profiled turns, 387 eligible steps / output cap 256







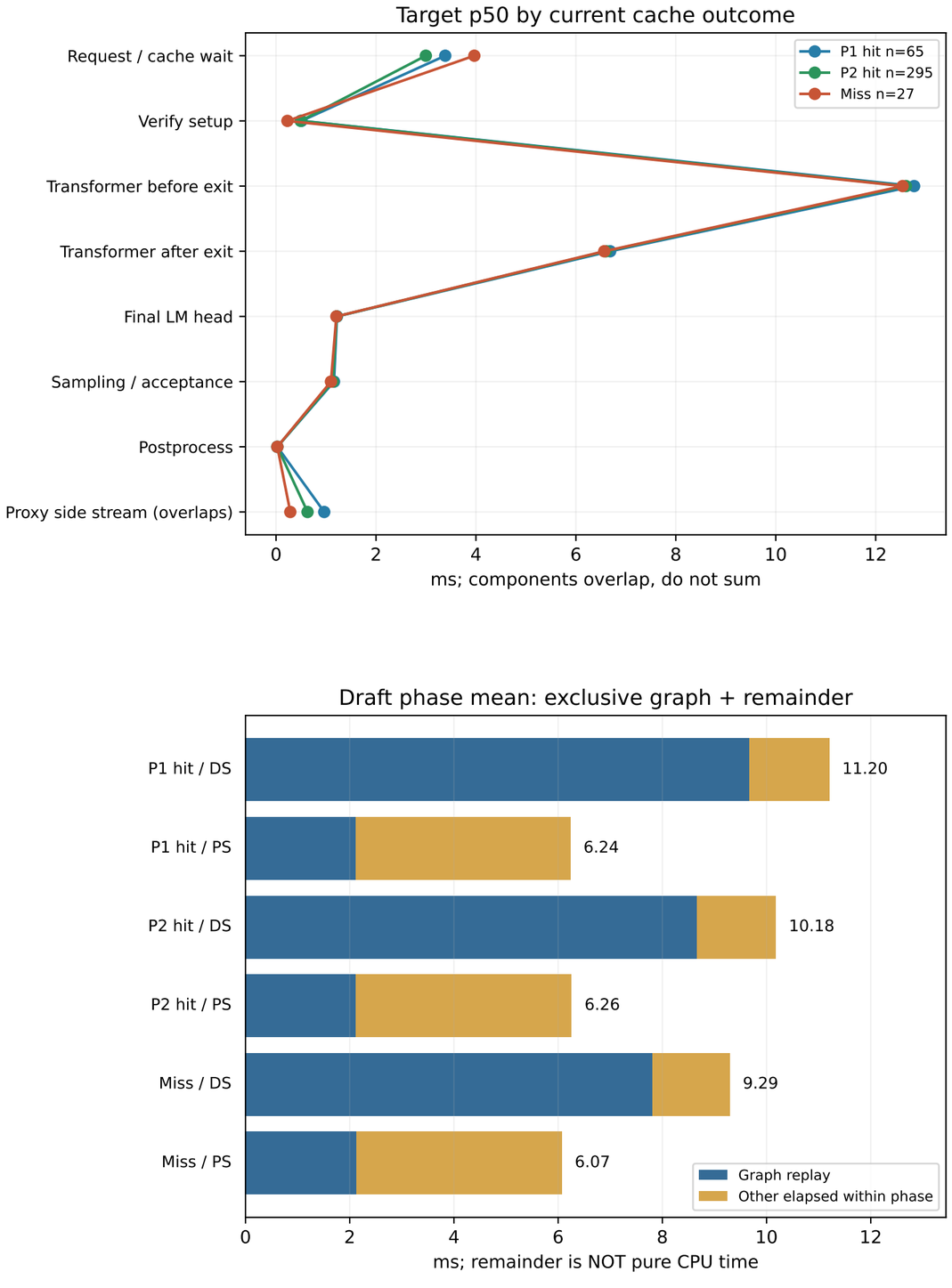


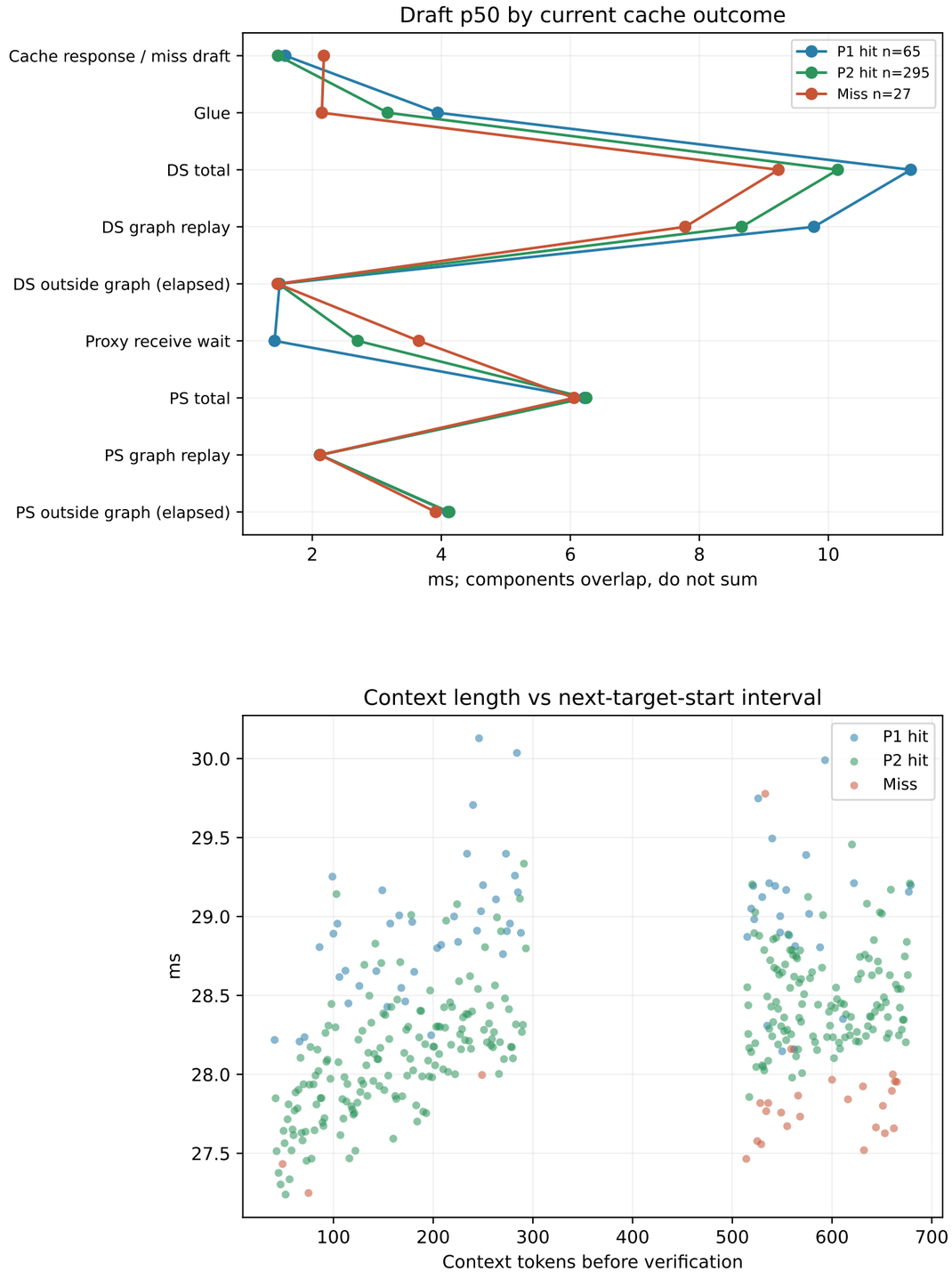
CUDA-event elapsed spans include queue/host stalls; not GPU busy time. Nested/overlapping components are NOT additive.

LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M
P1 hit / P2 hit / miss — component distributions and phase composition

exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 16dd4198 / eslab18 RTX3090 / 4 profiled turns, 387 eligible steps / output cap 256





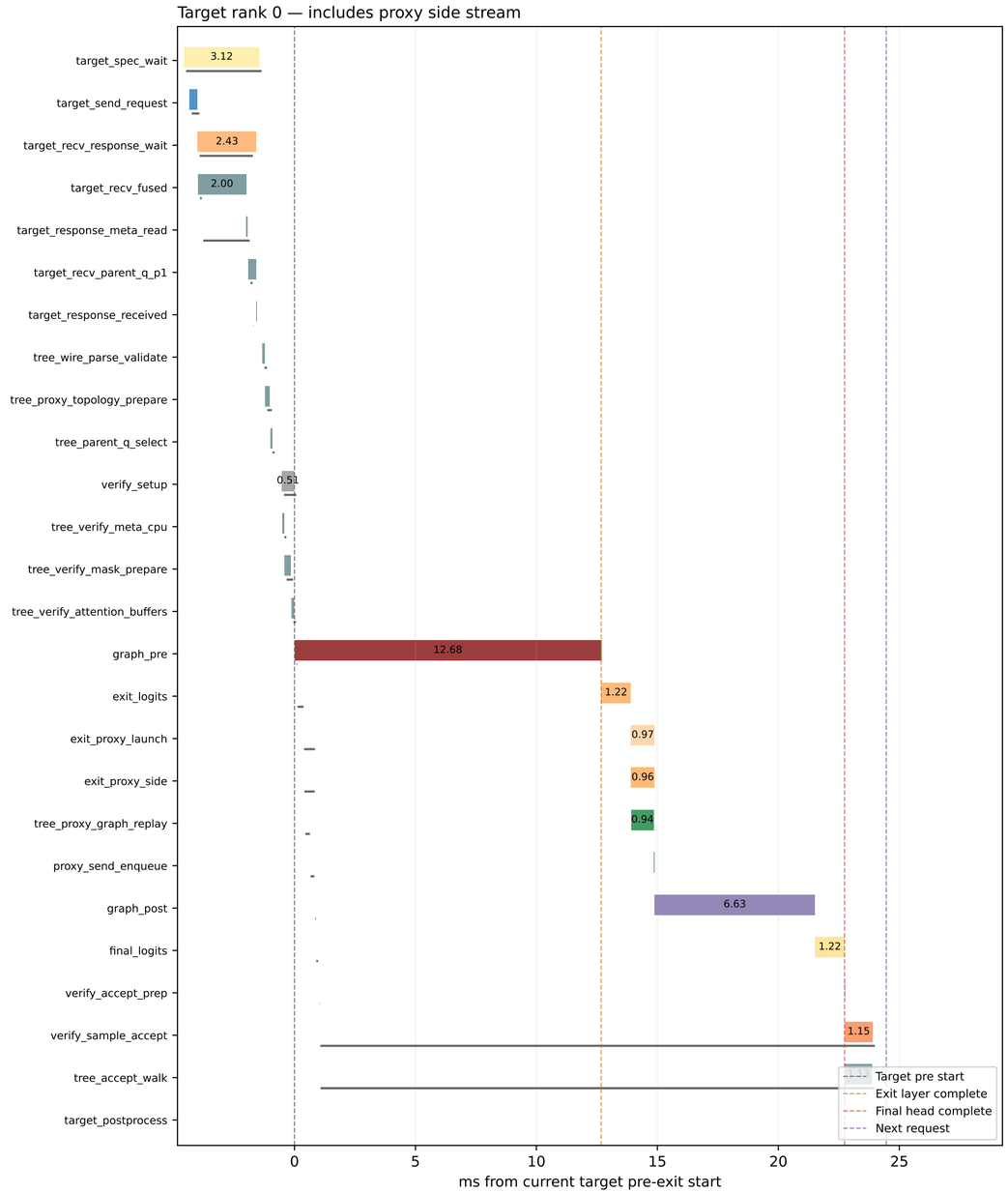
CUDA-event elapsed spans include queue/host stalls; not GPU busy time. Nested/overlapping components are NOT additive.

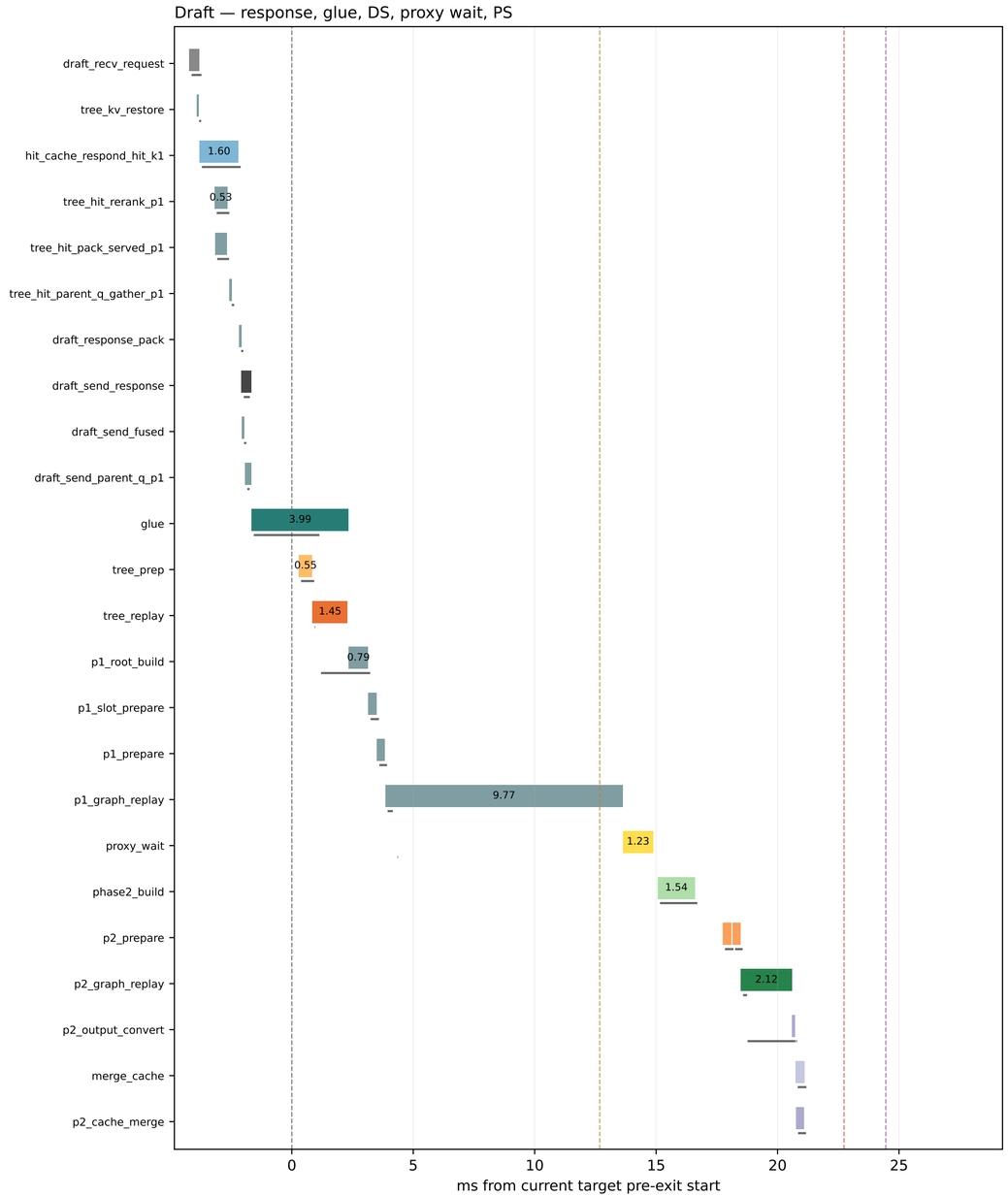
LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M
P1 hit / p25 target-step duration / request 1, UID 84_t0, step 113

exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 16dd4198 / eslab18 RTX3090 / 4 profiled turns, 387 eligible steps / output cap 256

valid proposal nodes=8, AL incl. recovery=1; DS margin +1.238 ms (estimated); PS margin +1.641 ms; next outcome P2 hit





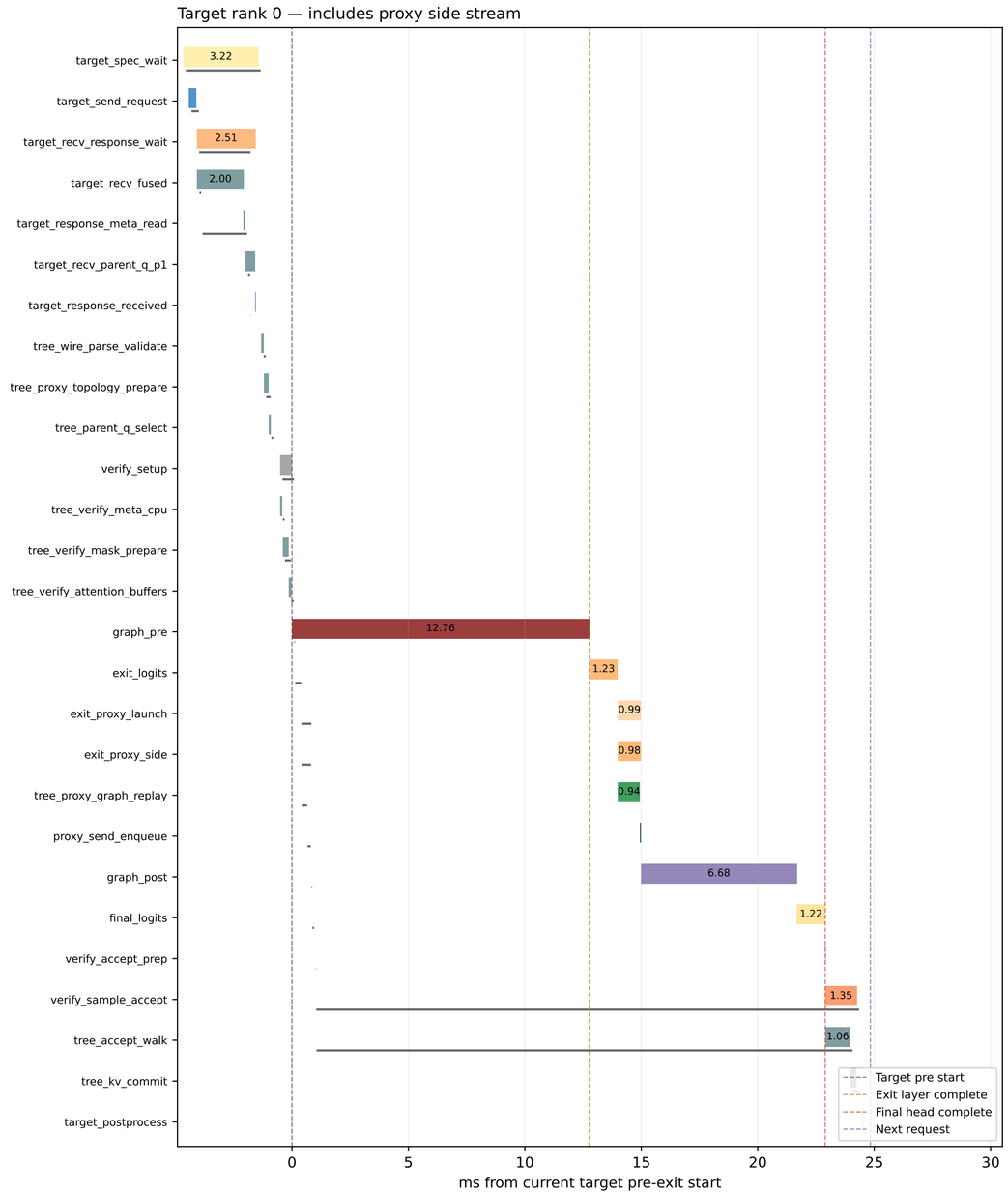
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

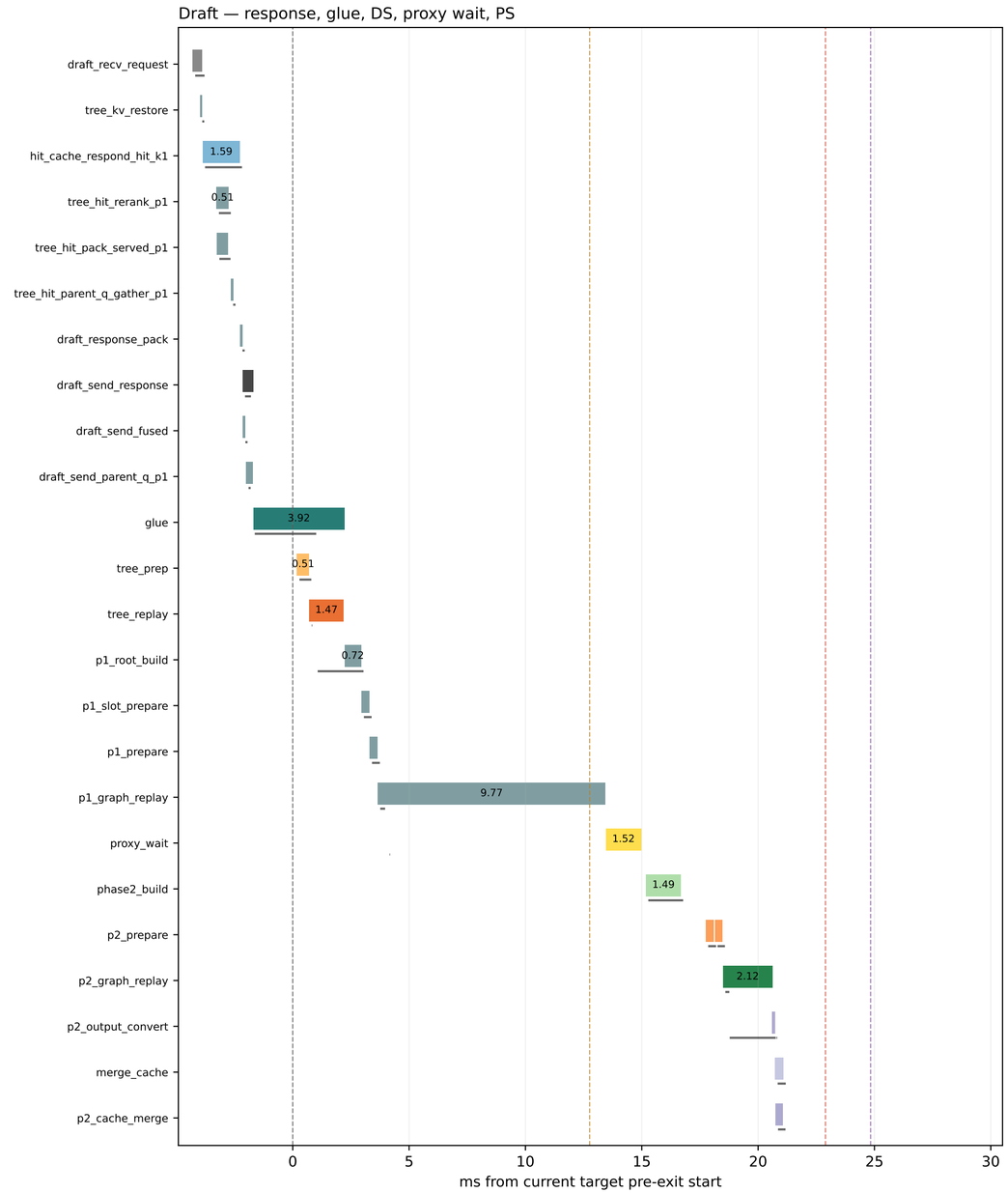
LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M
P1 hit / p50 target-step duration / request 1, UID 84_t0, step 170

exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 16dd4198 / eslab18 RTX3090 / 4 profiled turns, 387 eligible steps / output cap 256

valid proposal nodes=8, AL incl. recovery=4; DS margin +1.522 ms (estimated); PS margin +1.804 ms; next outcome P1 hit





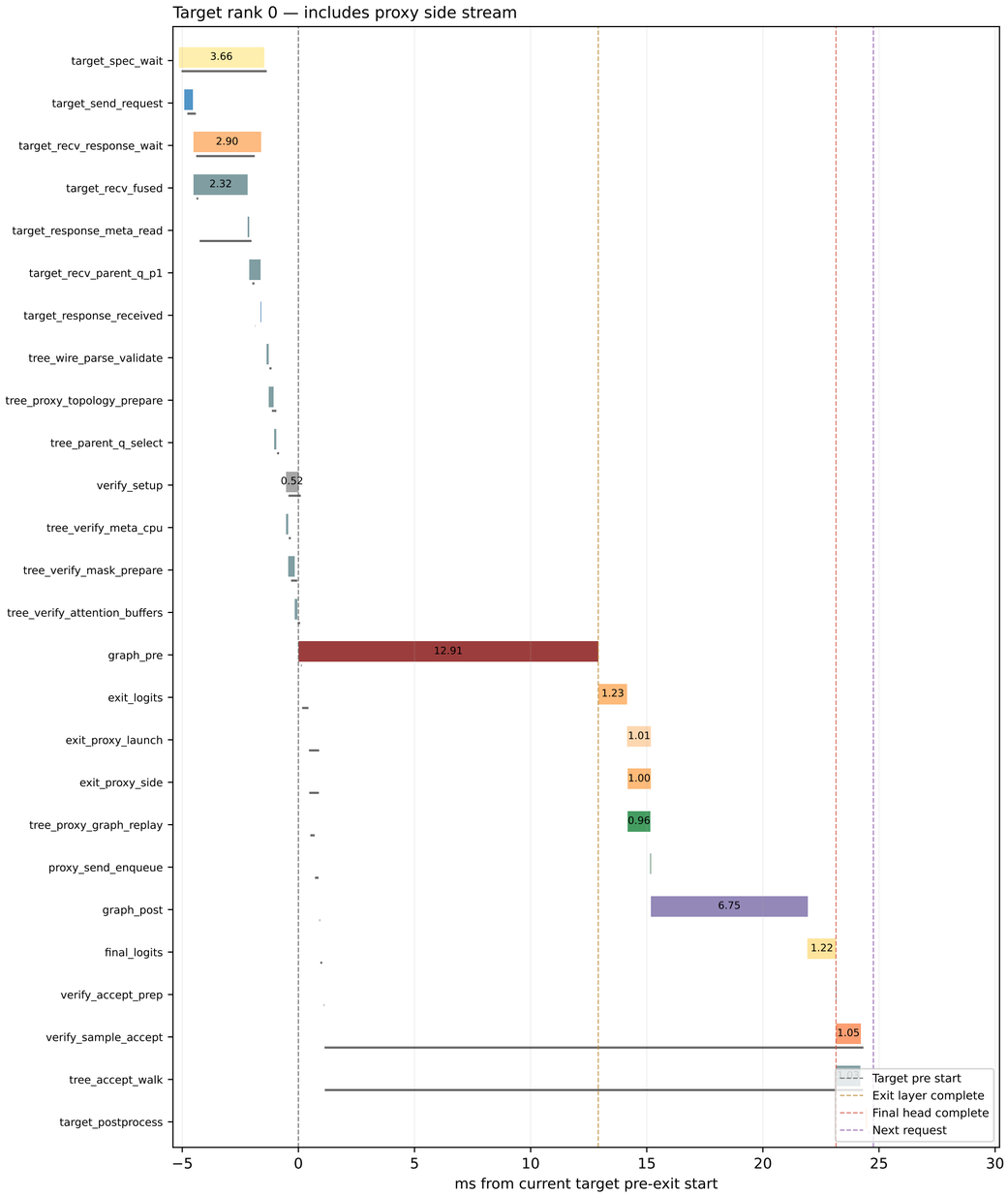
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

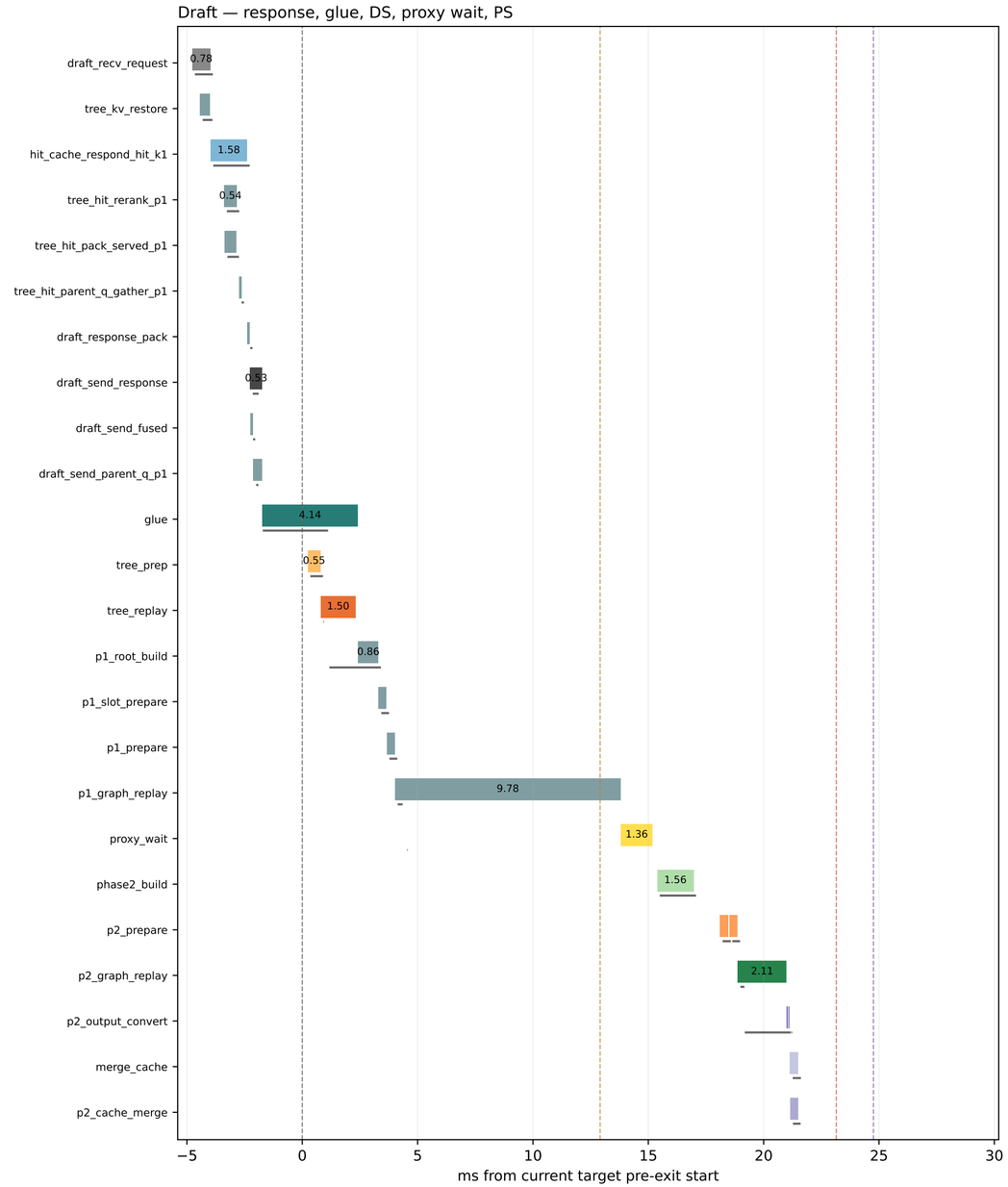
LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M
P1 hit / p75 target-step duration / request 2, UID 244_t0, step 48

exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 16dd4198 / eslab18 RTX3090 / 4 profiled turns, 387 eligible steps / output cap 256

valid proposal nodes=8, AL incl. recovery=2; DS margin +1.356 ms (estimated); PS margin +1.650 ms; next outcome P2 hit





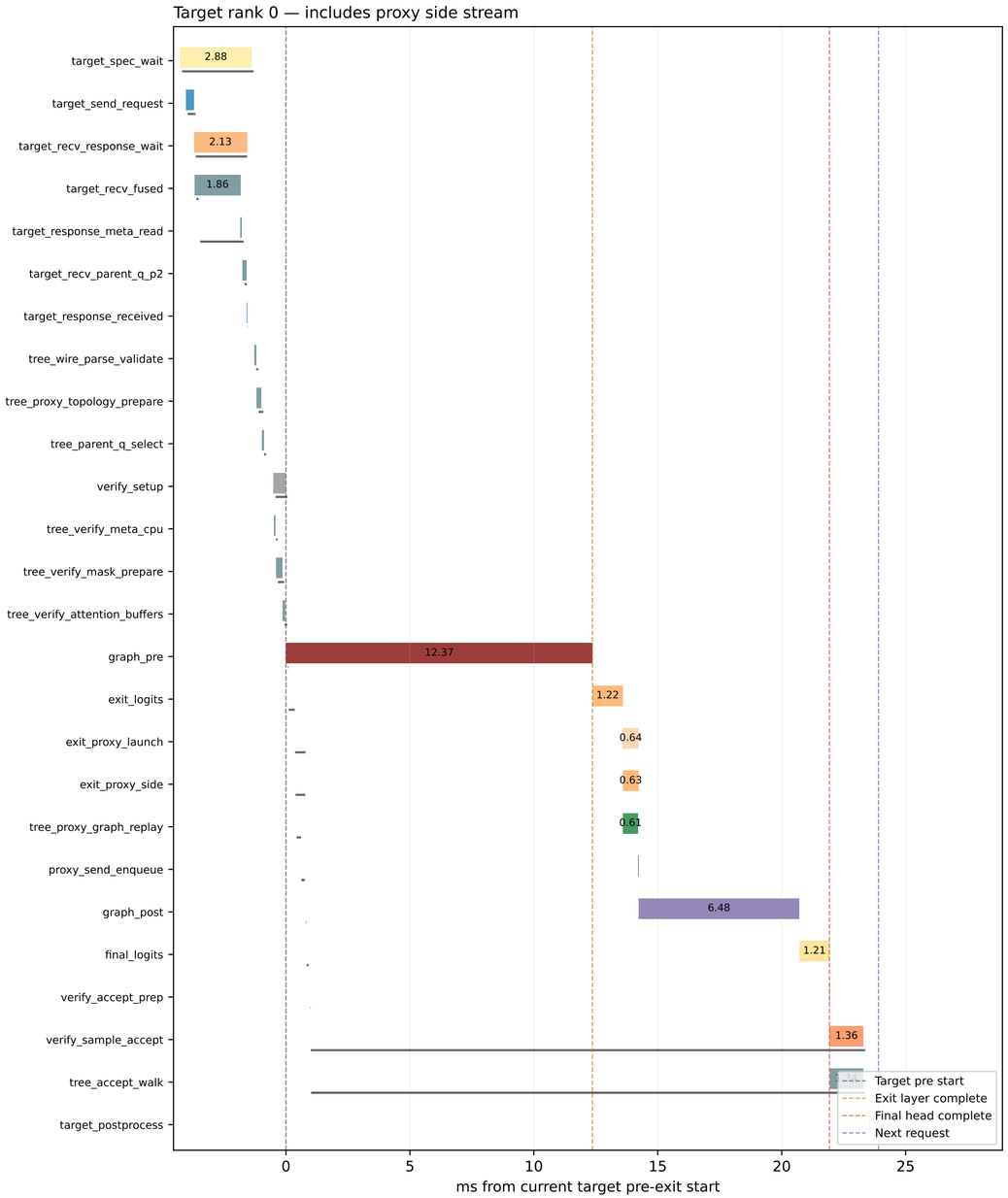
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

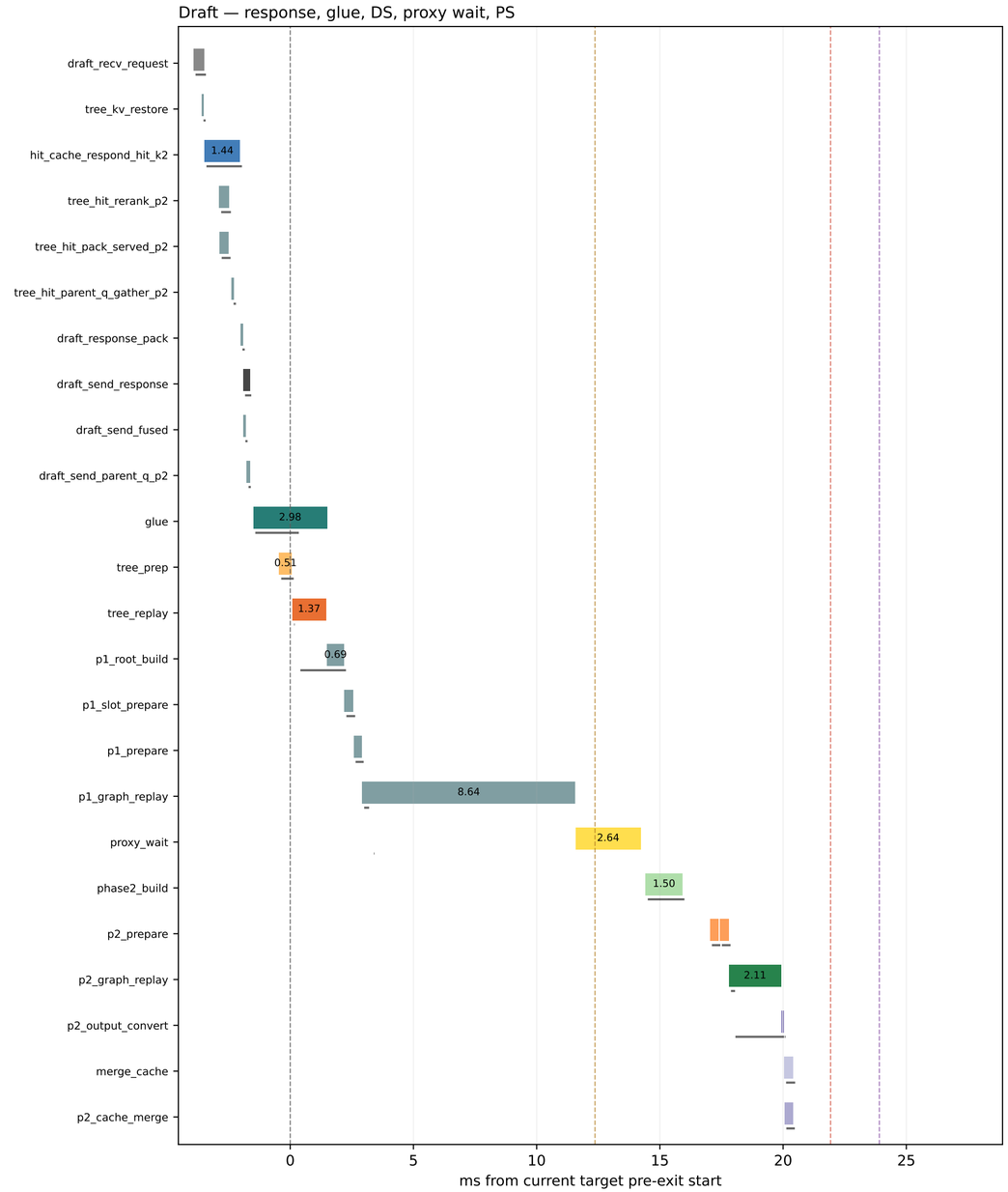
LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M
P2 hit / p25 target-step duration / request 1, UID 84_t0, step 23

exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 16dd4198 / eslab18 RTX3090 / 4 profiled turns, 387 eligible steps / output cap 256

valid proposal nodes=3, AL incl. recovery=1; DS margin +2.645 ms (estimated); PS margin +1.517 ms; next outcome P2 hit





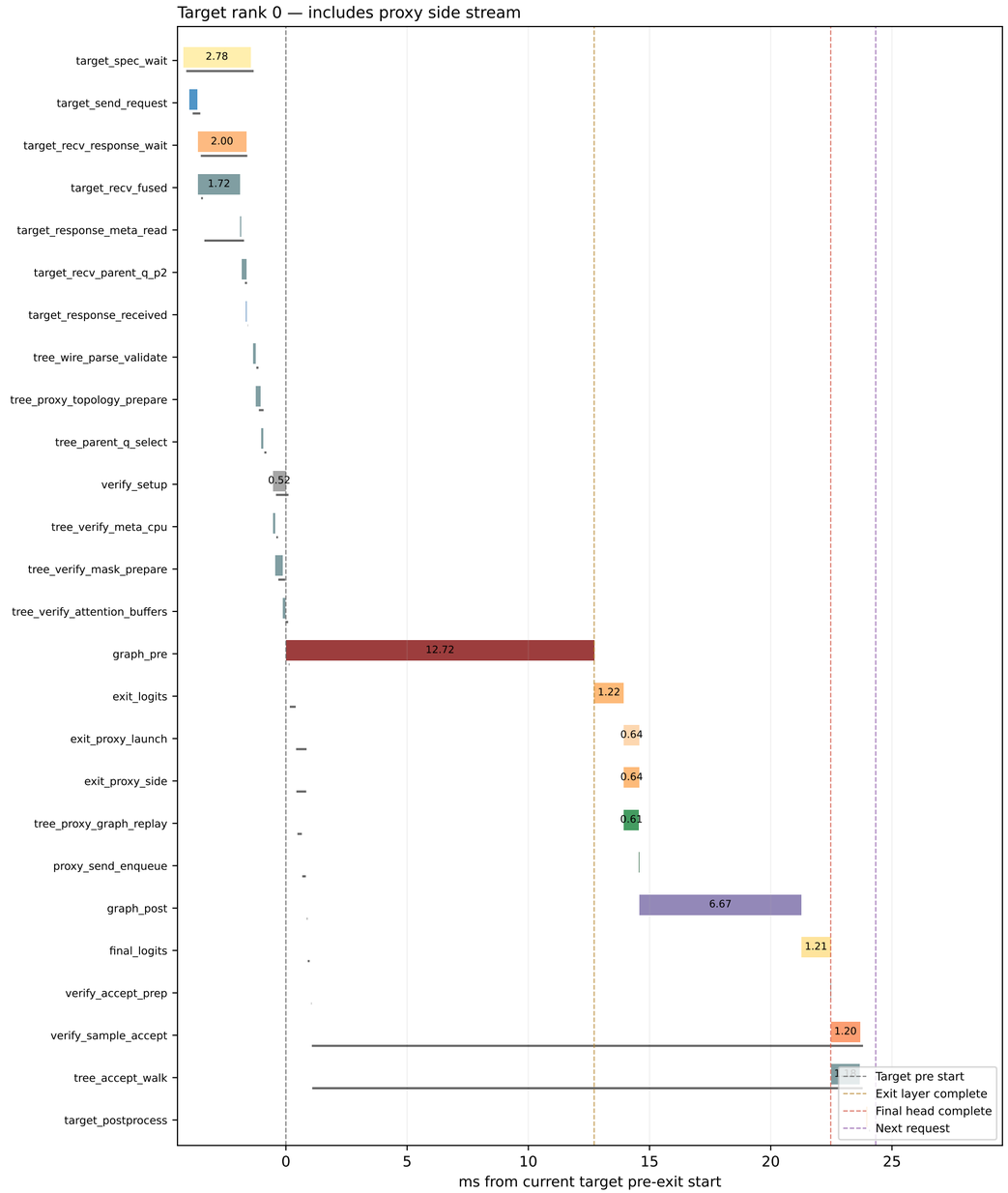
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

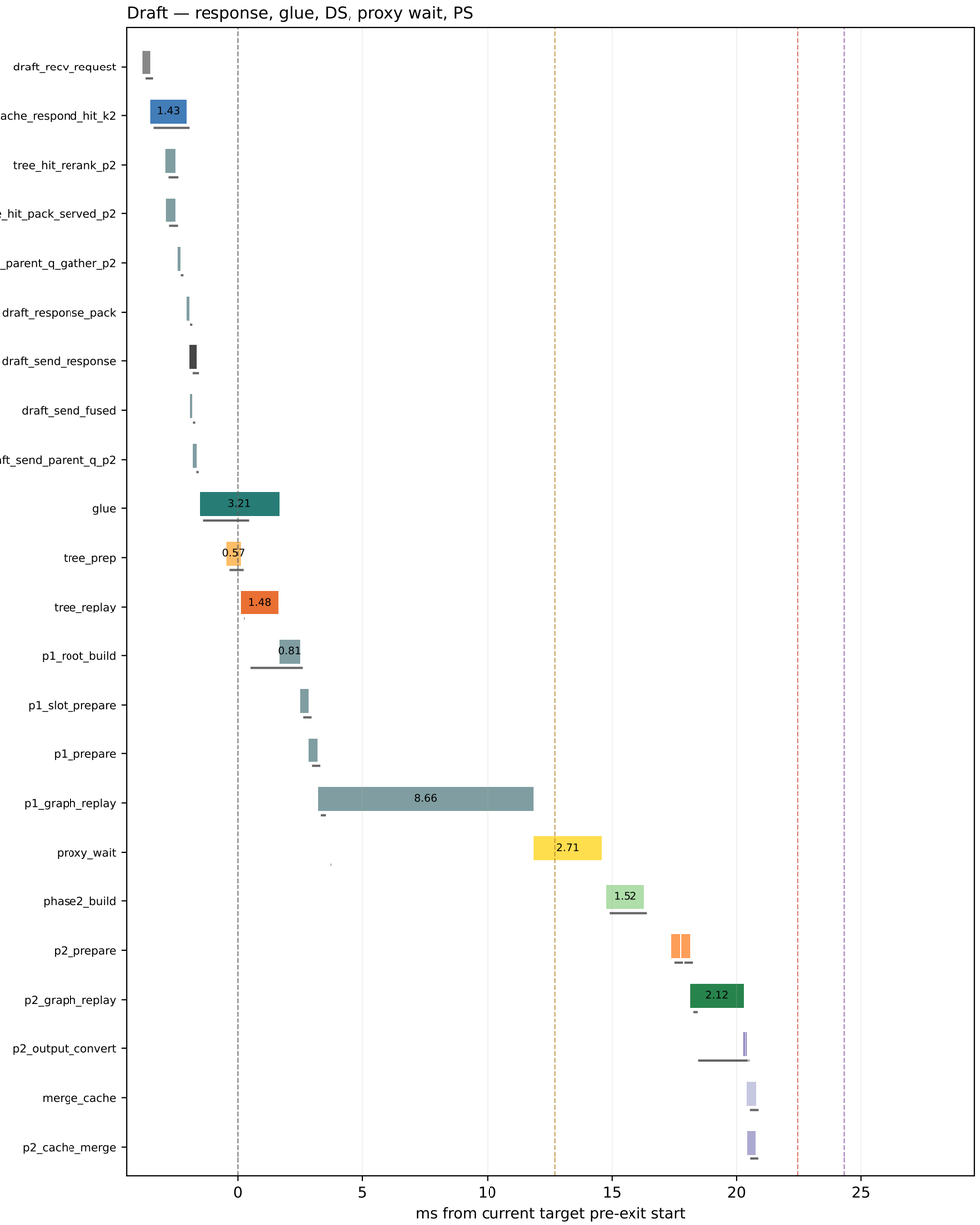
LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M
P2 hit / p50 target-step duration / request 2, UID 244_t0, step 92

exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 16dd4198 / eslab18 RTX3090 / 4 profiled turns, 387 eligible steps / output cap 256

valid proposal nodes=3, AL incl. recovery=1; DS margin +2.710 ms (estimated); PS margin +1.708 ms; next outcome P2 hit





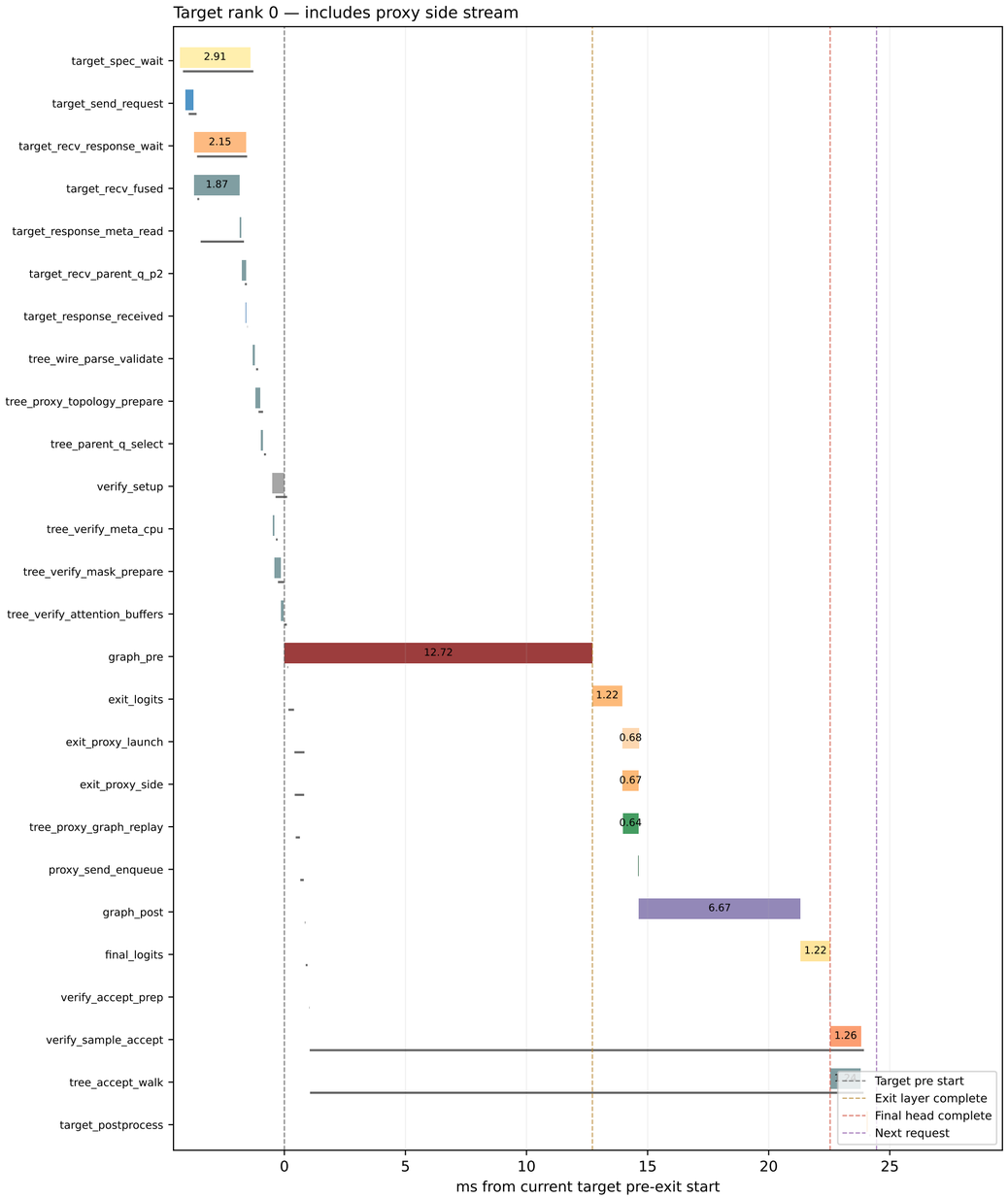
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

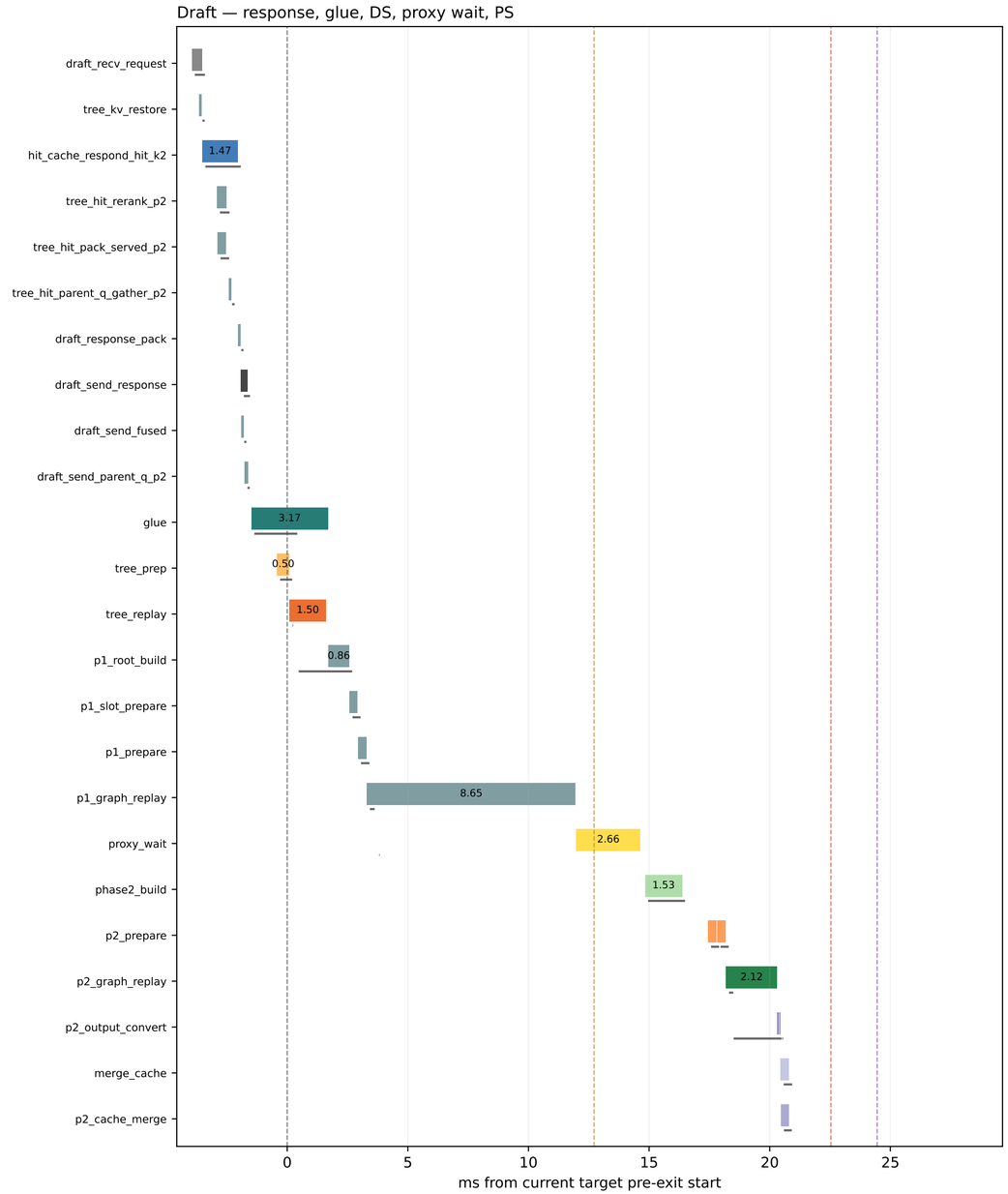
LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M
P2 hit / p75 target-step duration / request 2, UID 244_t0, step 101

exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 16dd4198 / eslab18 RTX3090 / 4 profiled turns, 387 eligible steps / output cap 256

valid proposal nodes=3, AL incl. recovery=1; DS margin +2.659 ms (estimated); PS margin +1.731 ms; next outcome P2 hit





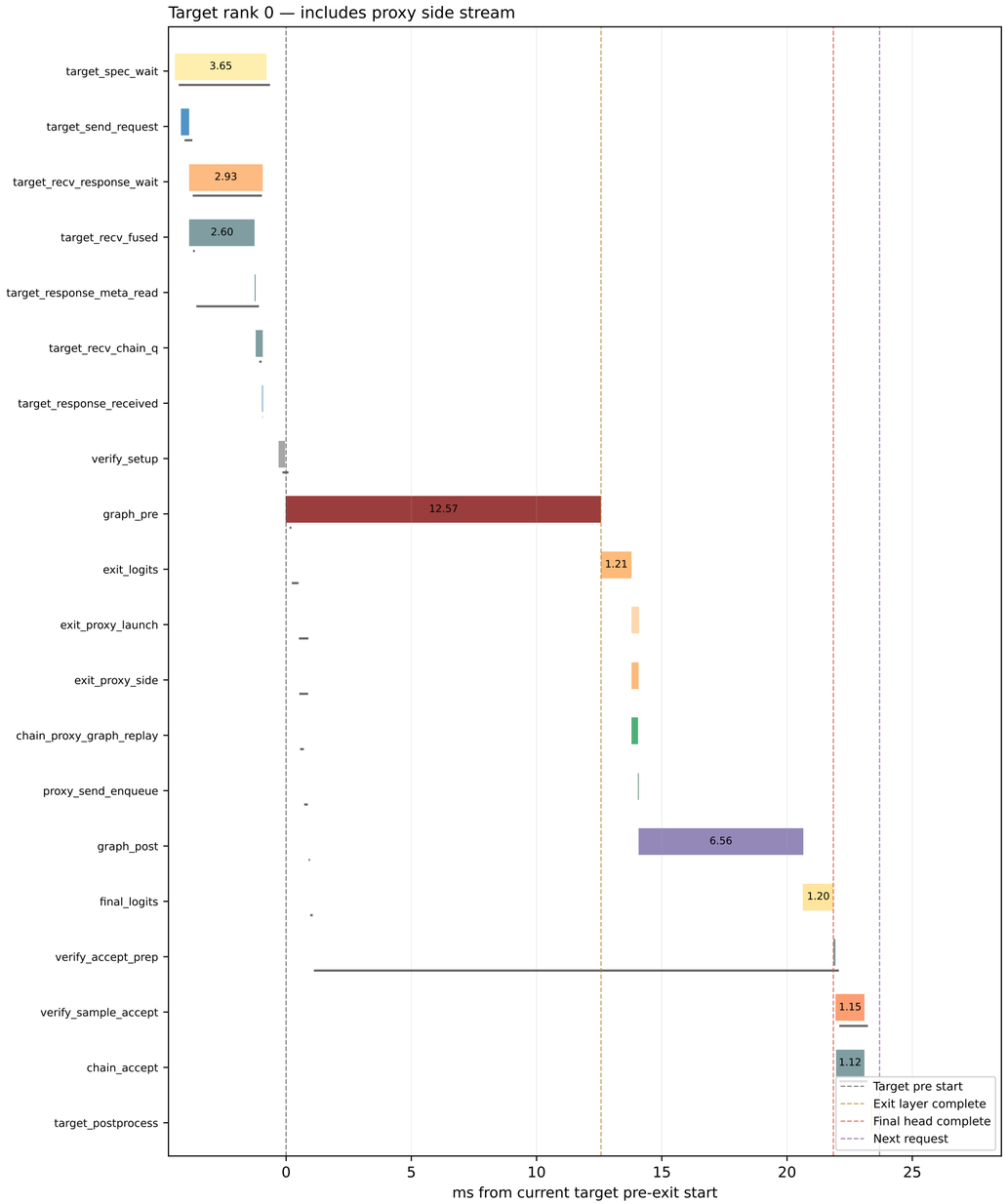
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

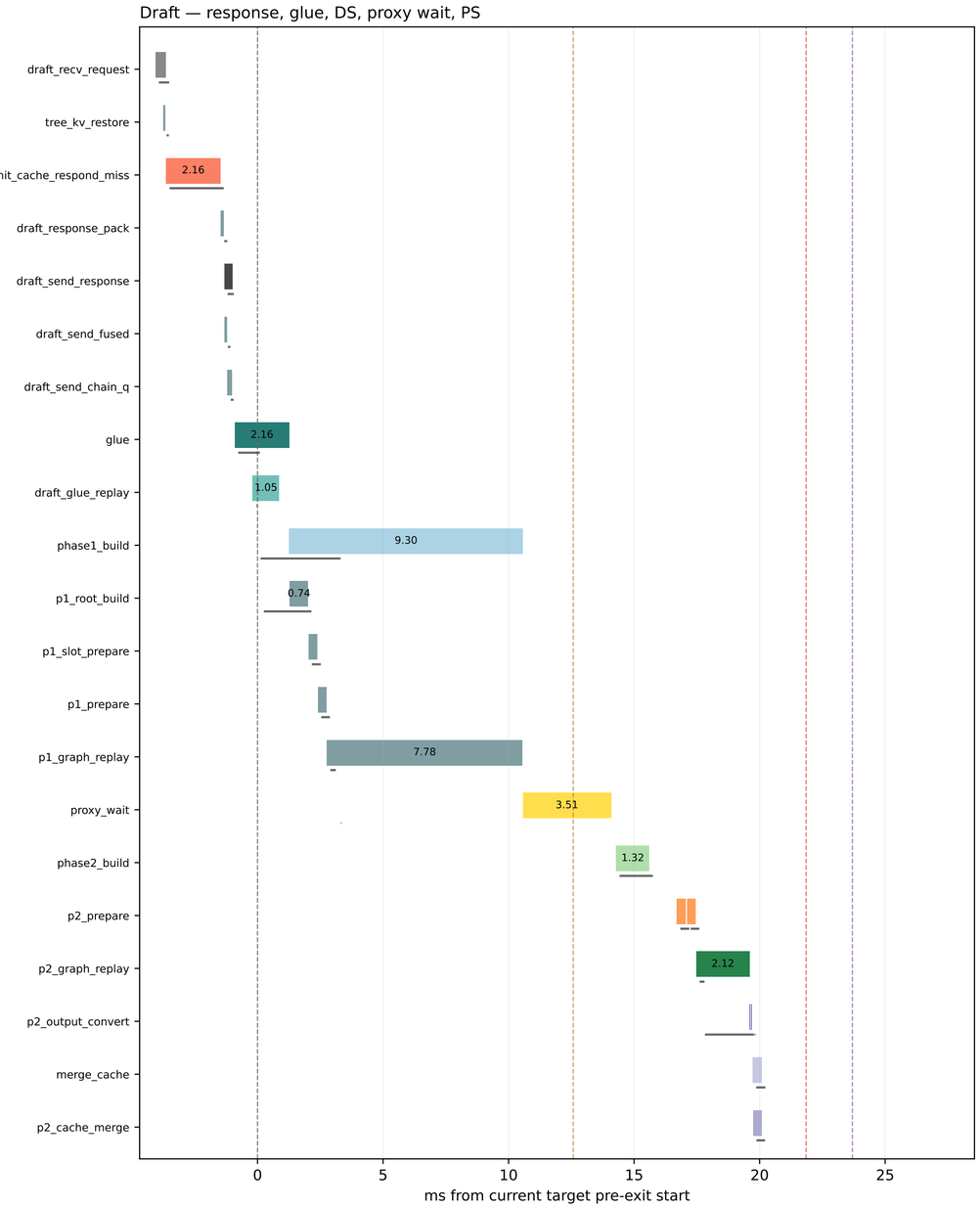
LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M
Miss / p25 target-step duration / request 4, UID 288_t0, step 26

exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 16dd4198 / eslab18 RTX3090 / 4 profiled turns, 387 eligible steps / output cap 256

valid proposal nodes=1, AL incl. recovery=1; DS margin +3.504 ms (estimated); PS margin +1.755 ms; next outcome P1 hit





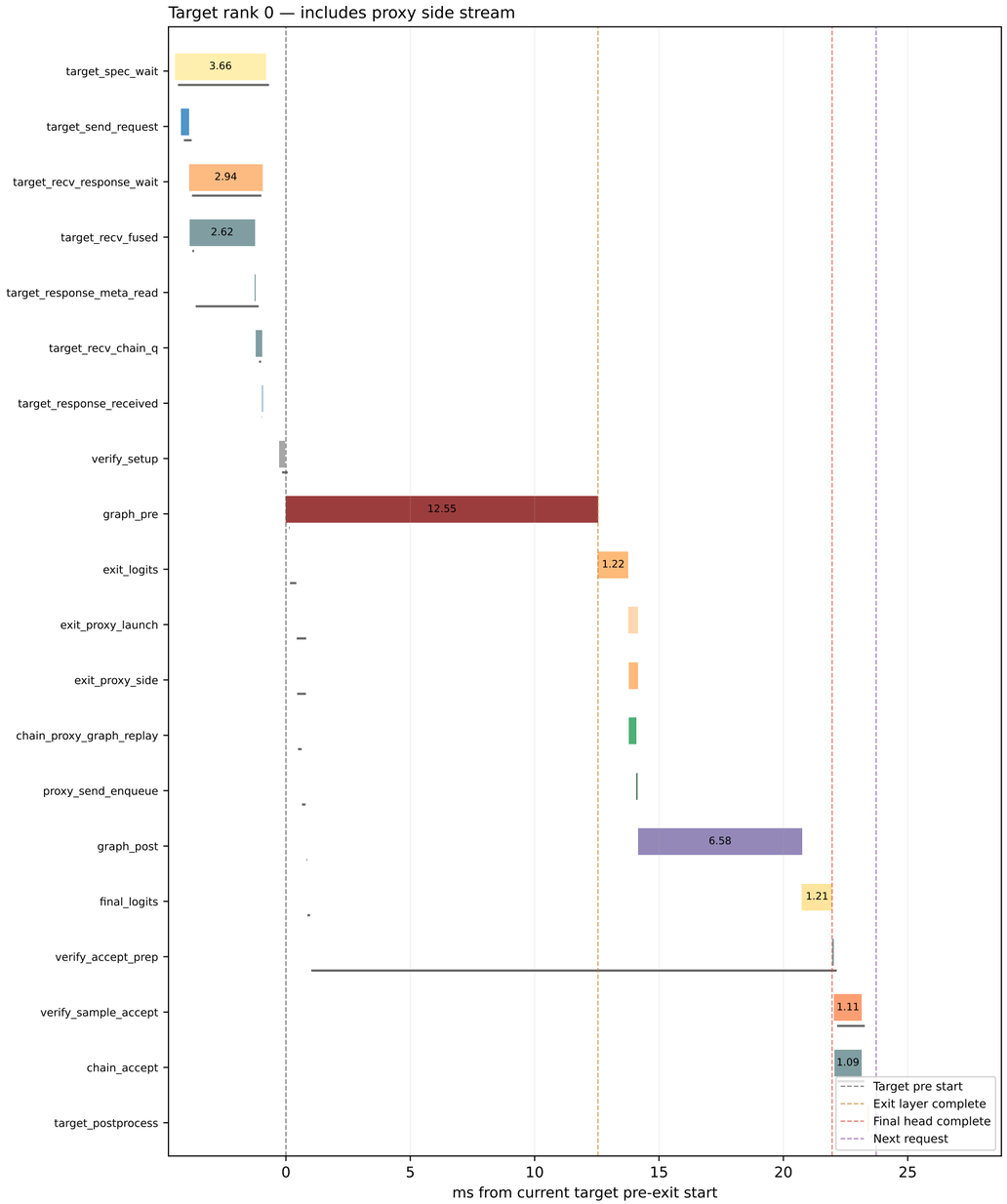
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

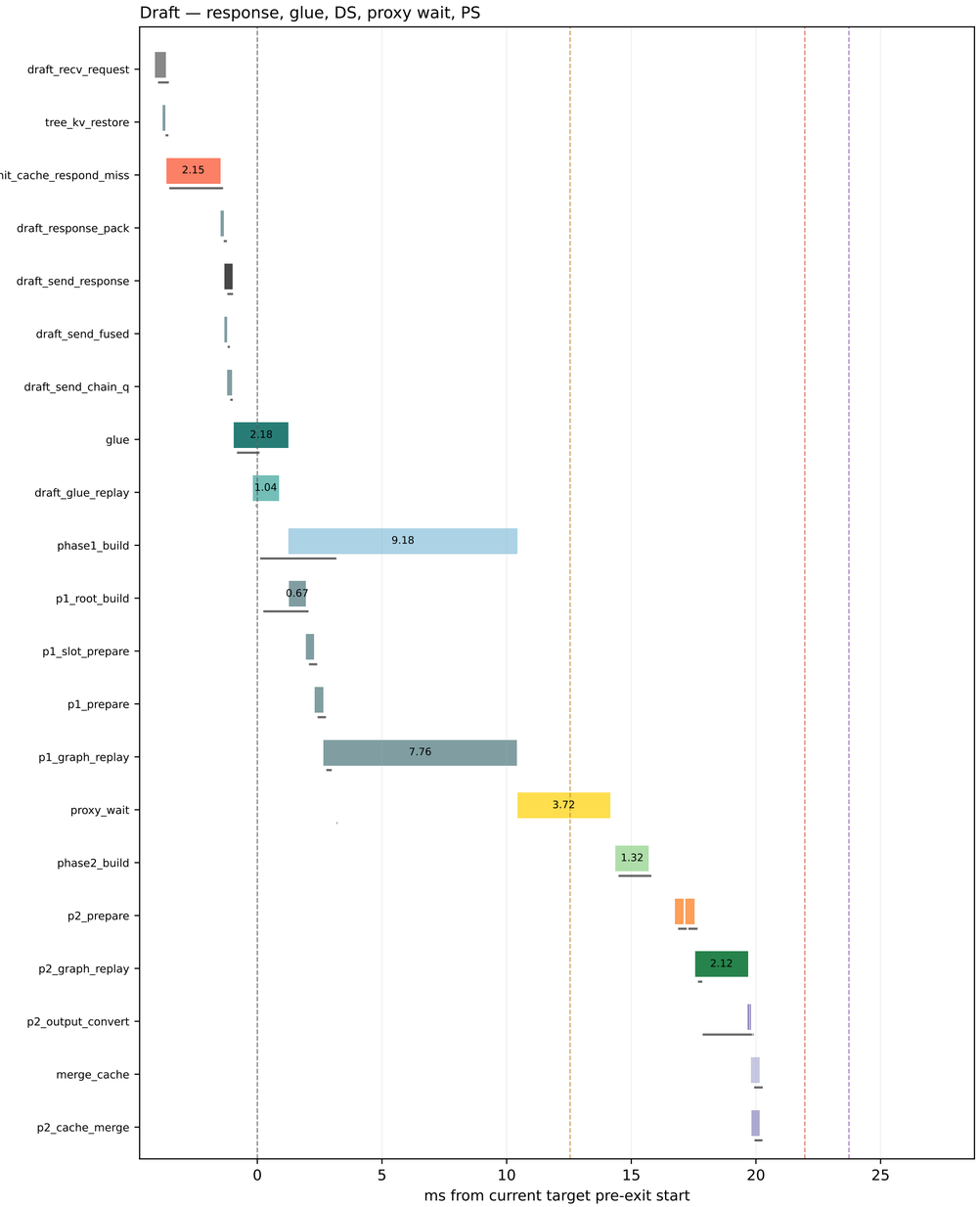
LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M
Miss / p50 target-step duration / request 2, UID 244_t0, step 65

exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 16dd4198 / eslab18 RTX3090 / 4 profiled turns, 387 eligible steps / output cap 256

valid proposal nodes=1, AL incl. recovery=1; DS margin +3.714 ms (estimated); PS margin +1.802 ms; next outcome P2 hit





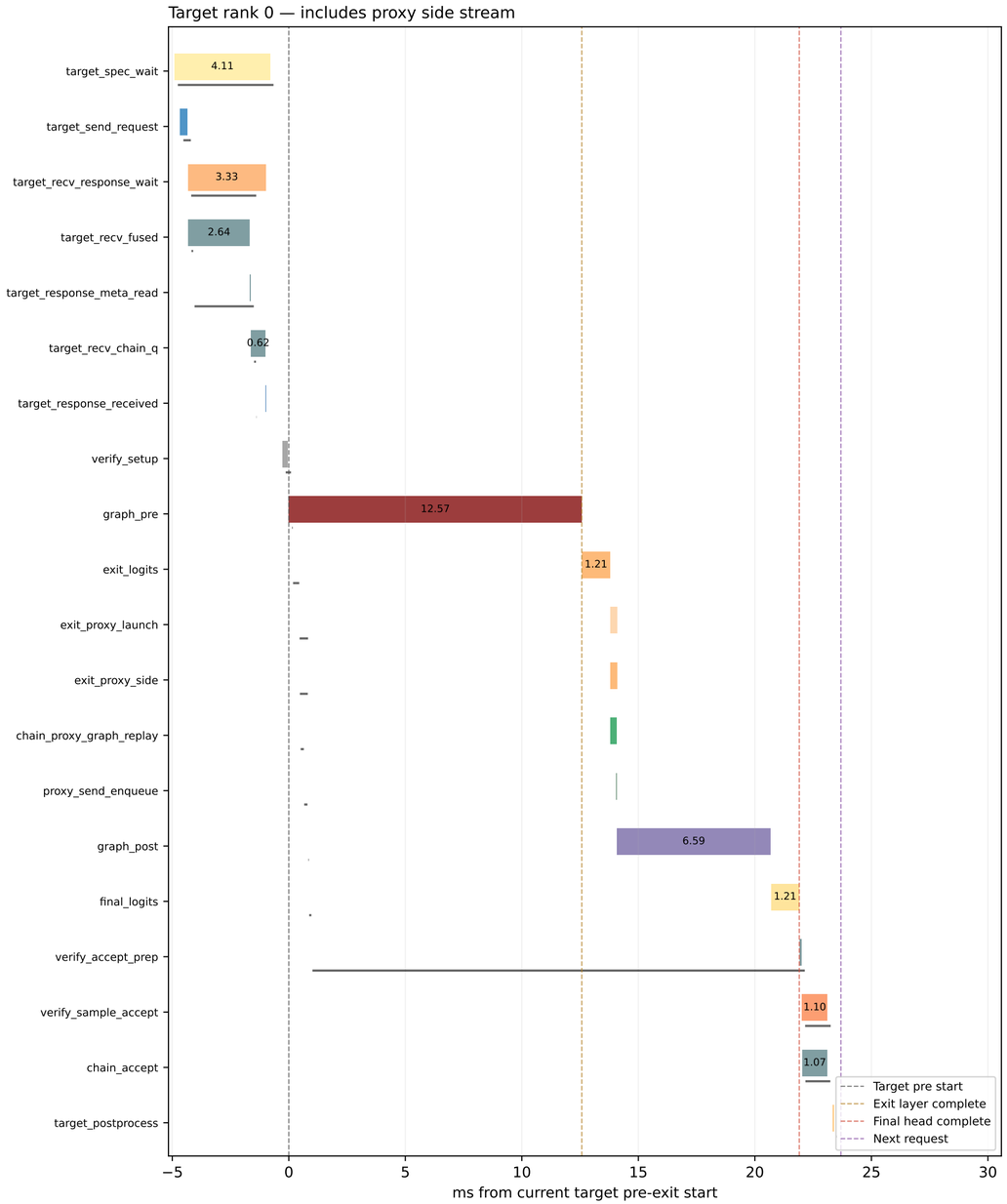
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

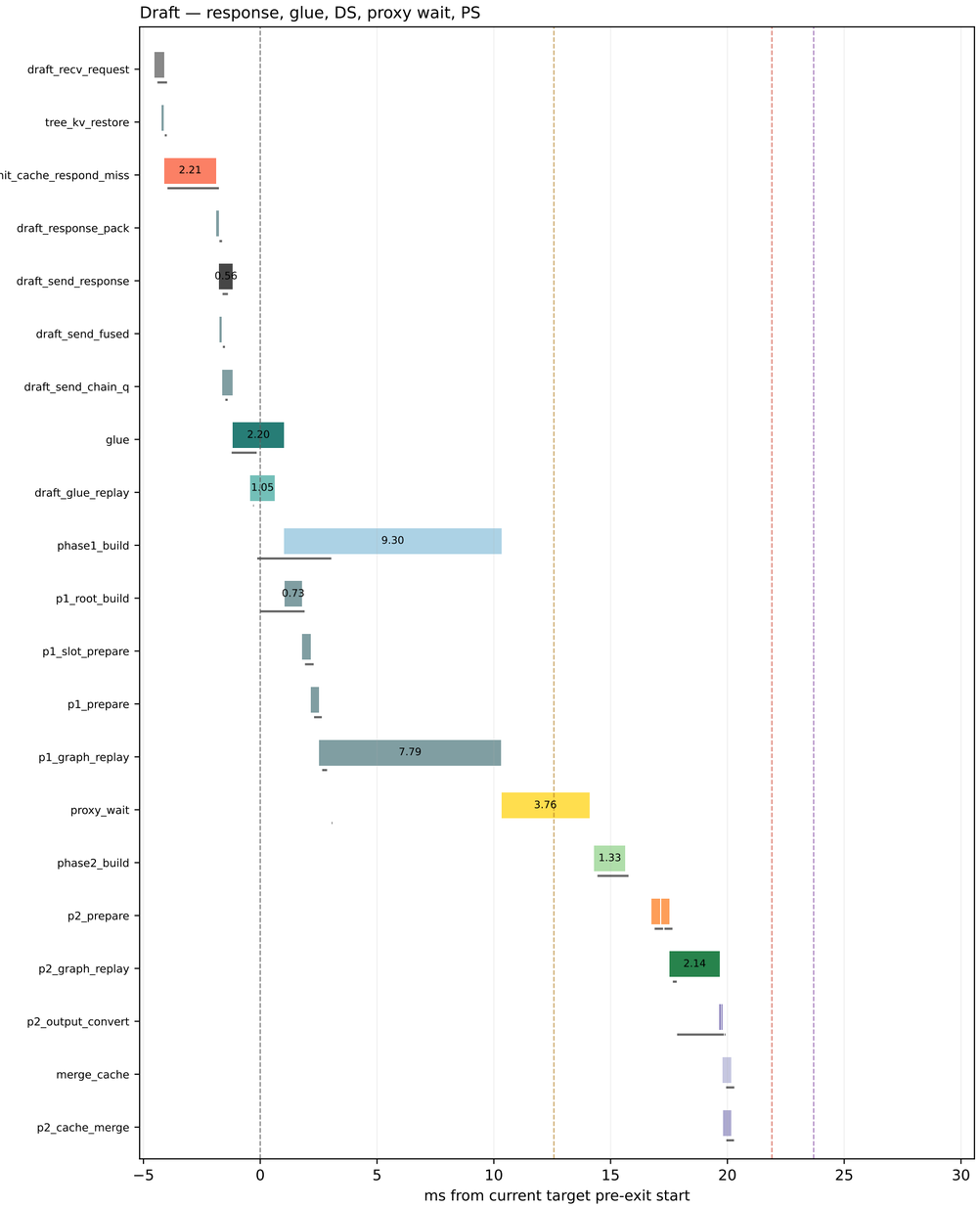
LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M
Miss / p75 target-step duration / request 4, UID 288_t0, step 15

exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 16dd4198 / eslab18 RTX3090 / 4 profiled turns, 387 eligible steps / output cap 256

valid proposal nodes=1, AL incl. recovery=2; DS margin +3.752 ms (estimated); PS margin +1.741 ms; next outcome P1 hit



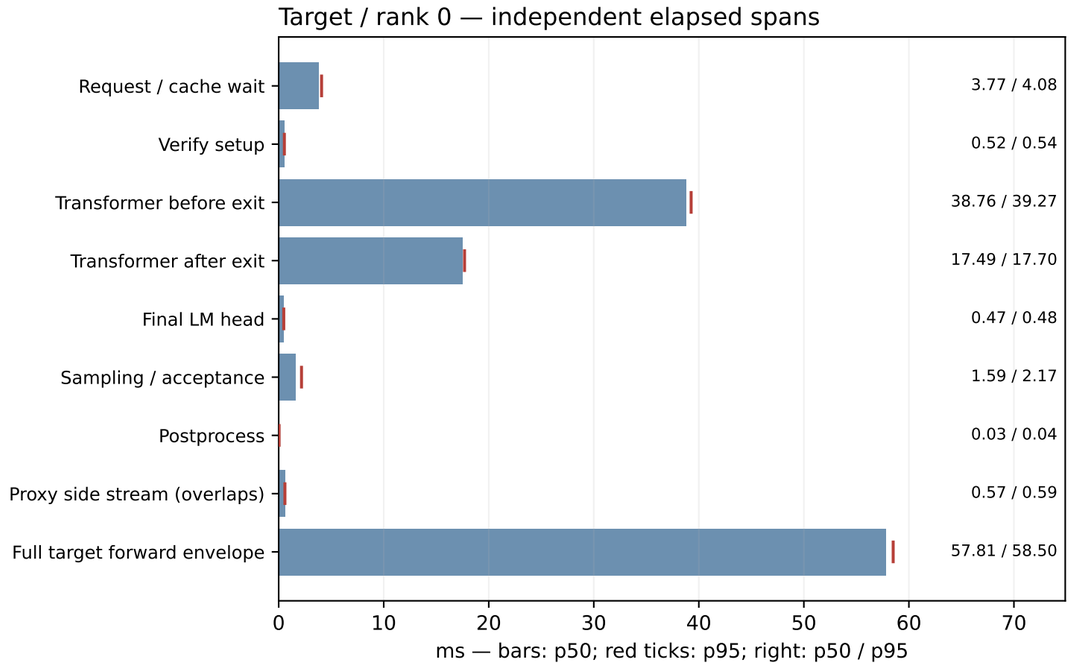


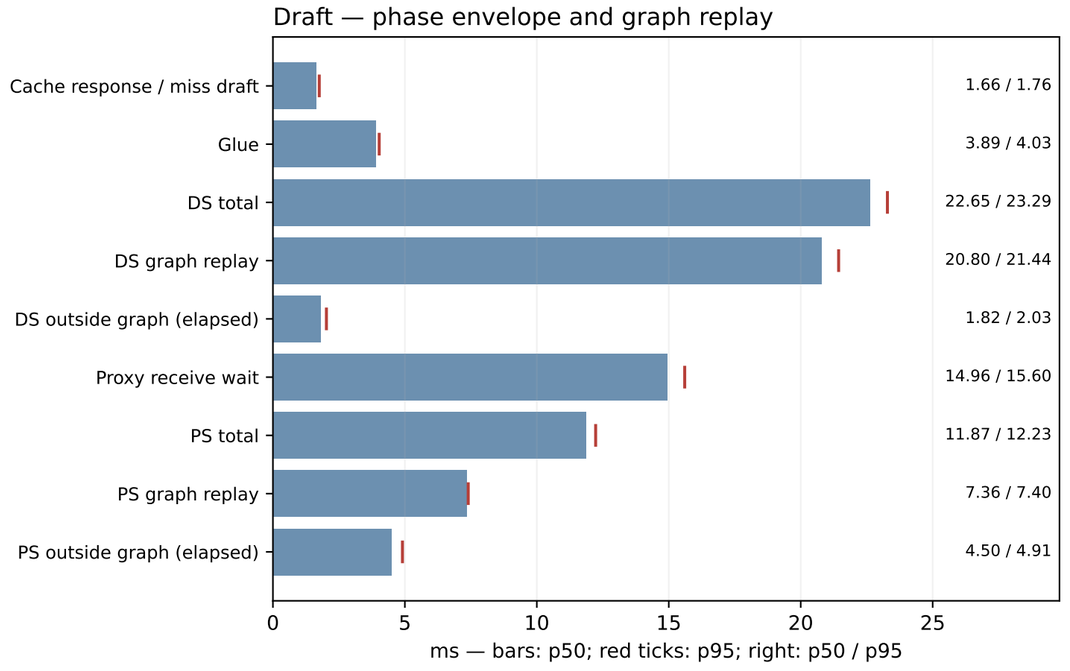
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

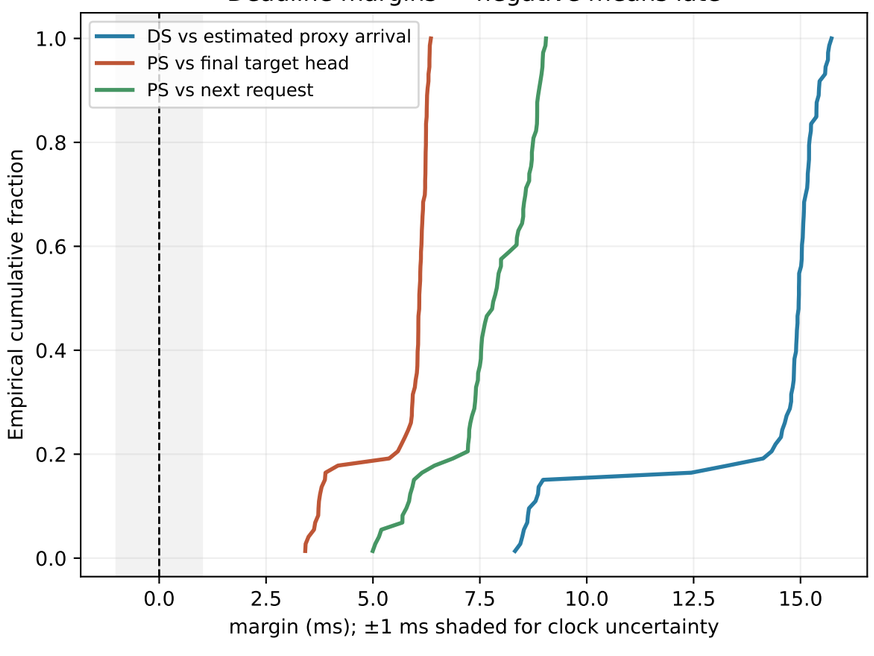
LayerSkip CodeLlama-34B + AMD-Llama-135M-code
Component latency, phase deadlines, and every measured step

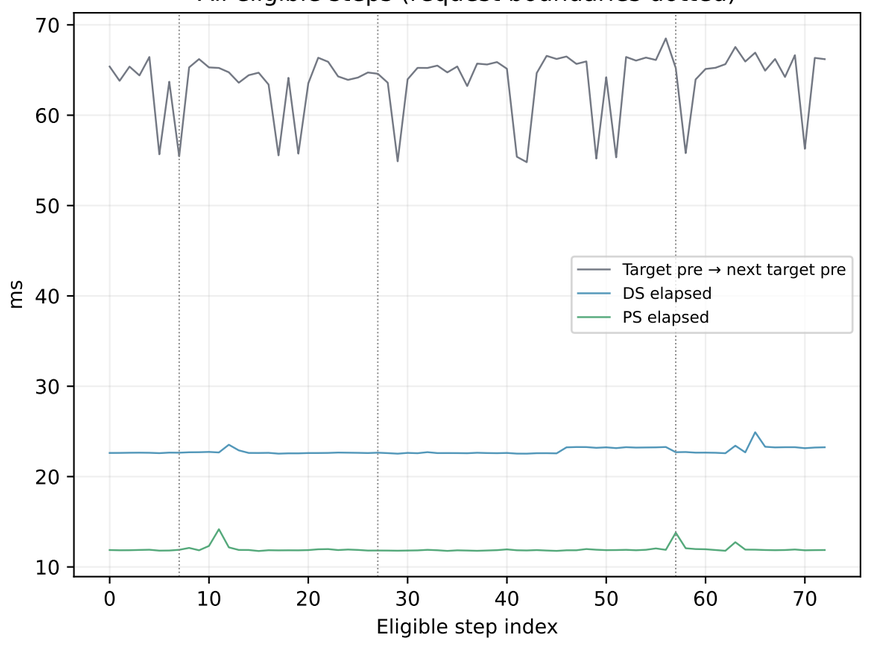
exit 30 (0-based) | K1/K2 12/6 | G=M DS/PS 12/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 59f46511 / eslab18 RTX3090 / 4 profiled turns, 73 eligible steps / output cap 256







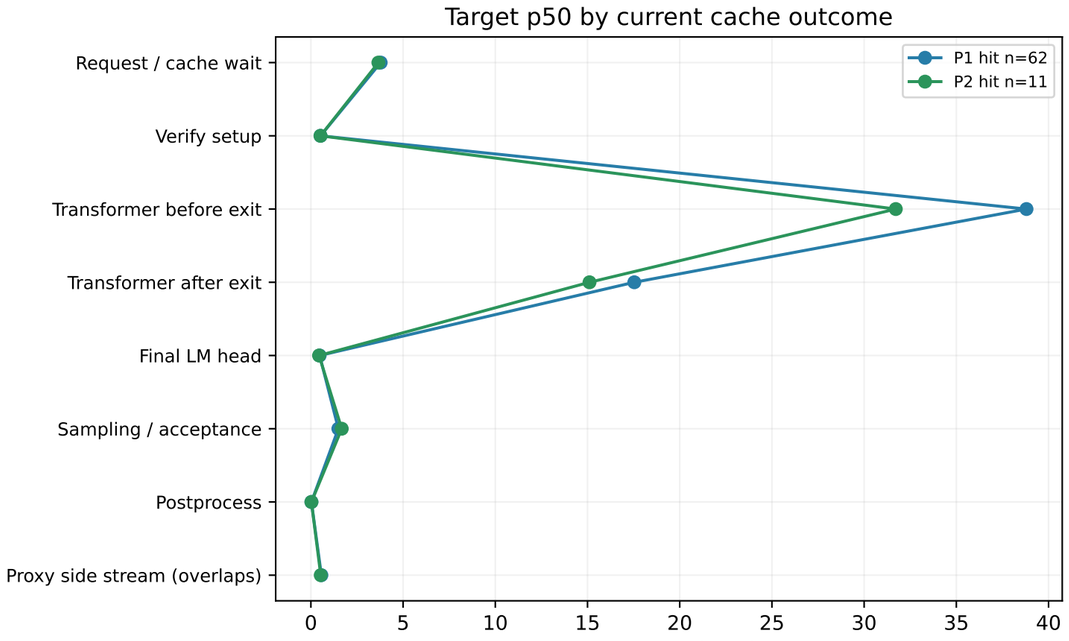


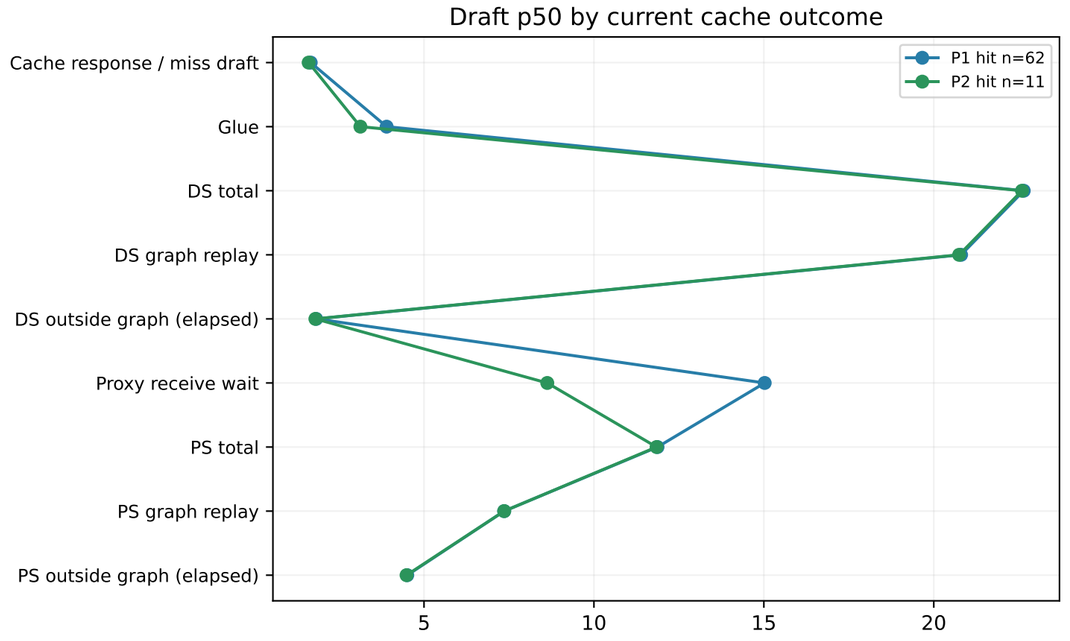
CUDA-event elapsed spans include queue/host stalls; not GPU busy time. Nested/overlapping components are NOT additive.

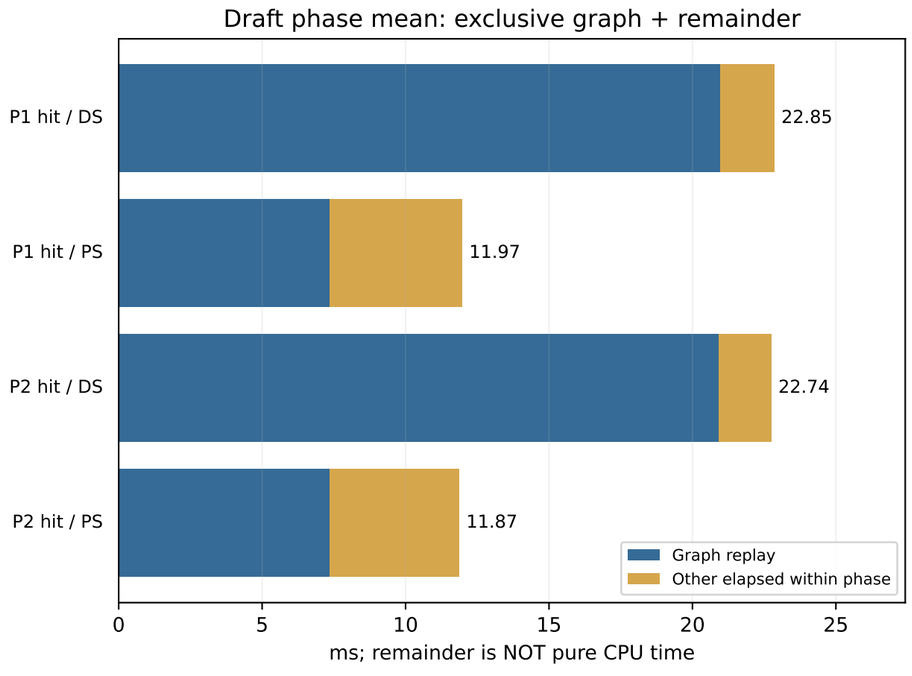
LayerSkip CodeLlama-34B + AMD-Llama-135M-code
P1 hit / P2 hit / miss — component distributions and phase composition

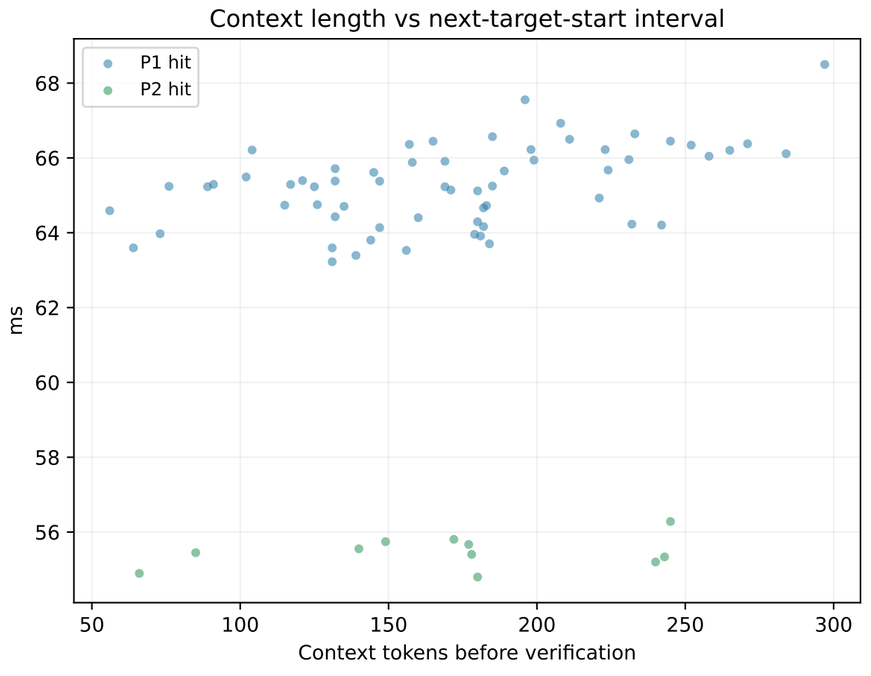
exit 30 (0-based) | K1/K2 12/6 | G=M DS/PS 12/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 59f46511 / eslab18 RTX3090 / 4 profiled turns, 73 eligible steps / output cap 256









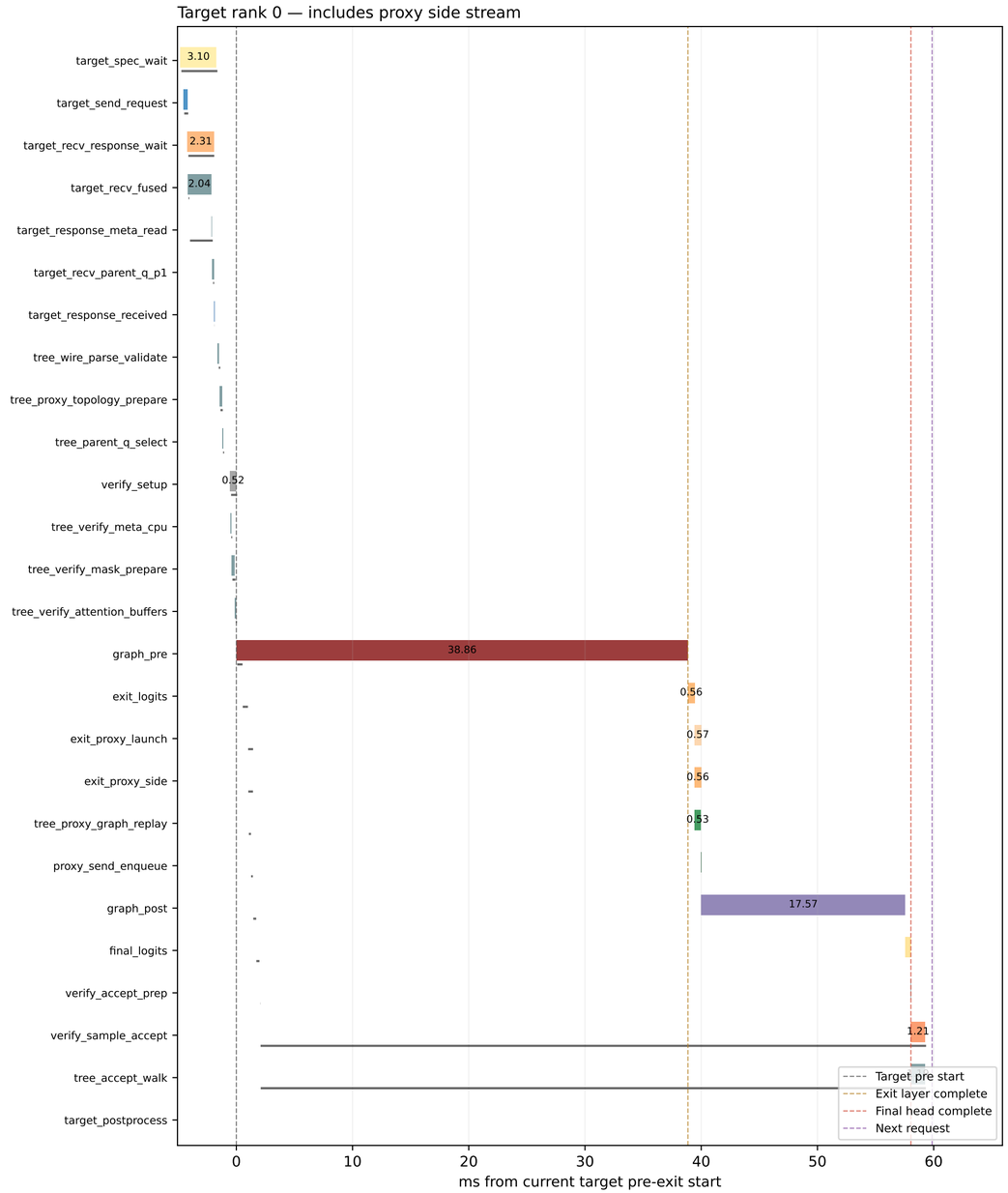
CUDA-event elapsed spans include queue/host stalls; not GPU busy time. Nested/overlapping components are NOT additive.

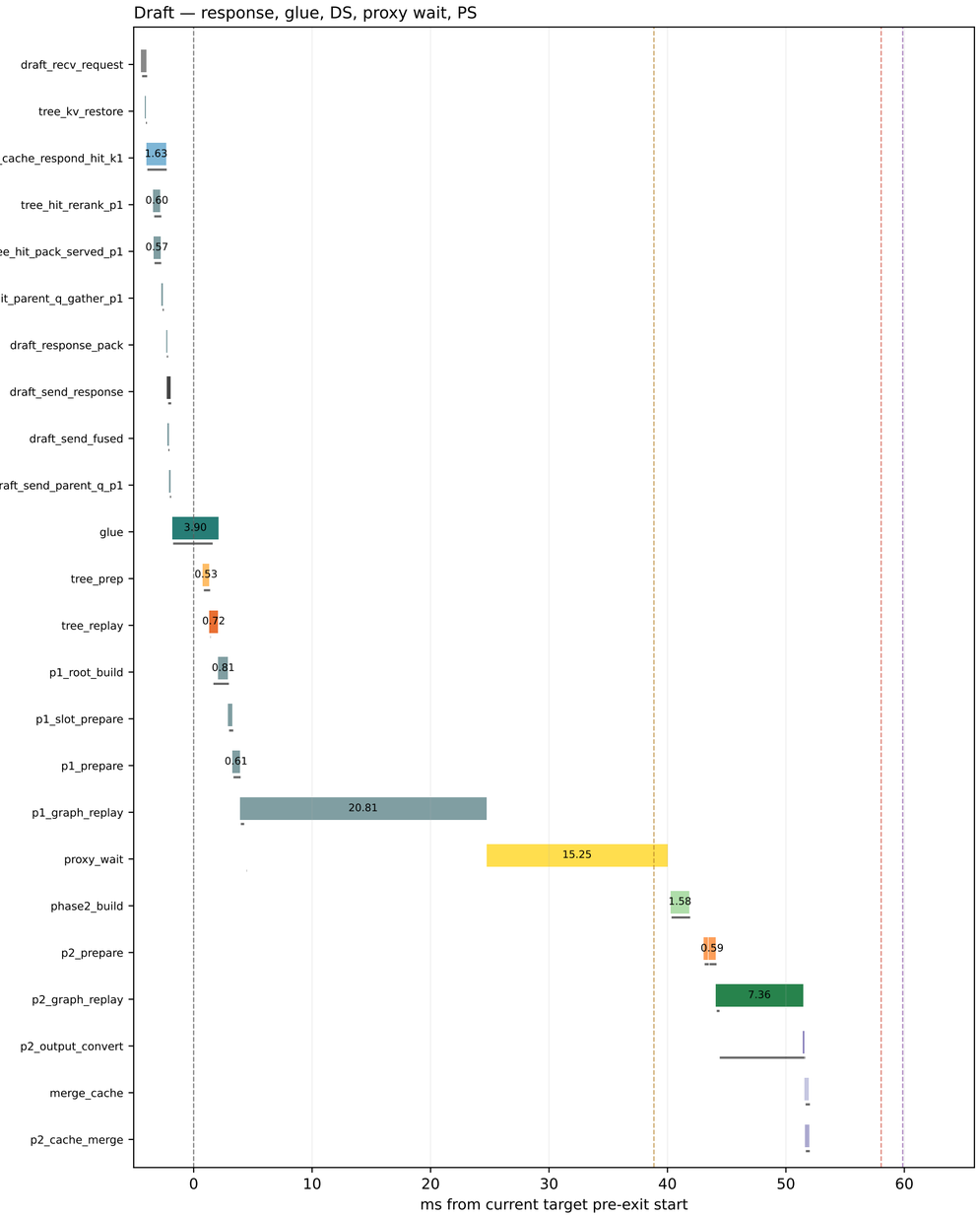
LayerSkip CodeLlama-34B + AMD-Llama-135M-code
P1 hit / p25 target-step duration / request 4, UID humaneval_local_033_t0, step 6

exit 30 (0-based) | K1/K2 12/6 | G=M DS/PS 12/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 59f46511 / eslab18 RTX3090 / 4 profiled turns, 73 eligible steps / output cap 256

valid proposal nodes=12, AL incl. recovery=5; DS margin +15.247 ms (estimated); PS margin +6.094 ms; next outcome P1 hit





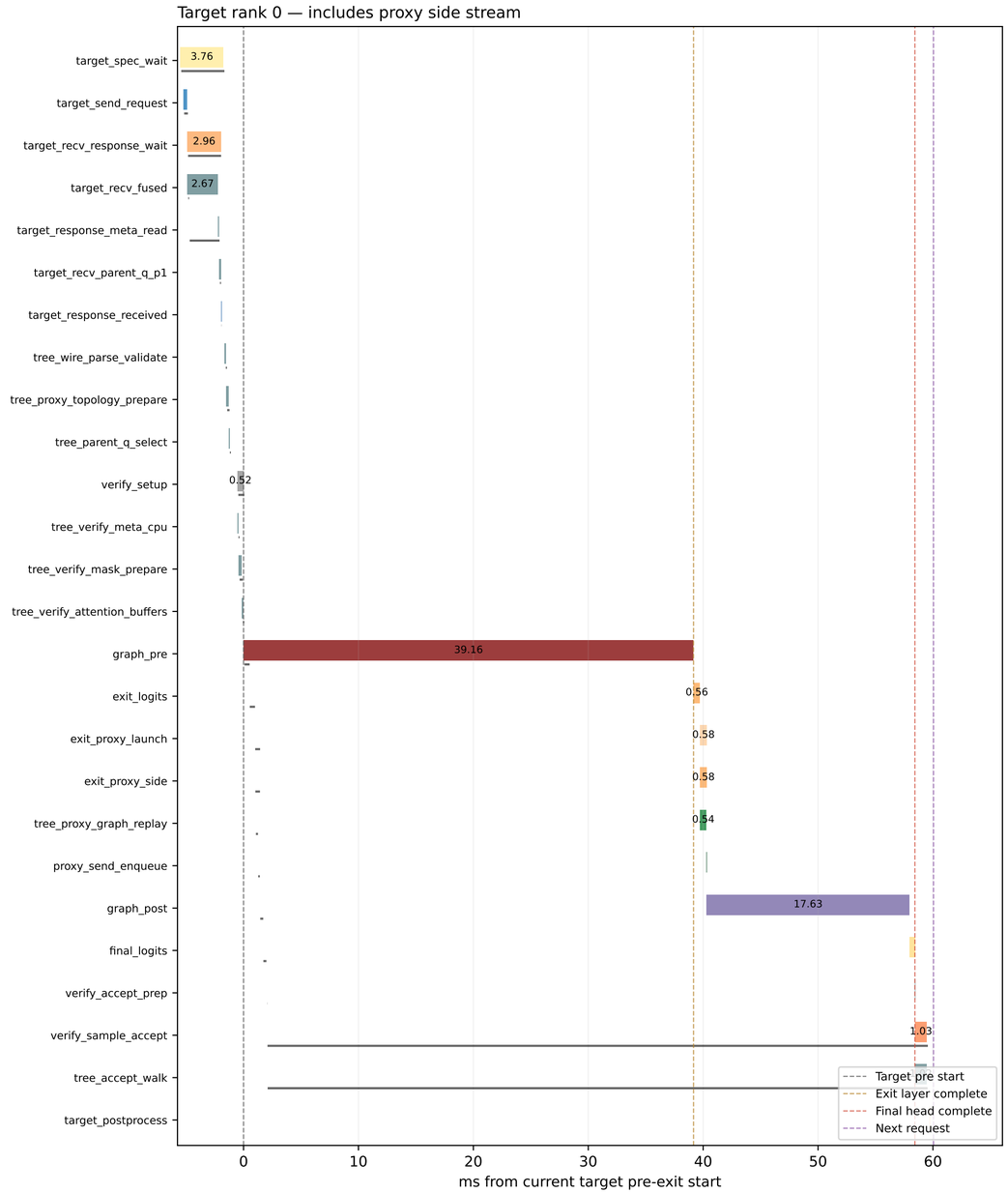
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

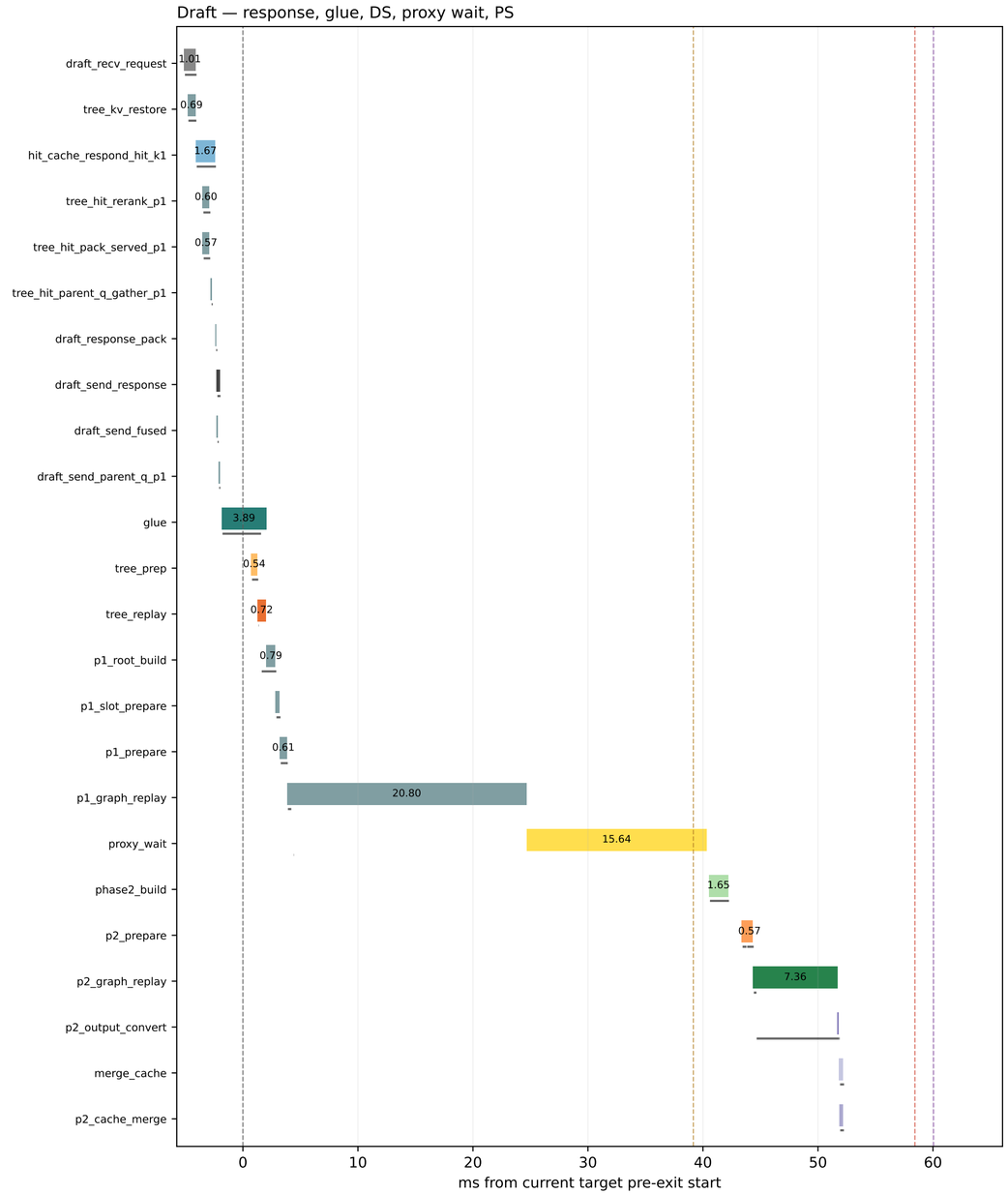
LayerSkip CodeLlama-34B + AMD-Llama-135M-code
P1 hit / p50 target-step duration / request 4, UID humaneval_local_033_t0, step 7

exit 30 (0-based) | K1/K2 12/6 | G=M DS/PS 12/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 59f46511 / eslab18 RTX3090 / 4 profiled turns, 73 eligible steps / output cap 256

valid proposal nodes=12, AL incl. recovery=4; DS margin +15.645 ms (estimated); PS margin +6.241 ms; next outcome P1 hit





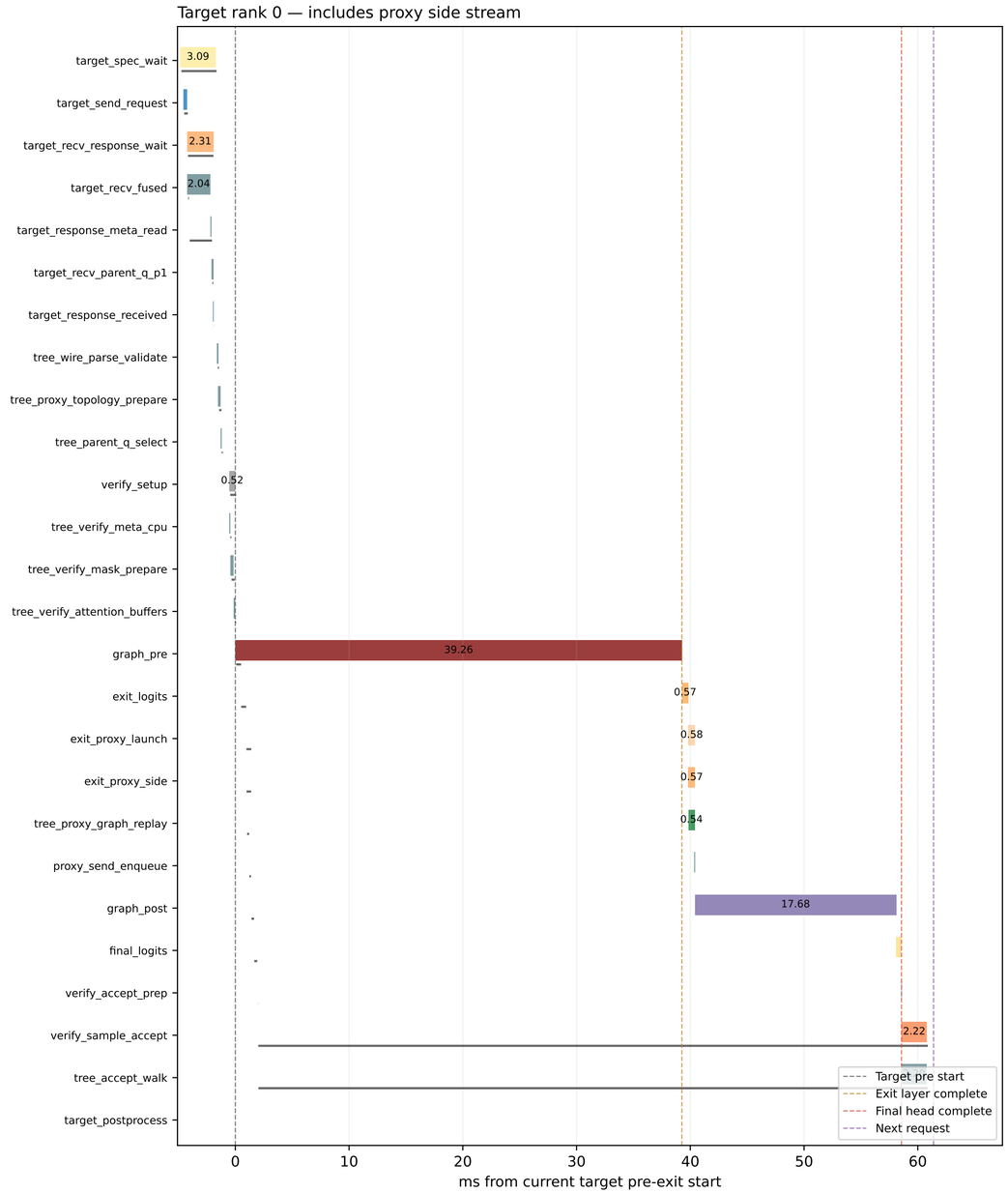
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

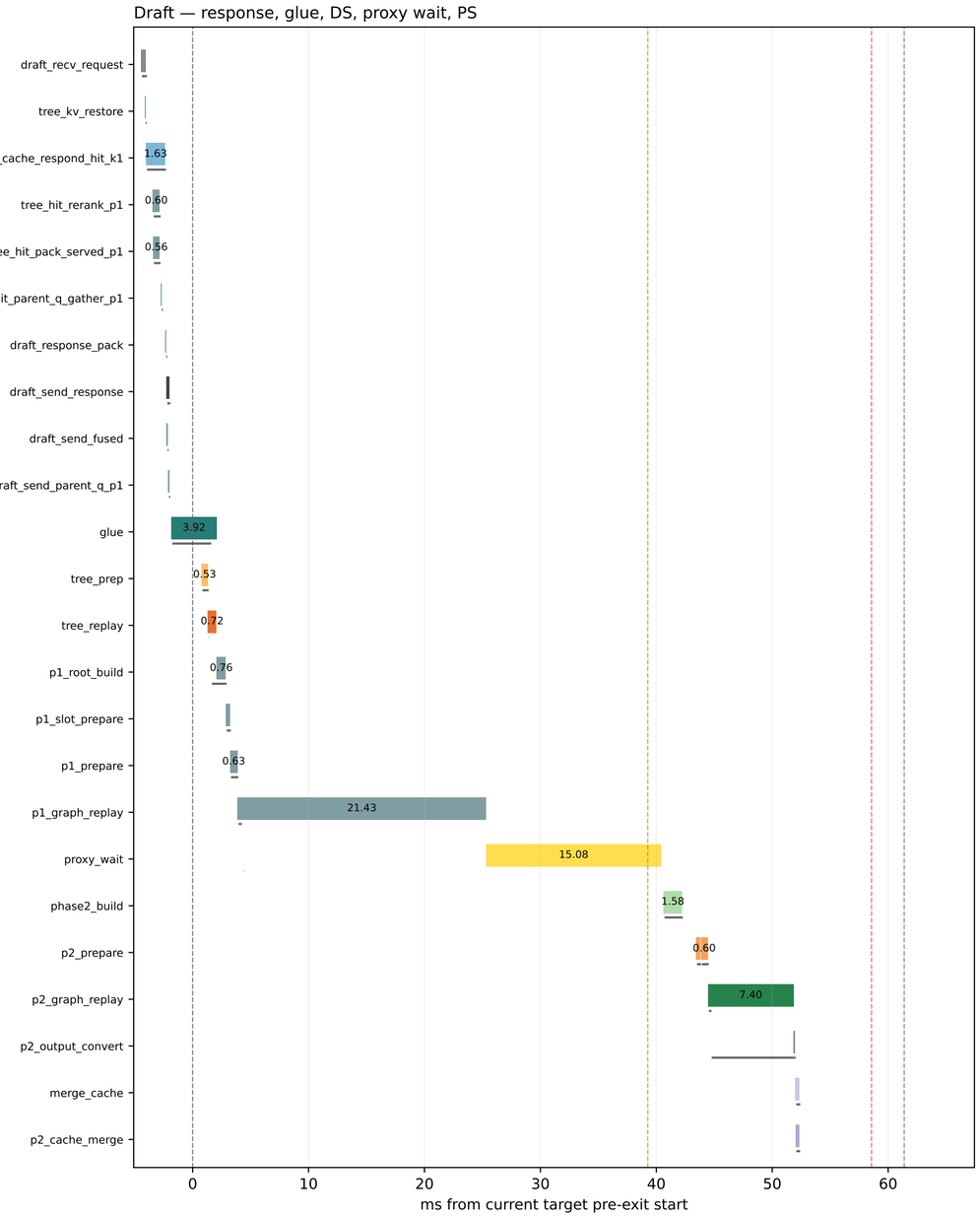
LayerSkip CodeLlama-34B + AMD-Llama-135M-code
P1 hit / p75 target-step duration / request 4, UID humaneval_local_033_t0, step 15

exit 30 (0-based) | K1/K2 12/6 | G=M DS/PS 12/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 59f46511 / eslab18 RTX3090 / 4 profiled turns, 73 eligible steps / output cap 256

valid proposal nodes=12, AL incl. recovery=12; DS margin +15.074 ms (estimated); PS margin +6.226 ms; next outcome P2 hit





Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

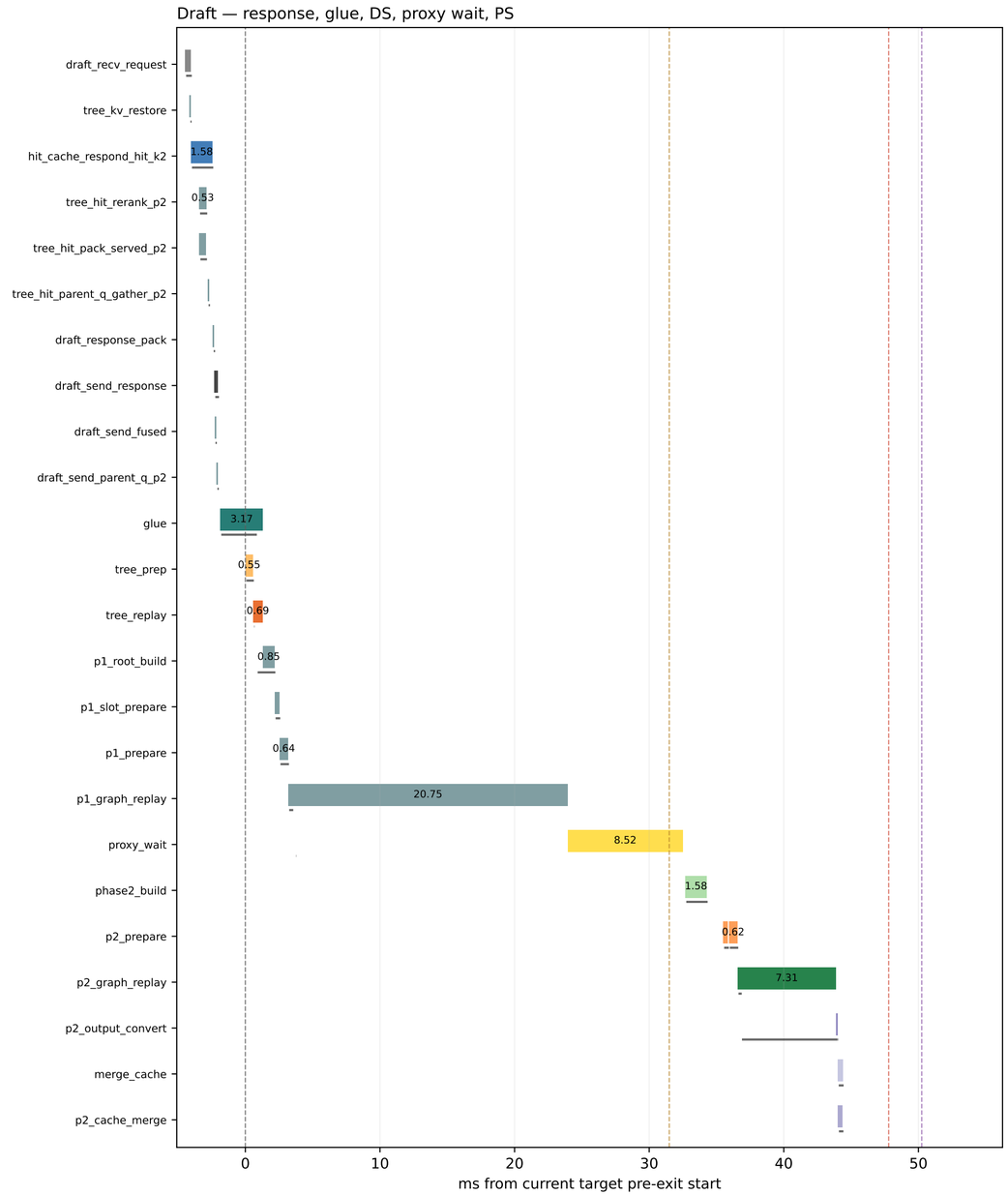
LayerSkip CodeLlama-34B + AMD-Llama-135M-code
P2 hit / p25 target-step duration / request 2, UID humaneval_local_013_t0, step 3

exit 30 (0-based) | K1/K2 12/6 | G=M DS/PS 12/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 59f46511 / eslab18 RTX3090 / 4 profiled turns, 73 eligible steps / output cap 256

valid proposal nodes=8, AL incl. recovery=6; DS margin +8.530 ms (estimated); PS margin +3.416 ms; next outcome P1 hit





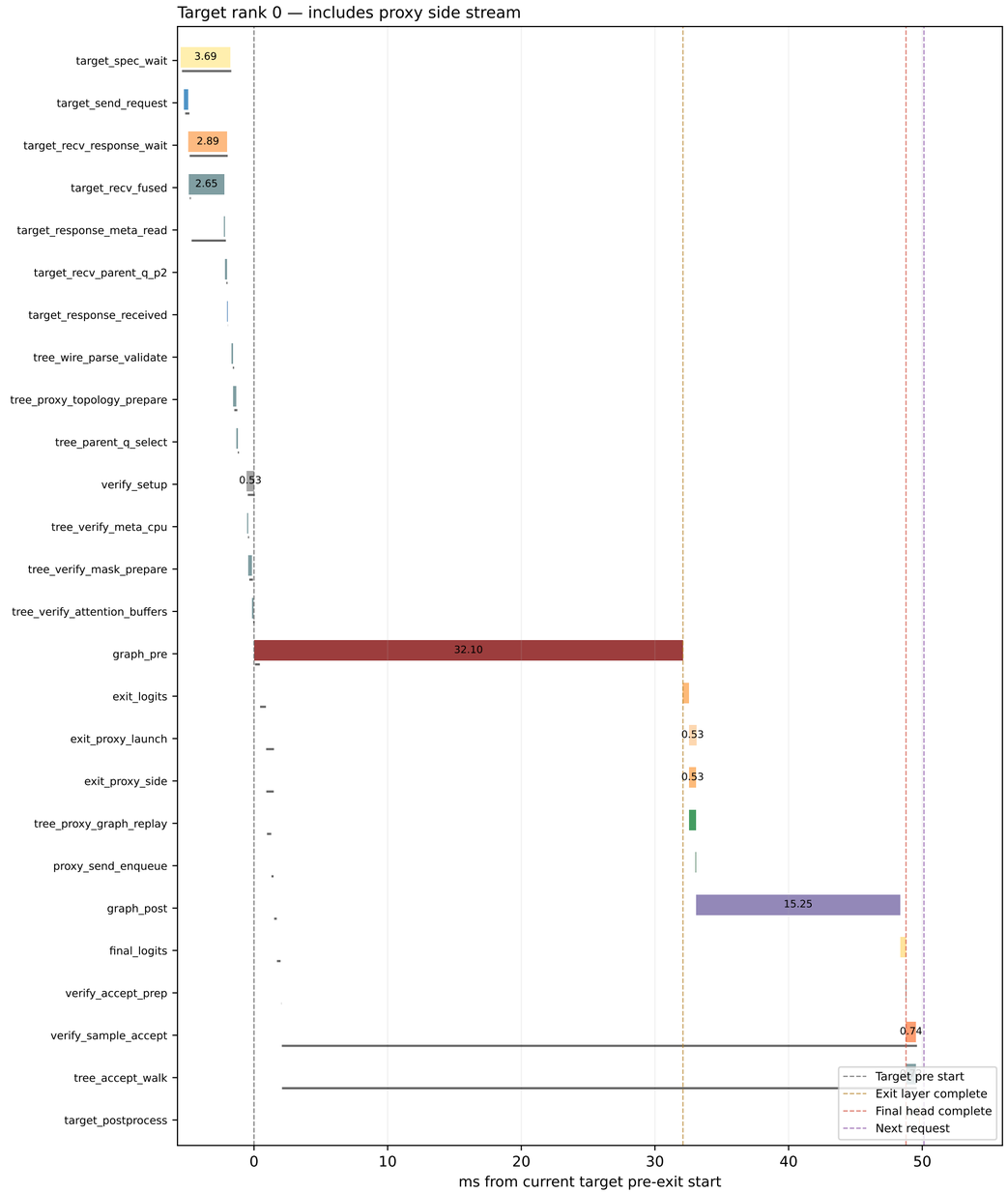
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

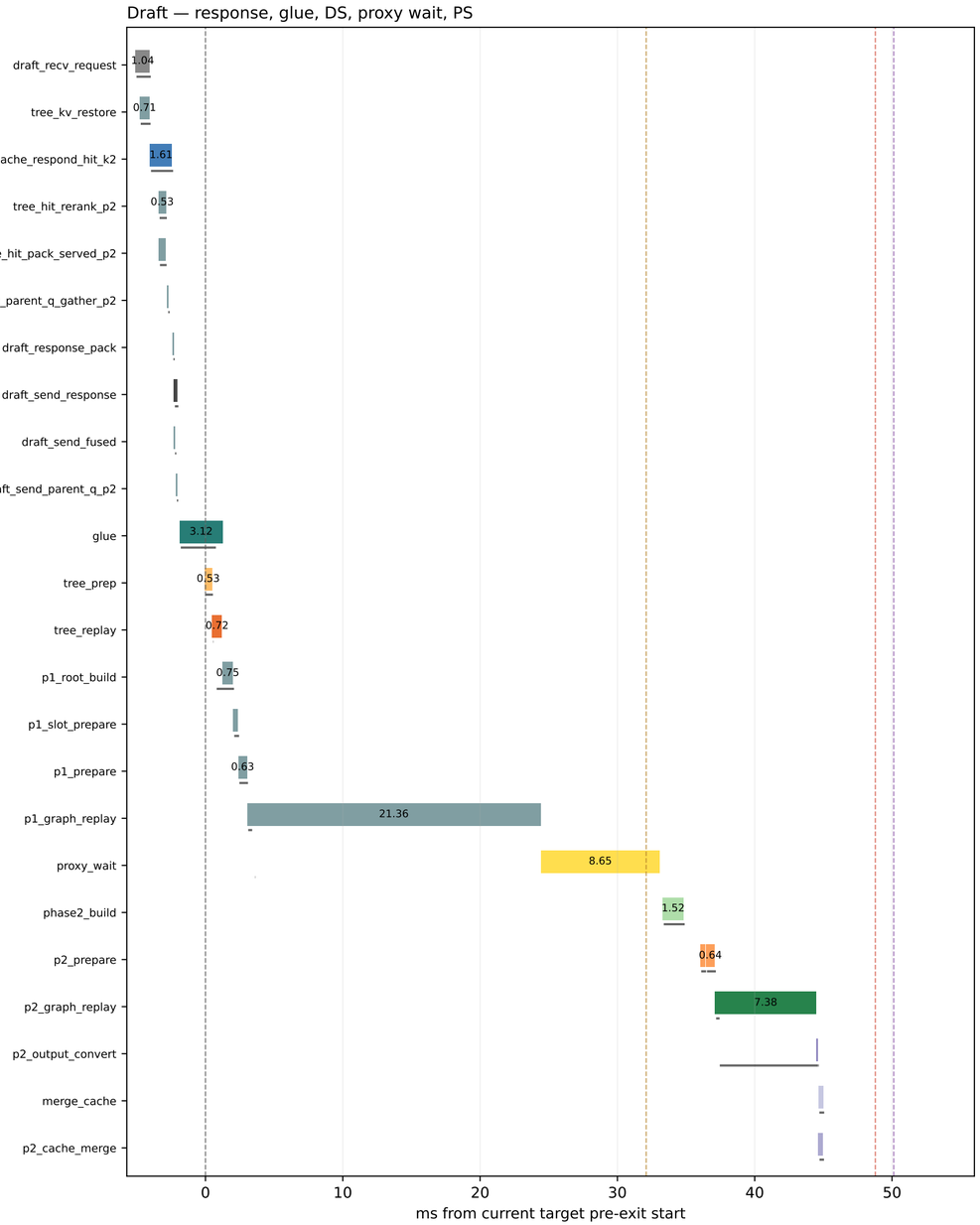
LayerSkip CodeLlama-34B + AMD-Llama-135M-code
P2 hit / p50 target-step duration / request 3, UID humaneval_local_023_t0, step 25

exit 30 (0-based) | K1/K2 12/6 | G=M DS/PS 12/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 59f46511 / eslab18 RTX3090 / 4 profiled turns, 73 eligible steps / output cap 256

valid proposal nodes=8, AL incl. recovery=2; DS margin +8.649 ms (estimated); PS margin +3.800 ms; next outcome P1 hit





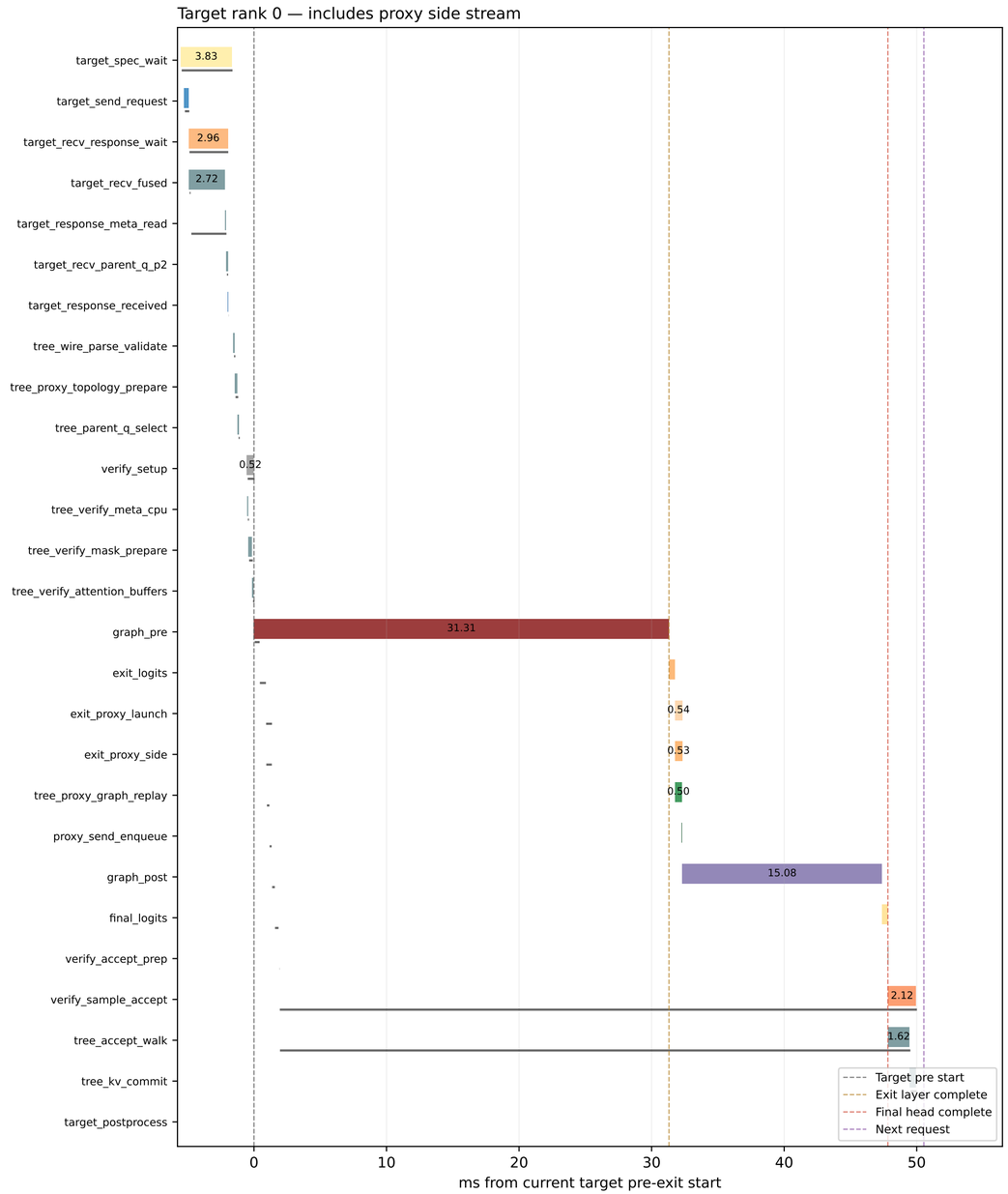
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

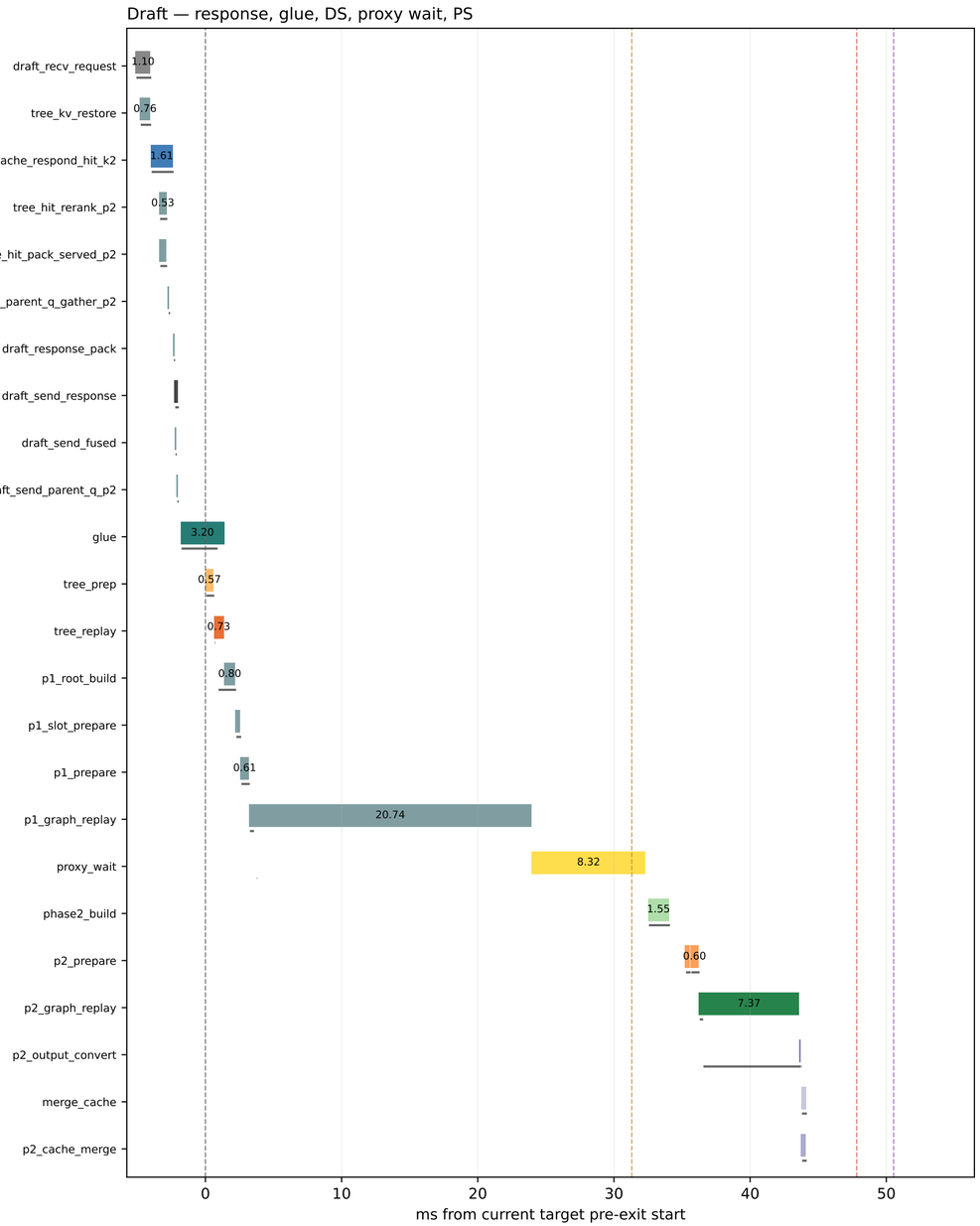
LayerSkip CodeLlama-34B + AMD-Llama-135M-code
P2 hit / p75 target-step duration / request 1, UID humaneval_local_003_t0, step 8

exit 30 (0-based) | K1/K2 12/6 | G=M DS/PS 12/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 59f46511 / eslab18 RTX3090 / 4 profiled turns, 73 eligible steps / output cap 256

valid proposal nodes=8, AL incl. recovery=7; DS margin +8.324 ms (estimated); PS margin +3.726 ms; next outcome P1 hit



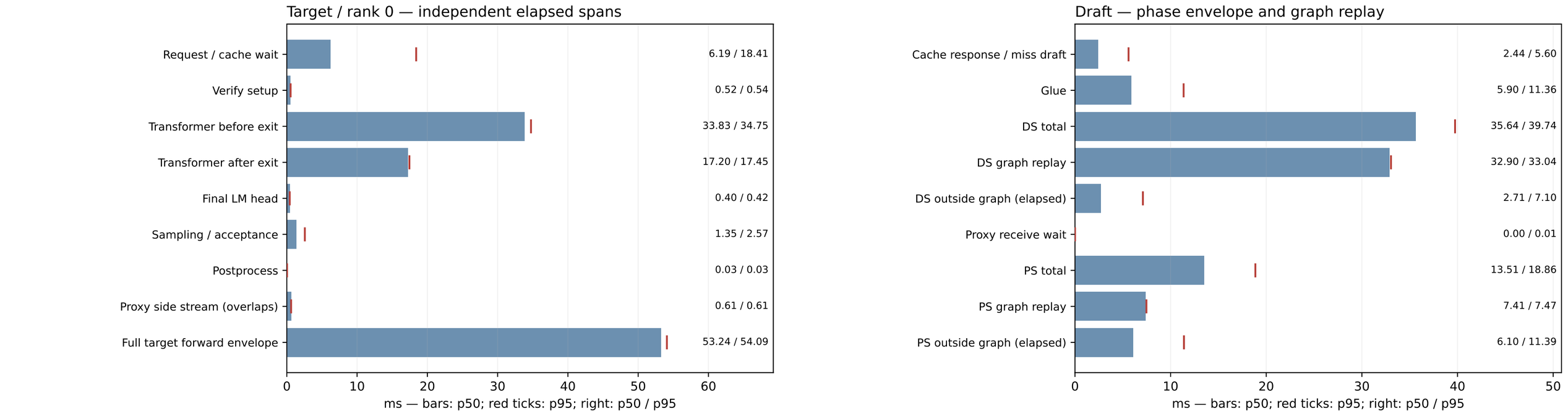


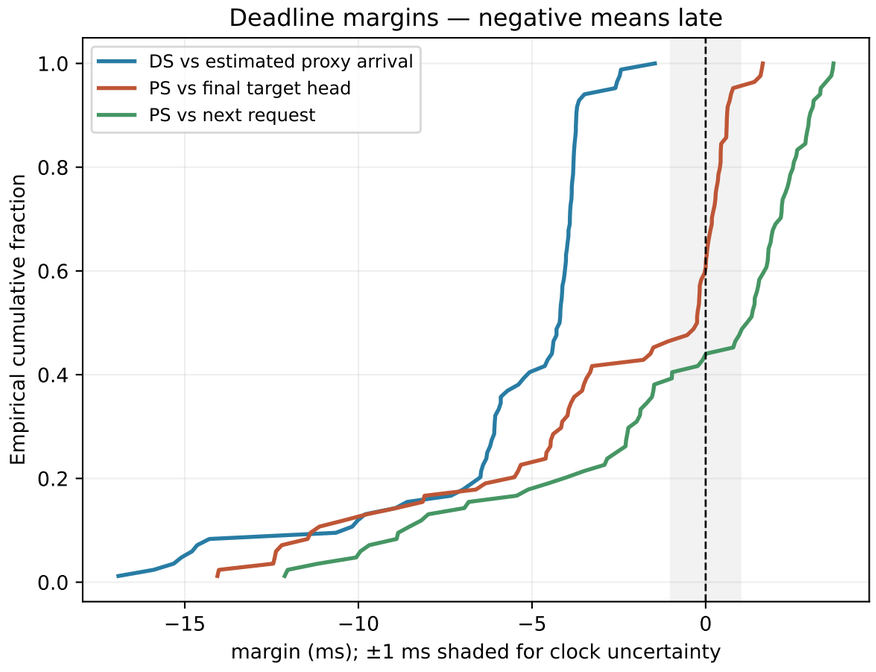
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

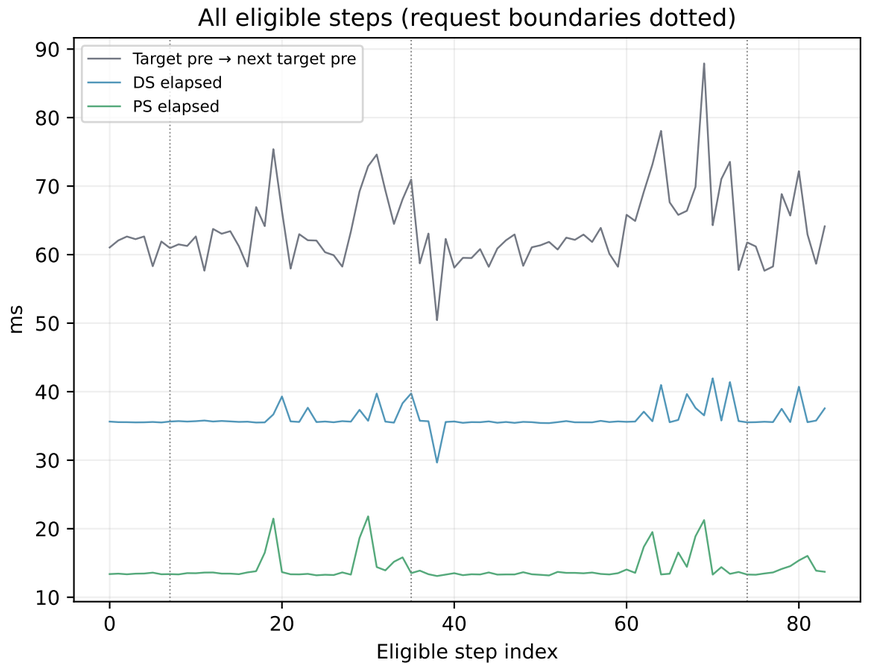
CodeLlama-34B + AMD-135M-code / throughput alternative
Component latency, phase deadlines, and every measured step

exit 30 (0-based) | K1/K2 16/6 | G=M DS/PS 16/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 59f46511 / eslab18 RTX3090 / 4 profiled turns, 84 eligible steps / output cap 256





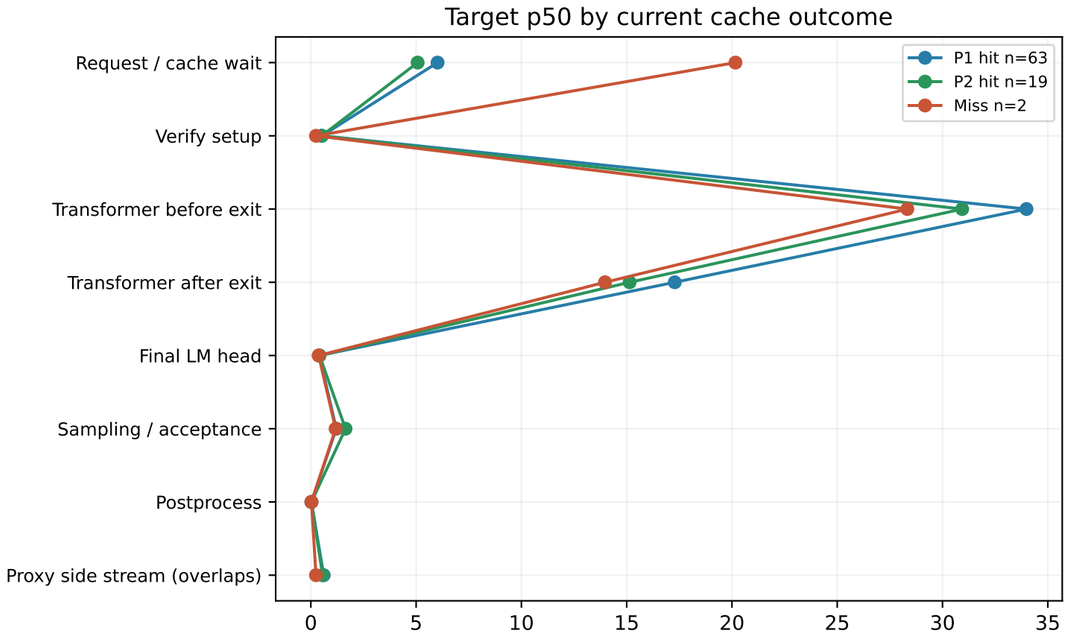


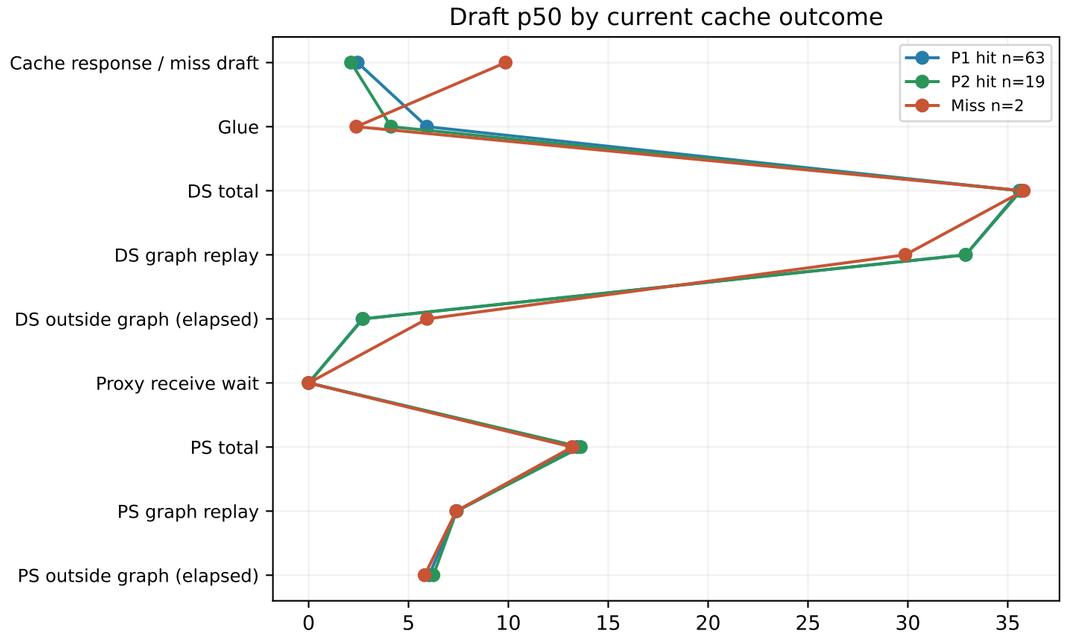
CUDA-event elapsed spans include queue/host stalls; not GPU busy time. Nested/overlapping components are NOT additive.

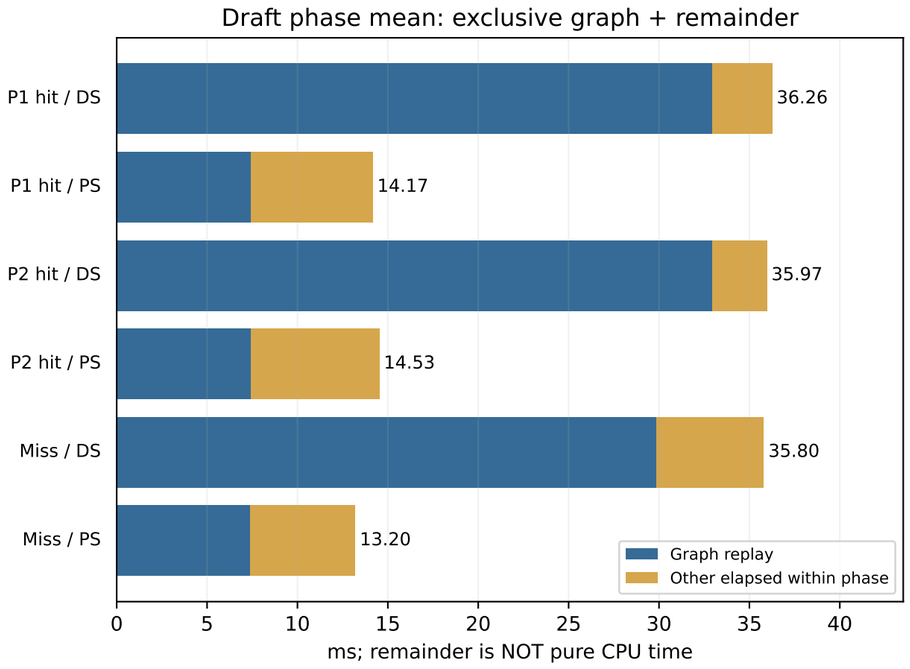
CodeLlama-34B + AMD-135M-code / throughput alternative
P1 hit / P2 hit / miss — component distributions and phase composition

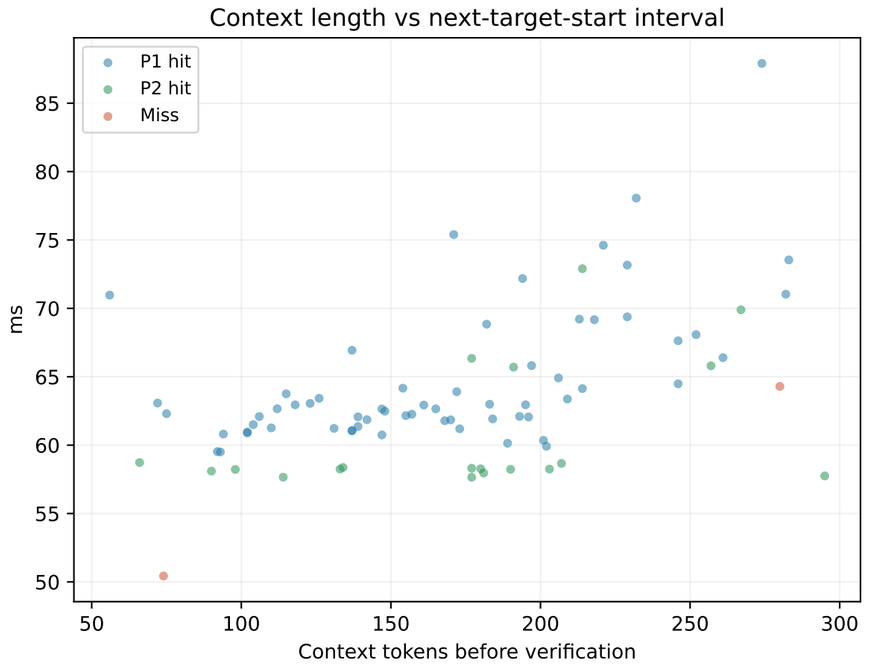
exit 30 (0-based) | K1/K2 16/6 | G=M DS/PS 16/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 59f46511 / eslab18 RTX3090 / 4 profiled turns, 84 eligible steps / output cap 256









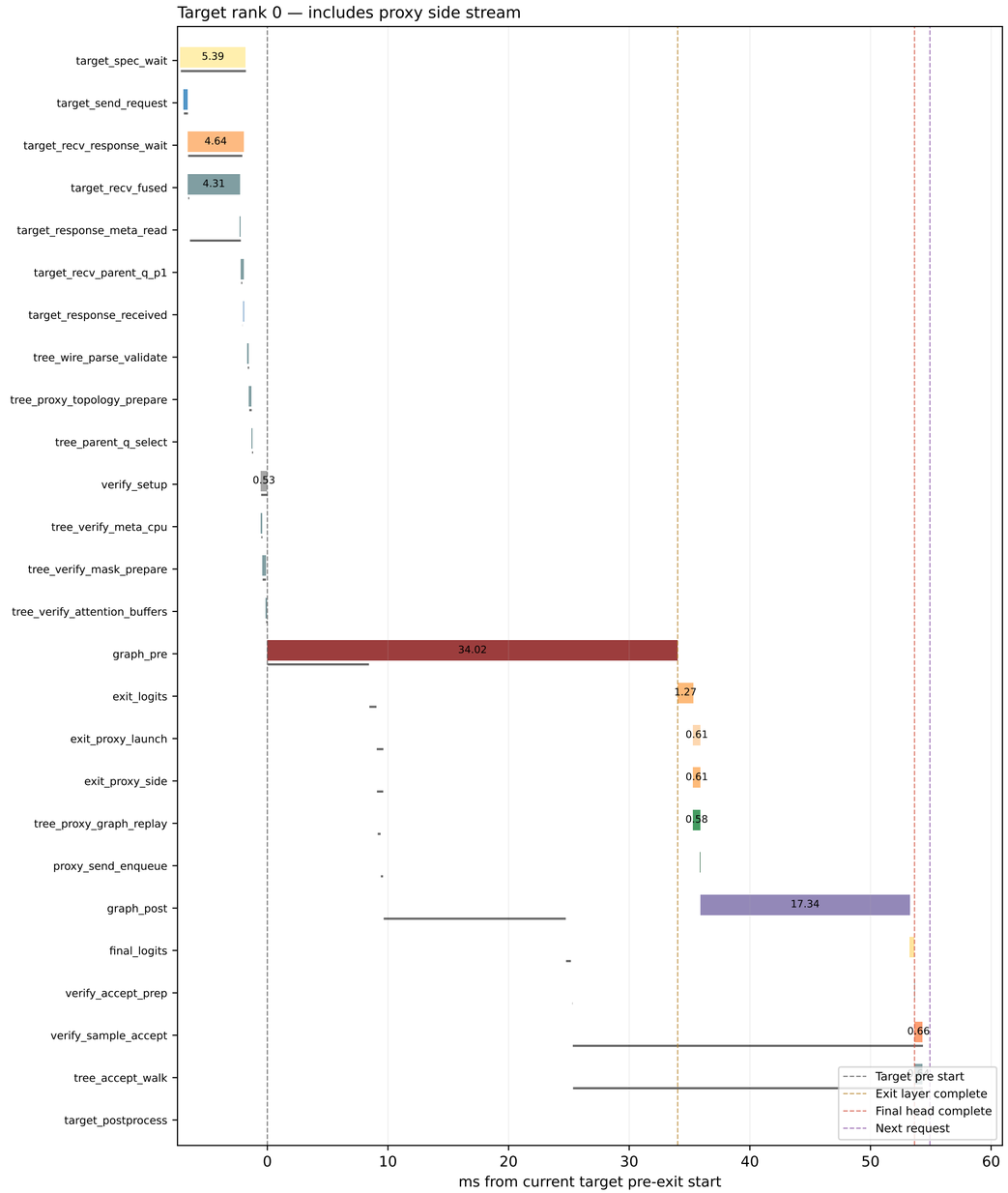
CUDA-event elapsed spans include queue/host stalls; not GPU busy time. Nested/overlapping components are NOT additive.

CodeLlama-34B + AMD-135M-code / throughput alternative
P1 hit / p25 target-step duration / request 3, UID humaneval_local_023_t0, step 26

exit 30 (0-based) | K1/K2 16/6 | G=M DS/PS 16/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 59f46511 / eslab18 RTX3090 / 4 profiled turns, 84 eligible steps / output cap 256

valid proposal nodes=16, AL incl. recovery=1; DS margin -3.798 ms (estimated); PS margin +0.620 ms; next outcome P2 hit





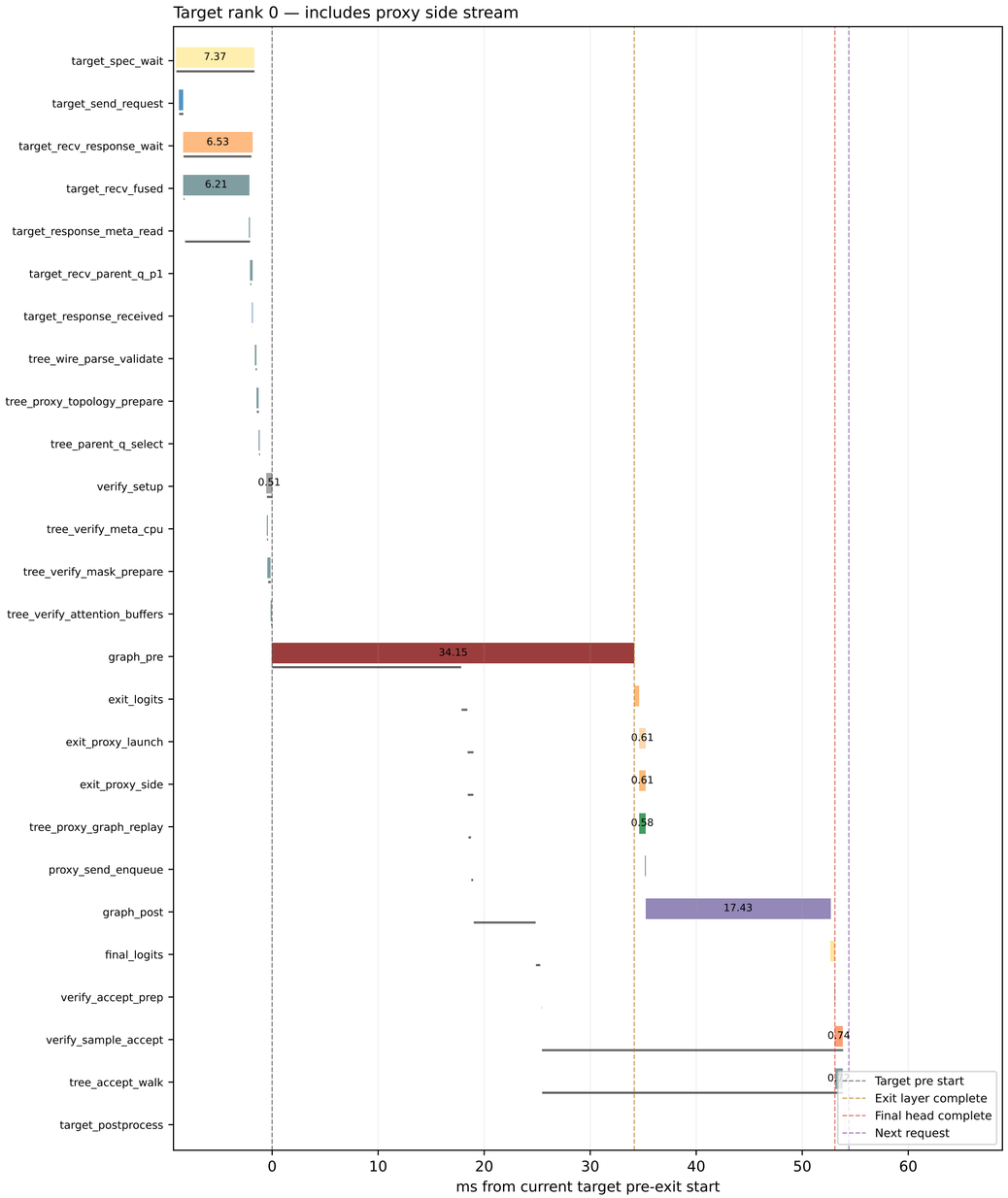
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

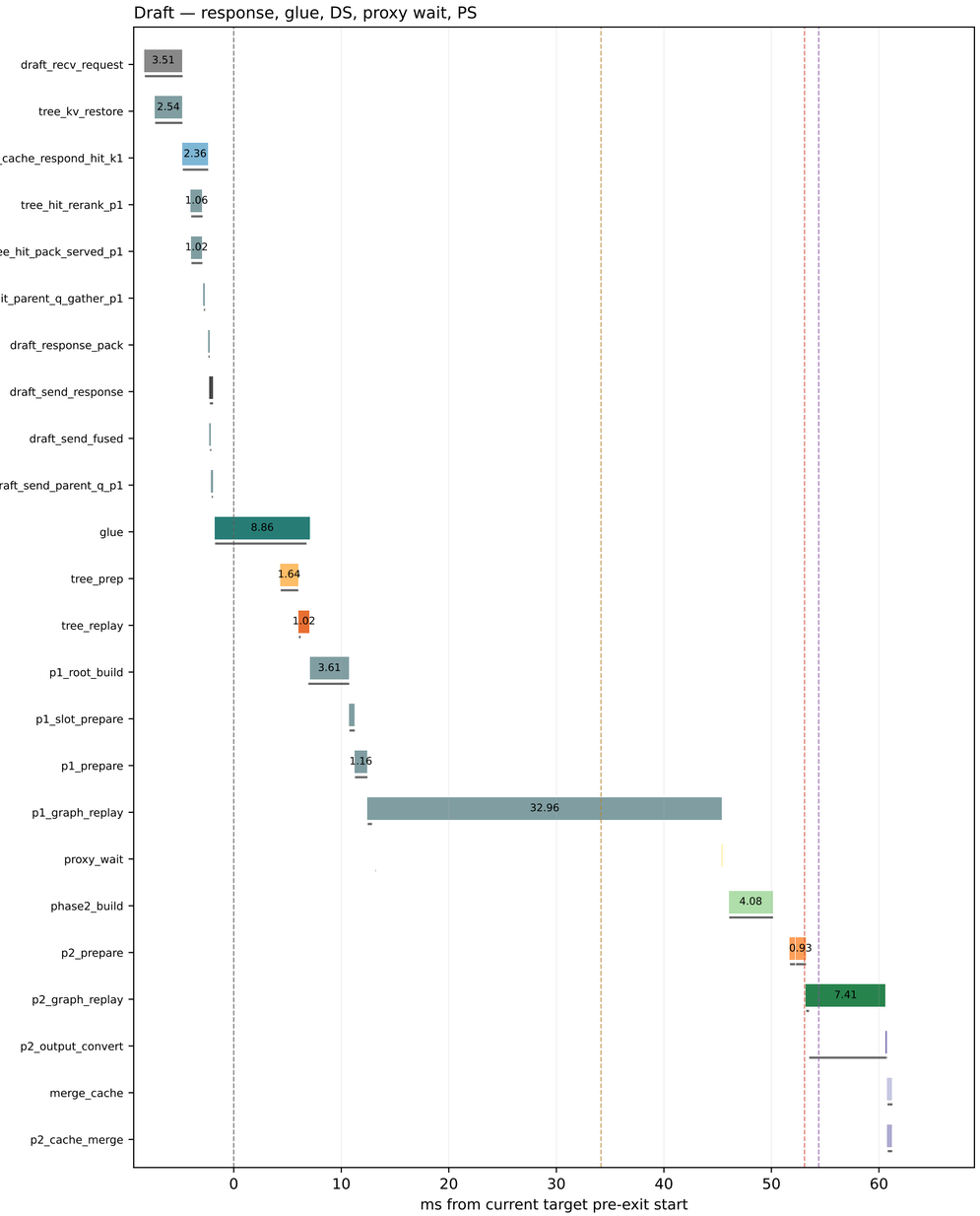
CodeLlama-34B + AMD-135M-code / throughput alternative
P1 hit / p50 target-step duration / request 2, UID humaneval_local_013_t0, step 30

exit 30 (0-based) | K1/K2 16/6 | G=M DS/PS 16/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 59f46511 / eslab18 RTX3090 / 4 profiled turns, 84 eligible steps / output cap 256

valid proposal nodes=16, AL incl. recovery=2; DS margin -10.177 ms (estimated); PS margin -8.147 ms; next outcome P1 hit





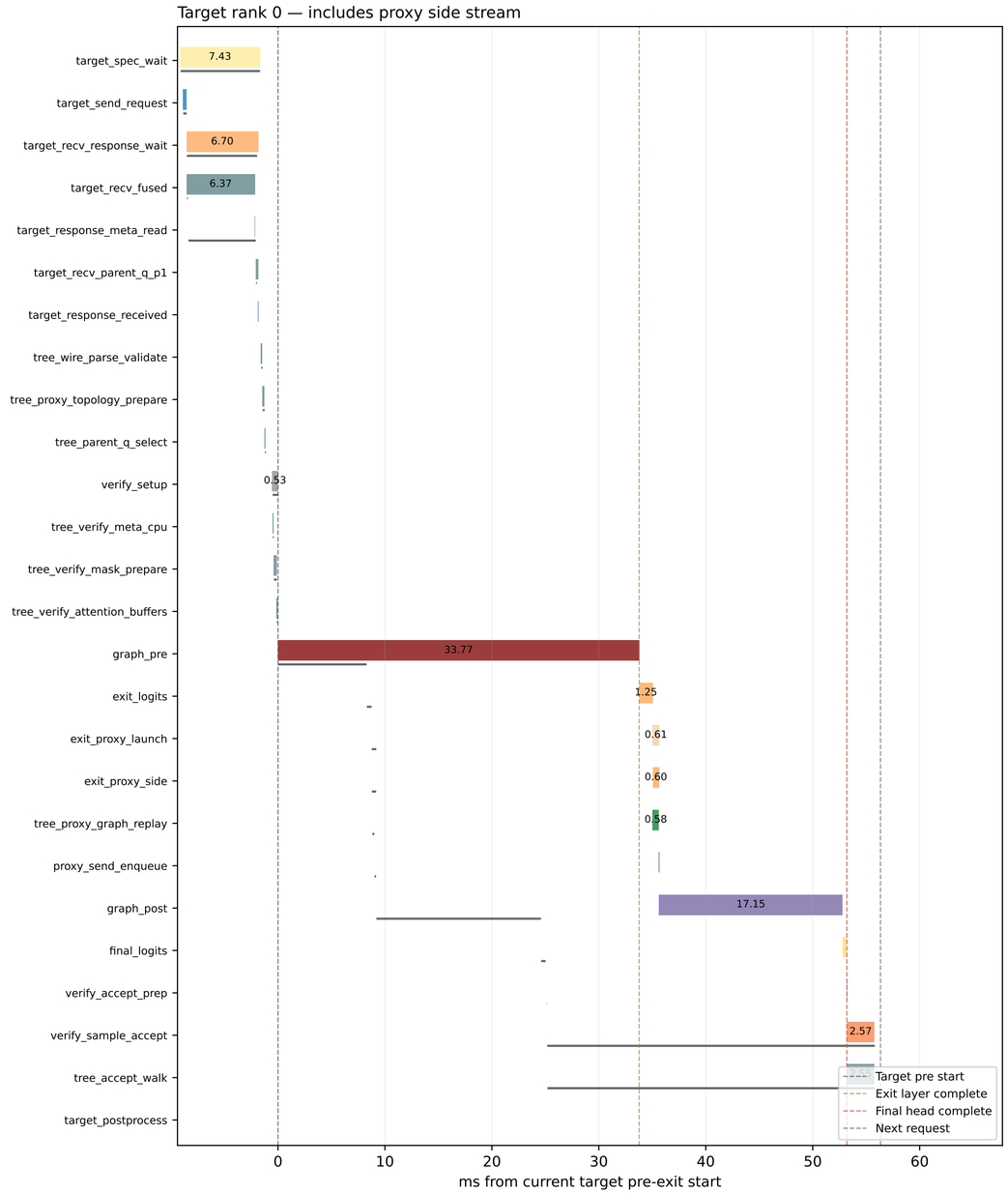
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

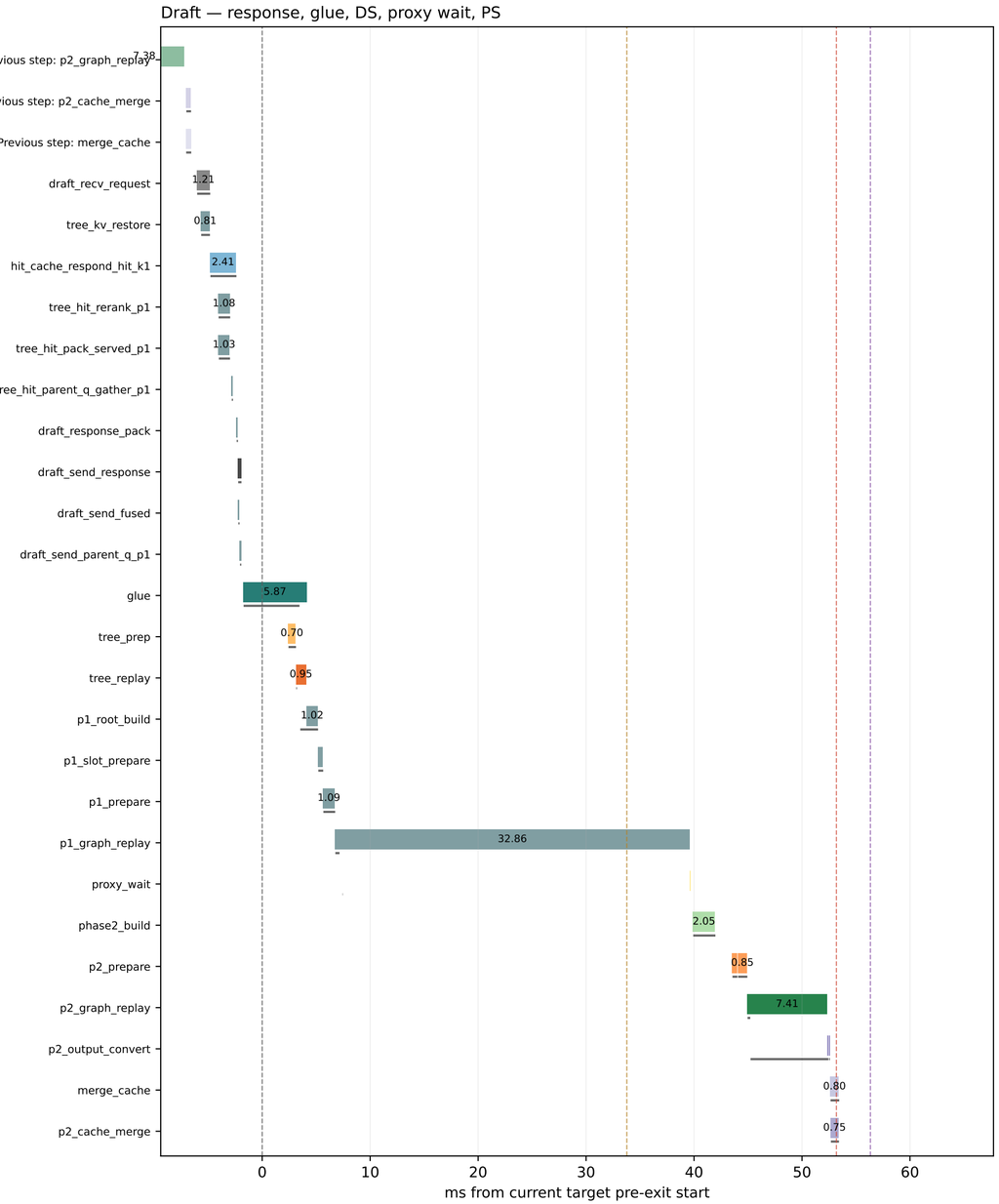
CodeLlama-34B + AMD-135M-code / throughput alternative
P1 hit / p75 target-step duration / request 2, UID humaneval_local_013_t0, step 13

exit 30 (0-based) | K1/K2 16/6 | G=M DS/PS 16/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 59f46511 / eslab18 RTX3090 / 4 profiled turns, 84 eligible steps / output cap 256

valid proposal nodes=16, AL incl. recovery=17; DS margin -3.975 ms (estimated); PS margin -0.225 ms; next outcome P1 hit





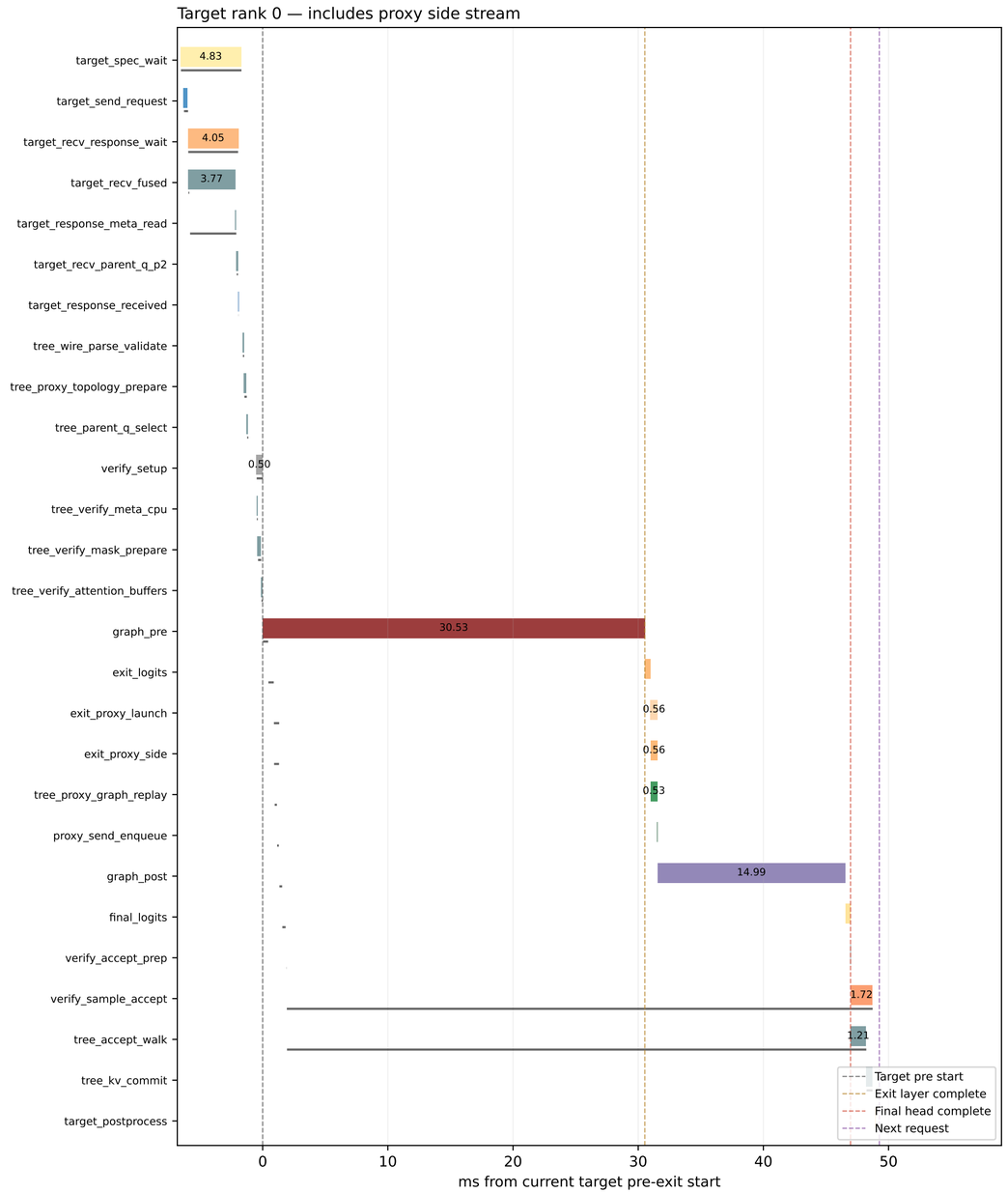
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

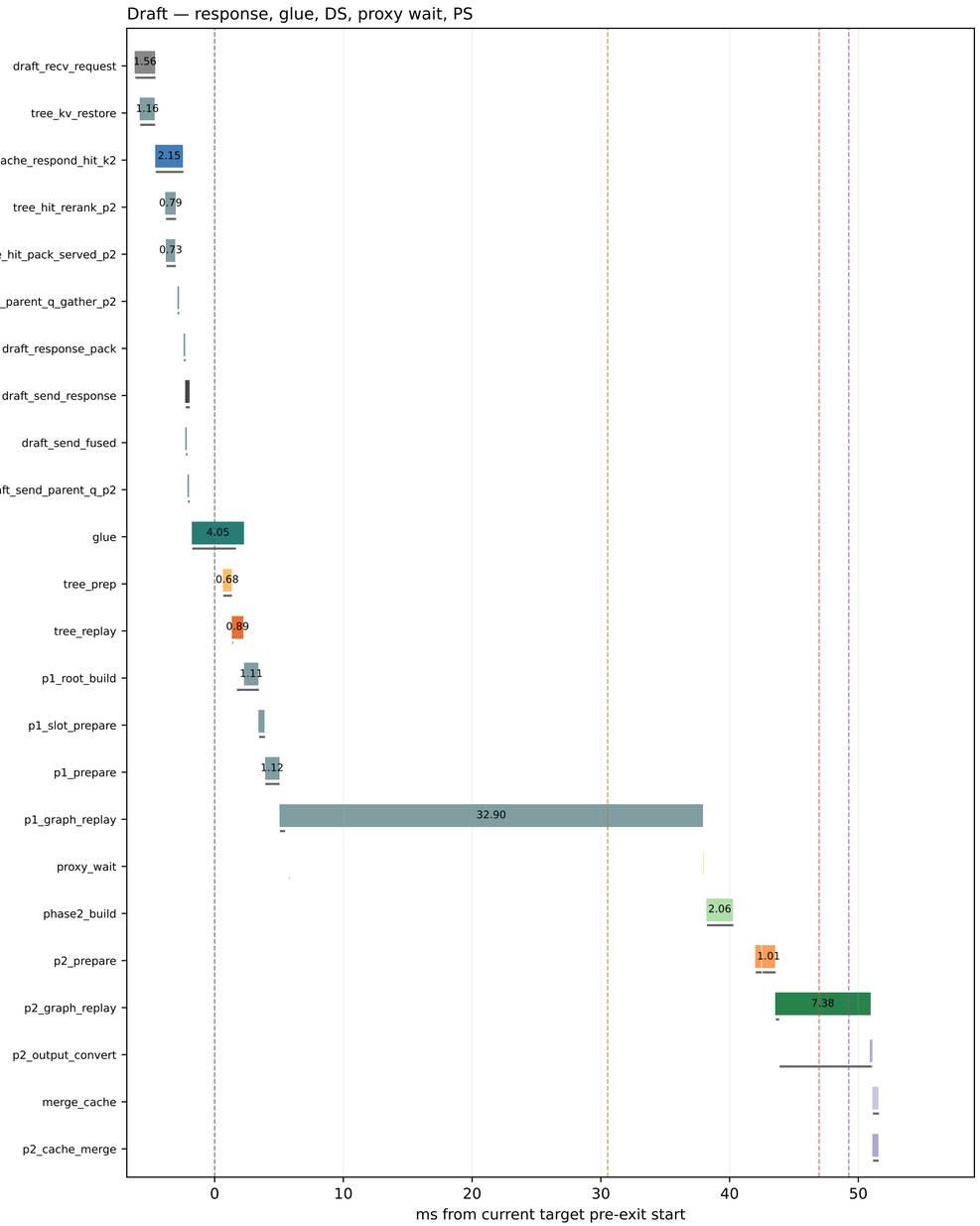
CodeLlama-34B + AMD-135M-code / throughput alternative
P2 hit / p25 target-step duration / request 3, UID humaneval_local_023_t0, step 12

exit 30 (0-based) | K1/K2 16/6 | G=M DS/PS 16/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 59f46511 / eslab18 RTX3090 / 4 profiled turns, 84 eligible steps / output cap 256

valid proposal nodes=8, AL incl. recovery=4; DS margin -6.409 ms (estimated); PS margin -4.607 ms; next outcome P1 hit





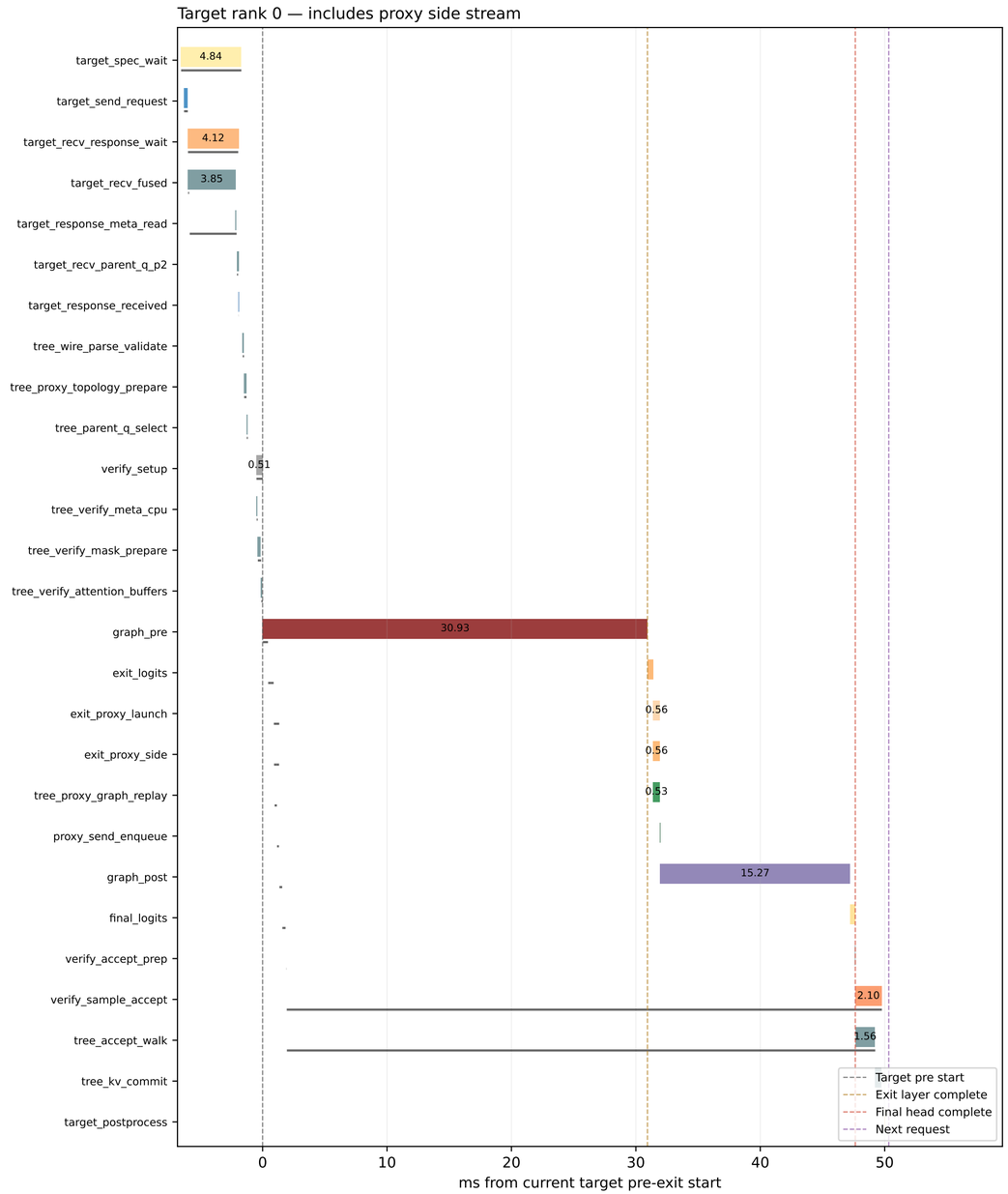
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

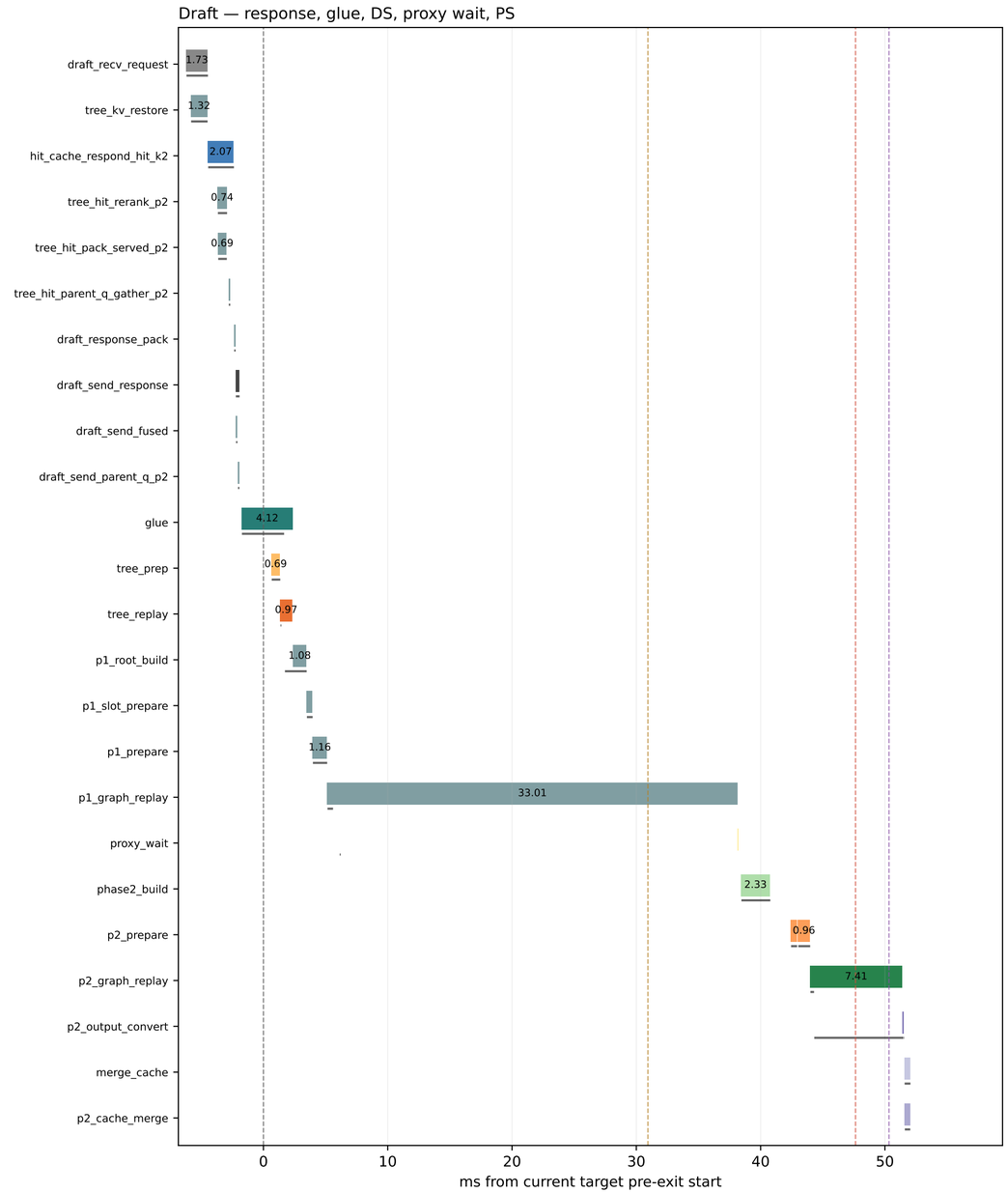
CodeLlama-34B + AMD-135M-code / throughput alternative
P2 hit / p50 target-step duration / request 4, UID humaneval_local_033_t0, step 11

exit 30 (0-based) | K1/K2 16/6 | G=M DS/PS 16/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 59f46511 / eslab18 RTX3090 / 4 profiled turns, 84 eligible steps / output cap 256

valid proposal nodes=8, AL incl. recovery=7; DS margin -6.208 ms (estimated); PS margin -4.390 ms; next outcome P1 hit





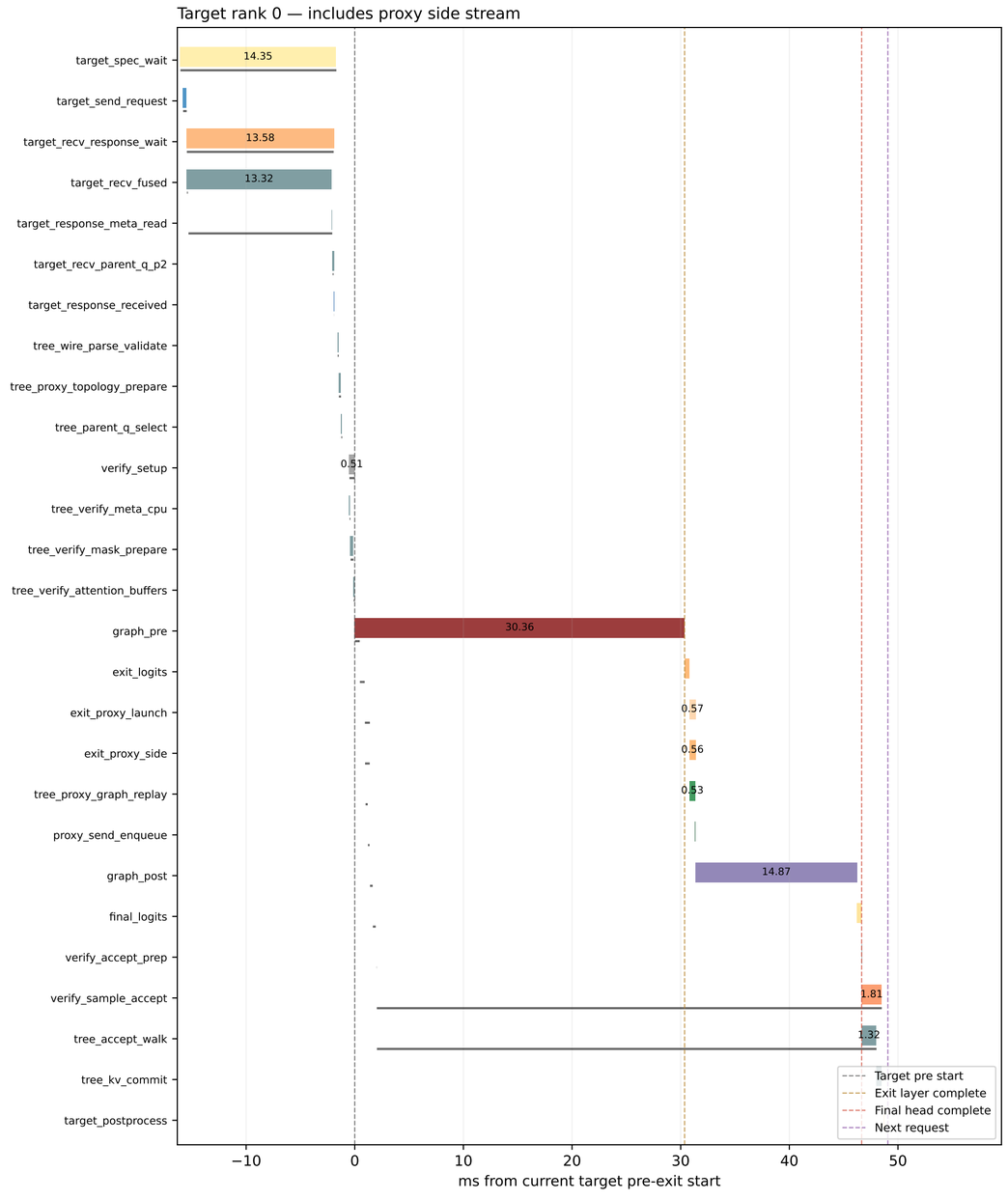
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

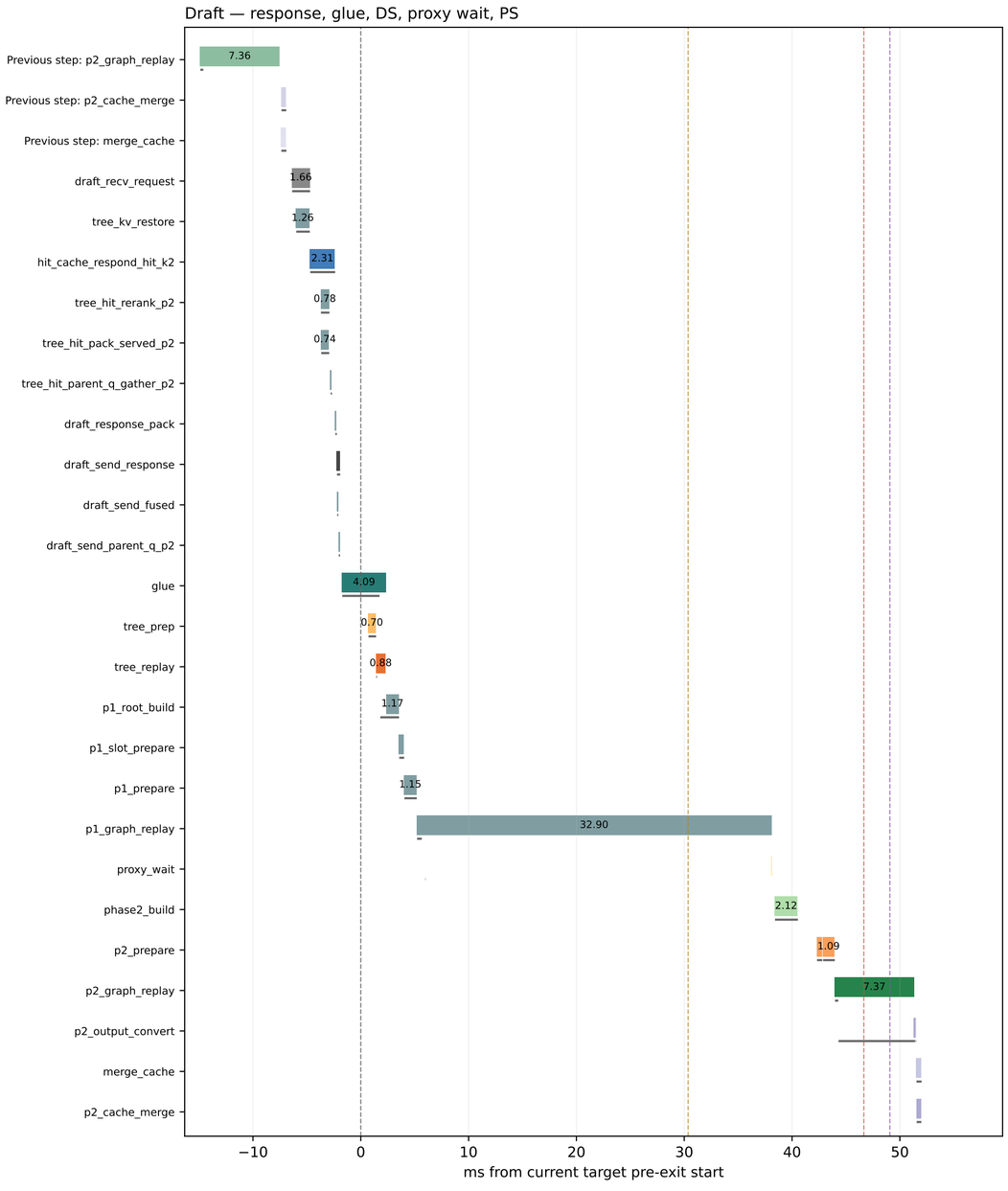
CodeLlama-34B + AMD-135M-code / throughput alternative
P2 hit / p75 target-step duration / request 3, UID humaneval_local_023_t0, step 4

exit 30 (0-based) | K1/K2 16/6 | G=M DS/PS 16/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 59f46511 / eslab18 RTX3090 / 4 profiled turns, 84 eligible steps / output cap 256

valid proposal nodes=8, AL incl. recovery=6; DS margin -6.734 ms (estimated); PS margin -5.324 ms; next outcome P1 hit





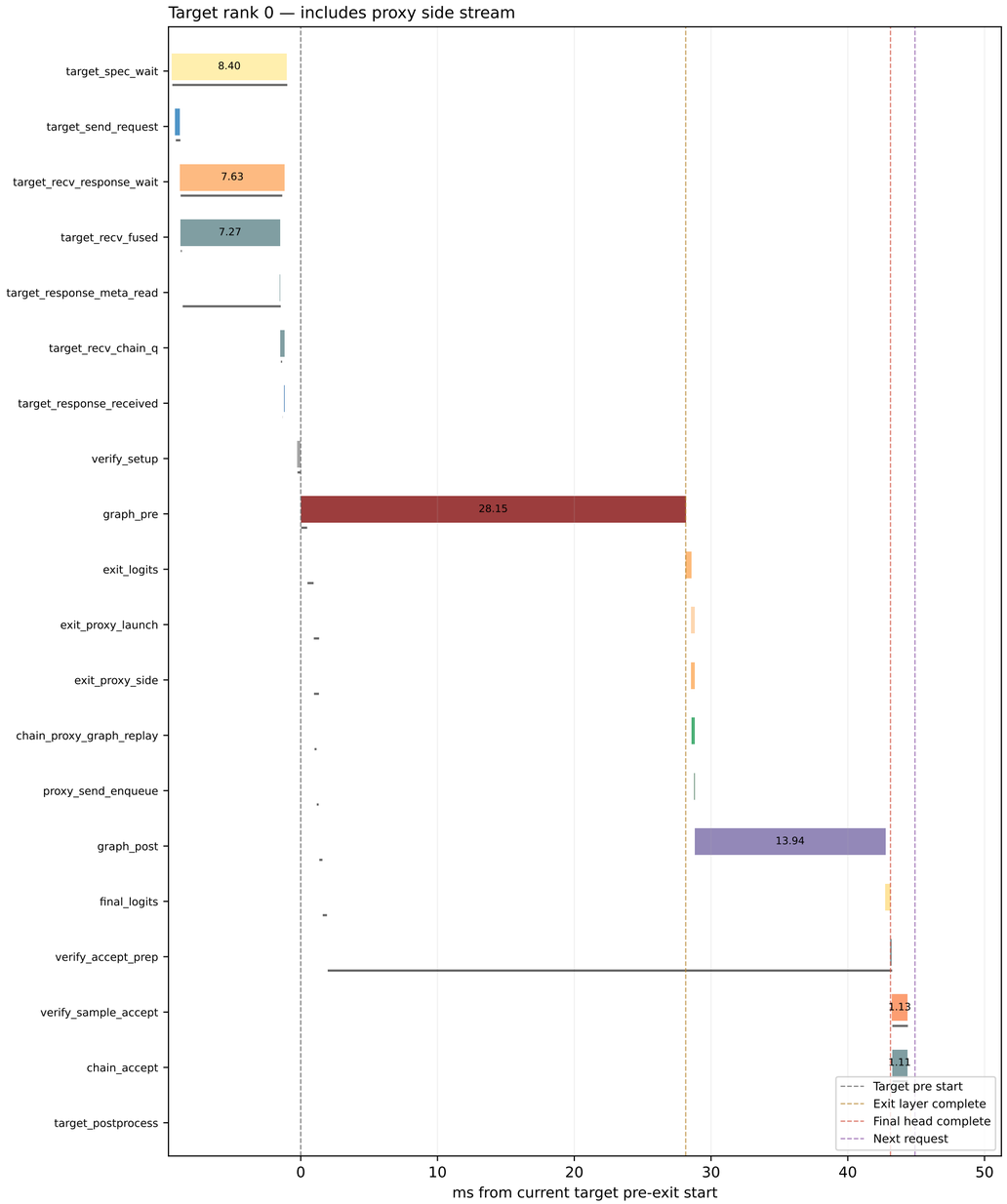
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

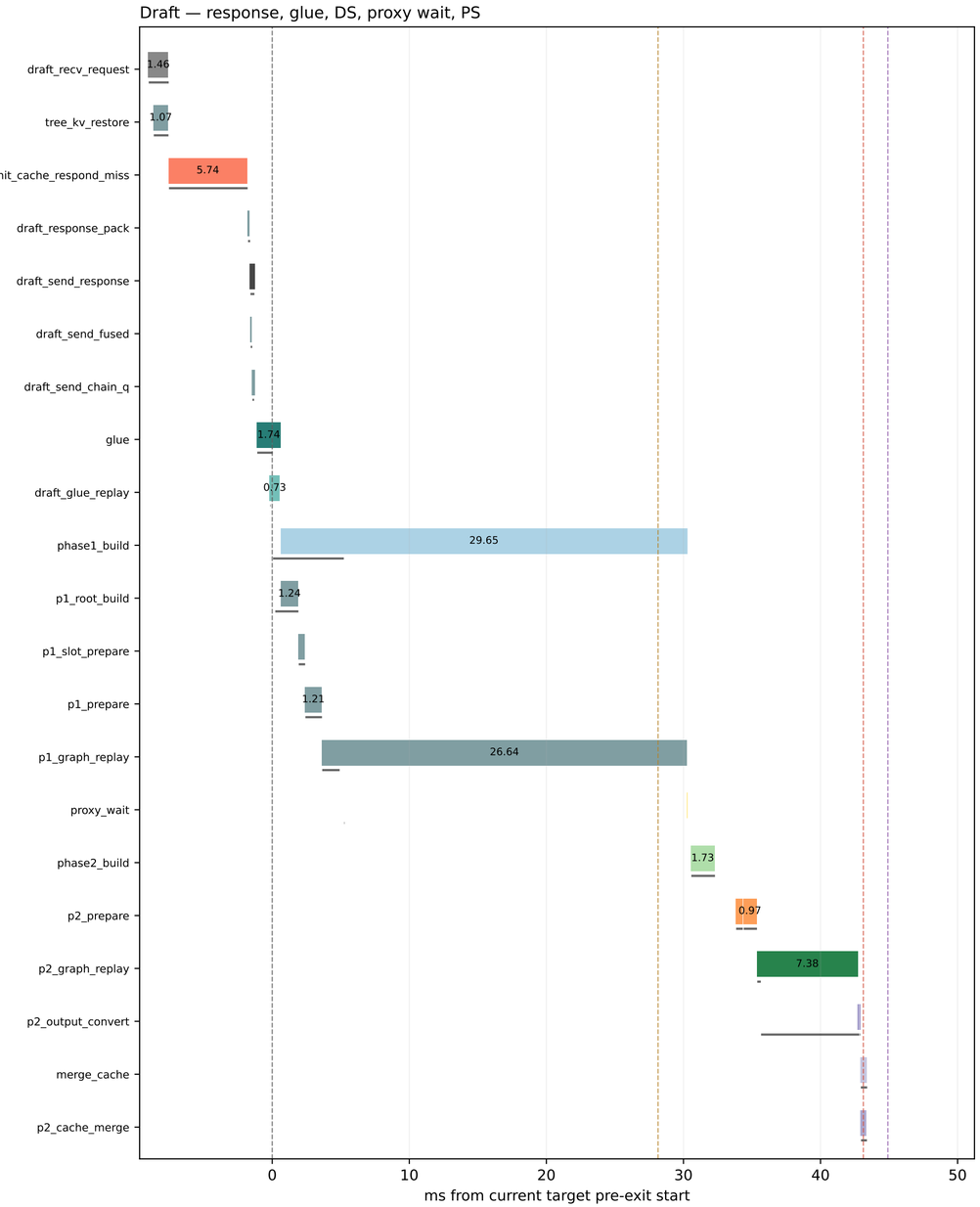
CodeLlama-34B + AMD-135M-code / throughput alternative
Miss / p25 target-step duration / request 3, UID humaneval_local_023_t0, step 6

exit 30 (0-based) | K1/K2 16/6 | G=M DS/PS 16/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 59f46511 / eslab18 RTX3090 / 4 profiled turns, 84 eligible steps / output cap 256

valid proposal nodes=6, AL incl. recovery=1; DS margin -1.462 ms (estimated); PS margin -0.247 ms; next outcome P1 hit





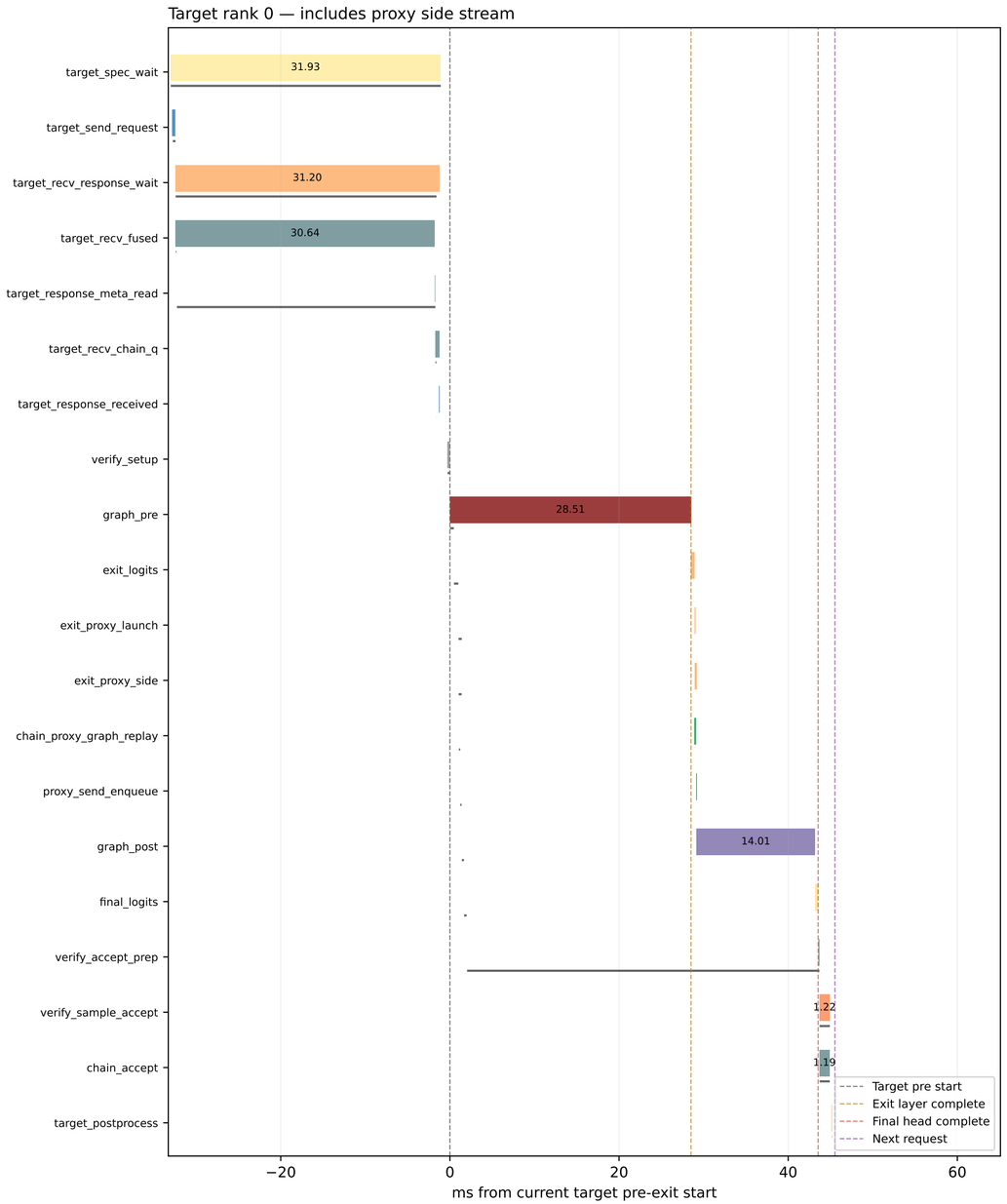
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

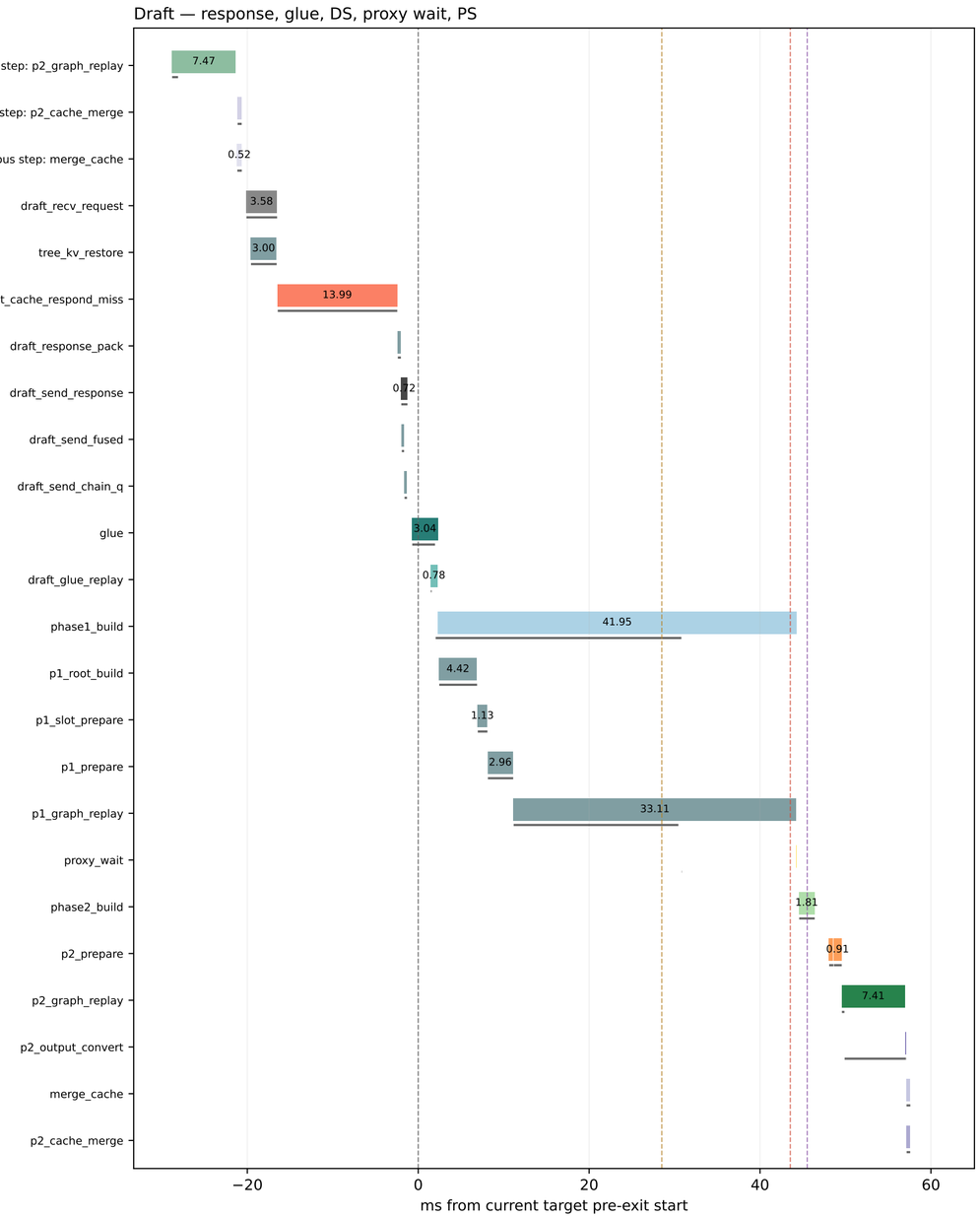
CodeLlama-34B + AMD-135M-code / throughput alternative
Miss / p75 target-step duration / request 3, UID humaneval_local_023_t0, step 38

exit 30 (0-based) | K1/K2 16/6 | G=M DS/PS 16/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 59f46511 / eslab18 RTX3090 / 4 profiled turns, 84 eligible steps / output cap 256

valid proposal nodes=6, AL incl. recovery=2; DS margin -15.098 ms (estimated); PS margin -14.018 ms; next outcome P1 hit





Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.